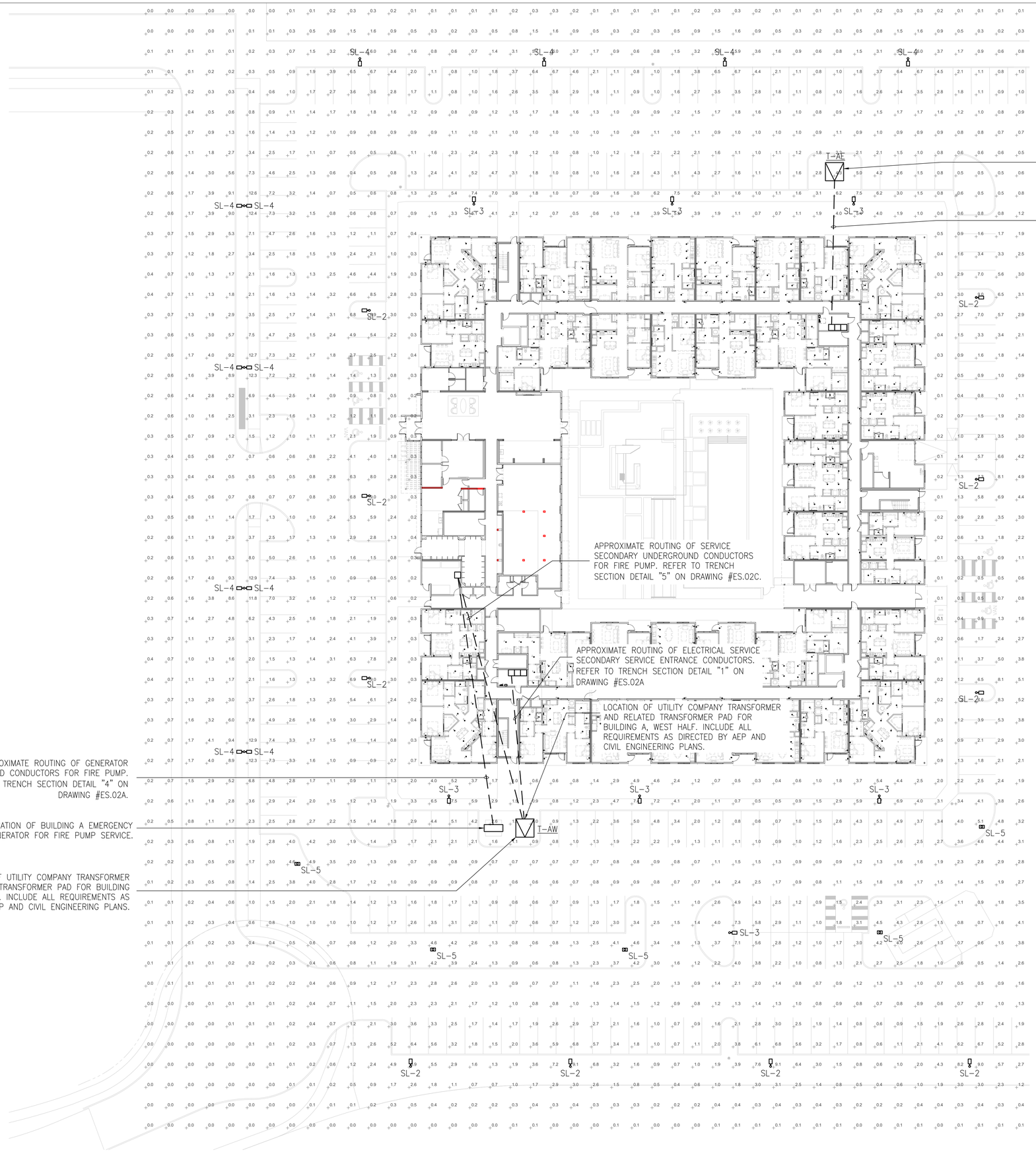

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

whs engineering
2012 West 25th Street, Suite 900
Cleveland, OH 44113
Phone: 216 227 8505

OSU AIRPORT WEST

BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title

BUILDING A SITE ELECTRICAL PLAN

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number
Checked by: KDS	ES.01A
Approved: GSS	

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OSU
AIRPORT WEST

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3455 W. Dublin Granville Rd,
Columbus OH, 43235

Drawing Title

BUILDING A SITE
ELECTRICAL DETAILS

Date	Project Number
03.05.2026	24-065

Drawn by	Drawing Number
DJS	ES.02A
Checked by: KDS	
Approved: GSS	

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PART 1 – ELECTRICAL GENERAL PROVISIONS

1.1 RELATED DOCUMENTS

A. THE PROVISIONS OF THE INSTRUCTIONS TO BIDDERS, GENERAL CONDITIONS, SUPPLEMENTARY CONDITIONS, ALTERNATES, ADDENDA'S, AND DIVISION 1 ARE A PART OF THIS SPECIFICATION. CONTRACTORS AND SUBCONTRACTORS SHALL EXAMINE SAME AS WELL AS OTHER DIVISIONS OF THE SPECIFICATIONS WHICH AFFECT WORK UNDER THIS DIVISION.

1.2 SUMMARY

A. THIS SECTION INCLUDES GENERAL ADMINISTRATIVE AND PROCEDURAL REQUIREMENTS FOR ELECTRICAL INSTALLATIONS COMMON TO ALL SECTIONS OF DIVISION 16. THE ADMINISTRATIVE AND PROCEDURAL REQUIREMENTS IN THIS SECTION EXPAND AND SUPPLEMENT THE REQUIREMENTS SPECIFIED IN DIVISION 1.

1.3 DESCRIPTION OF WORK

A. ELECTRICAL, ARCHITECTURAL, MECHANICAL, PLUMBING, FIRE PROTECTION, STRUCTURAL, ETC., AND ALL OTHER DRAWINGS AS WELL AS THE SPECIFICATIONS FOR ALL THE DIVISIONS ARE A PART OF THE CONTRACT DOCUMENTS.
B. DRAWINGS AND SPECIFICATIONS ARE TO BE CONSIDERED AS SUPPLEMENTING EACH OTHER. WORK SPECIFIED BUT NOT SHOWN, OR SHOWN BUT NOT SPECIFIED, SHALL BE PERFORMED OR FURNISHED AS THOUGH MENTIONED IN BOTH SPECIFICATIONS AND DRAWINGS.
C. VISIT THE SITE OF THE WORK AND BECOME FAMILIAR WITH CONDITIONS AFFECTING THE INSTALLATION. SUBMISSION OF A PROPOSAL SHALL PRESUPPOSE KNOWLEDGE OF SUCH CONDITIONS AND NO ADDITIONAL COMPENSATION SHALL BE ALLOWED WHERE EXTRA LABOR OR MATERIALS ARE REQUIRED BECAUSE OF IGNORANCE OF THESE CONDITIONS.

1.4 WORK INCLUDES

A. INCLUDE ALL LABOR, MATERIAL, EQUIPMENT, SERVICES, AND PERMITS NECESSARY FOR THE PROPER COMPLETION OF ALL ELECTRICAL WORK SHOWN. ITEMS OMITTED, BUT NECESSARY TO MAKE THE ELECTRICAL SYSTEM COMPLETE AND WORKABLE, SHALL BE UNDERSTOOD TO FORM PART OF THE WORK.
B. IT IS THE PURPOSE OF THE ELECTRICAL DRAWINGS TO INDICATE THE APPROXIMATE LOCATION OF ALL EQUIPMENT, OUTLETS, ETC. ASCERTAIN EXACT LOCATIONS AND ARRANGE WORK ACCORDINGLY. THE RIGHT IS RESERVED TO EFFECT REASONABLE CHANGES IN THE LOCATION OF OUTLETS UP TO THE TIME OF ROUGHING–IN, WITHOUT ADDITIONAL COST TO THE OWNER. CHANGES IN LOCATION OF OUTLETS OR EQUIPMENT NECESSITATED BY INTERFERENCE WITH THE WORK OF OTHER TRADES SHALL BE MADE ONLY WITH THE CONSENT OF THE ARCHITECT AND ENGINEER OR OWNER'S REPRESENTATIVE, AND AT NO ADDITIONAL COST.
C. AS USED IN THIS SPECIFICATION, "PROVIDE" MEANS "FURNISH AND INSTALL" AND "HVAC" MEANS "HEATING, VENTILATING AND AIR CONDITIONING" AND "POS" MEANS "PROVIDED UNDER OTHER SECTIONS". "FURNISH" MEANS TO PURCHASE AND DELIVER TO THE PROJECT SITE COMPLETE WITH EVERY NECESSARY APPURTENANCE AND SUPPORT," AND "INSTALL" MEANS "TO UNLOAD AT THE DELIVERY POINT AT THE SITE AND PERFORM EVERY OPERATION NECESSARY FOR PROPER INSTALLATION PER CODES AND MANUFACTURERS REQUIREMENTS, TO ESTABLISH SECURE MOUNTING AND CORRECT OPERATION AT THE PROPER LOCATION IN THE PROJECT."
D. WORK INCLUDES, BUT IS NOT LIMITED TO:
   1. NEW PANELBOARDS AND CIRCUIT BREAKERS.
   2. FEEDERS AND BRANCH CIRCUIT WIRING.
   3. HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS.
   4. ELECTRICAL IDENTIFICATION.
   5. CONDUCTORS AND CABLES.
   6. GROUNDING AND BONDING.
   7. RACEWAYS AND BOXES.
   8. WIRING DEVICES.
   9. LIGHTING CONTROL DEVICES.
   10. LUMINAIRES, INCLUDING NEW LAMPS, BALLASTS, AND DRIVERS.
   11. **NEW** FIRE ALARM SYSTEM.
   12. FIRE STOPPING.
   13. ELECTRICAL CONNECTIONS TO MECHANICAL, HVAC, AND PLUMBING EQUIPMENT.
   14. IN–BUILDING TWO–WAY EMERGENCY RESPONDER COMMUNICATION ENHANCEMENT SYSTEM, IF REQUIRED AFTER INITIAL SURVEY.
   15. METER SOCKETS AND C.T. CABINETS.
   16. NAMEPLATES, LABELS, AND TAGS.
   17. COORDINATION DRAWINGS.
   18. SHOP DRAWINGS.
   19. OPERATION AND MAINTENANCE INSTRUCTIONS AND MANUALS.
   20. TESTING.
E. THE ELECTRICAL DESIGN IS BASED ON THE CURRENT ADOPTED EDITION NFPA 70 – "THE NATIONAL ELECTRICAL CODE". THE CONTRACTOR SHALL INCLUDE IN THEIR BID THE COST OF INSTALLING MATERIALS AND EQUIPMENT NECESSARY TO SATISFY ALL LOCAL, AND/OR STATE CODES.

1.5 WORK OR MATERIALS NOT INCLUDED

A. THE EXACT WIRING REQUIREMENTS SHALL BE AS RECOMMENDED BY THE MANUFACTURER OF THE EQUIPMENT AND SHALL BE VERIFIED BY THE ELECTRICAL CONTRACTOR WITH THE EQUIPMENT MANUFACTURER BEFORE SUBMITTING THE BID.
B. STARTERS SUPPLIED AS AN INTEGRAL PART OF THE EQUIPMENT SHALL BE FURNISHED UNDER THE DIVISION PROVIDING THE EQUIPMENT. POWER WIRING DISCONNECT SHALL BE UNDER DIVISION 16. ALL OTHER STARTERS AND AUXILIARY CONTROL EQUIPMENT SHALL BE SUPPLIED AND WIRED UNDER DIVISION 16, UNLESS OTHERWISE SHOWN.

1.6 RELATED WORK SPECIFIED ELSEWHERE

A. DIVISION 13 – SPECIAL CONSTRUCTION
B. DIVISION 15 – MECHANICAL

1.7 CODES, PERMITS, AND FEES

A. INSTALL WORK IN FULL ACCORDANCE WITH RULES AND REGULATIONS OF STATE, COUNTY, AND CITY AUTHORITIES HAVING JURISDICTION (AHJ) OVER PREMISES. THIS SHALL INCLUDE SAFETY REQUIREMENTS OF THE STATE OF **OHIO** DEPARTMENTS OF COMMERCE AND NATURAL RESOURCES. DO NOT CONSTRUE THIS AS RELIEVING CONTRACTOR FROM COMPLIANCE WITH ANY REQUIREMENTS OF SPECIFICATION WHICH ARE IN EXCESS OF CODE REQUIREMENTS AND NOT IN CONFLICT THEREWITH. WORK SHALL BE INSTALLED IN ACCORDANCE WITH ALL APPLICABLE PROVISIONS OF THE NATIONAL ELECTRICAL CODE, AS INTERPRETED BY THE LOCAL AUTHORITY HAVING JURISDICTION, AS WELL AS ANY FURTHER MODIFICATIONS OR REGULATIONS PUBLISHED BY LOCAL OR STATE AUTHORITIES.
B. GIVE PROPER AUTHORITIES NOTICE AS REQUIRED BY LAW RELATIVE TO THE WORK IN THEIR CHARGE. COMPLY WITH THE REGULATIONS REGARDING TEMPORARY ENCLOSURES, OBSTRUCTIONS, OR EXCAVATIONS AND PAY ALL LEGAL FEES INVOLVED.
C. SECURE AND PAY FOR PERMITS AND CERTIFICATES OF INSPECTIONS INCIDENTAL TO THIS WORK, AS REQUIRED BY ALL FOREGOING AUTHORITIES. BE RESPONSIBLE FOR PAYMENTS TO ALL PUBLIC UTILITIES FOR WORK PERFORMED BY THEM IN CONNECTION WITH PROVISION OF SERVICE CONNECTIONS REQUIRED UNDER THIS DIVISION OF SPECIFICATIONS. TURN OVER CERTIFICATES OF APPROVAL TO THE CONSTRUCTION MANAGER AND/OR OWNER PROMPTLY WHEN RECEIVED, AND BEFORE PAYMENT IS MADE FOR THE WORK. DELIVER ALL CERTIFICATES TO ARCHITECT IN DUPLICATE.
D. PROVISIONS OF THE LATEST REVISIONS TO THE FOLLOWING CODES AND STANDARDS SHALL BE FOLLOWED WHERE APPLICABLE:
   1. **OHIO** ADOPTED STATE BUILDING CODES.
   2. CITY OF **COLUMBUS** BUILDING CODES.
   3. **FRANKLIN** COUNTY BUILDING CODES.
   4. NFPA 70 – NATIONAL ELECTRICAL CODE (NEC).
   5. NFPA 101 – LIFE SAFETY CODE.
   6. TIA/EIA – 568, 569, AND J–STD–607.
   7. AMERICANS WITH DISABILITIES ACT (ADA).
   8. FEDERAL OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA).

1.8 COORDINATION WITH OTHER TRADES

A. PRIOR TO ROUGH–IN, CONSULT THE DRAWINGS, PRODUCT DATA, AND SHOP DRAWING SUBMITTALS COVERING THE WORK FOR OTHER TRADES, THE FIELD LAYOUTS OF THE CONTRACTORS FOR THE OTHER TRADES AND MAKE ADJUSTMENTS ACCORDINGLY IN LAYING OUT AND INSTALLING THE ELECTRICAL WORK.
B. KEEP FULLY INFORMED OF THE PROGRESS OF THE GENERAL CONSTRUCTION. INSTALL WORK THAT IS TO BE CONCEALED WITHIN THE BUILDING CONSTRUCTION IN SUFFICIENT TIME TO SECURE PROPER LOCATION WITHOUT DELAY TO THE WORK OF OTHER TRADES. ALL CONDUIT AND OUTLET BOXES CONCEALED IN MASONRY CONSTRUCTION SHALL BE INSTALLED DURING WALL CONSTRUCTION. ATTEND TO ELECTRICAL WORK DURING THE PROGRESS OF BUILDING–IN TO PREVENT MISALIGNMENTS AND DAMAGES TO THE ELECTRICAL WORK.
C. EXAMINE THE WORK OF OTHER TRADES WHEN IT COMES IN CONTACT WITH, OR IS COVERED BY WORK IN THIS DIVISION. DO NOT ATTACH TO, COVER UP, OR FINISH AGAINST ANY DEFECTIVE WORK, OR INSTALL WORK IN A MANNER WHICH WILL PREVENT PROPER INSTALLATION OF THE WORK OF OTHER TRADES.
D. ALL OUTLETS, SWITCHES, AND RECEPTACLES SHALL BE CENTERED WITH REGARD TO PANELING, WALL COVERINGS, TRIM, EQUIPMENT, ETC., AND SHALL LINE UP WITH EITHER BOTTOM OR TOP OF MASONRY COURSES. CHANGES TO THE SPECIFIED MOUNTING HEIGHTS OF ANY DEVICE SHALL BE APPROVED BY THE OWNER'S REPRESENTATIVE BEFORE ROUGH–IN.
E. TAKE ALL FIELD MEASUREMENTS NECESSARY AND ASSUME RESPONSIBILITY FOR THEIR ACCURACY.
F. BEFORE BEGINNING CONSTRUCTION OF THE PROJECT, THE CONTRACTOR SHALL PROVIDE TO THE MECHANICAL CONTRACTOR, MARKED–UP PRINTS INDICATING ALL ELECTRICAL ITEMS WHICH AFFECT THE LOCATION OF HEATING, VENTILATING, AIR CONDITIONING, PLUMBING, PIPING, AND DUCTWORK. THESE SHALL INCLUDE BUT NOT BE LIMITED TO PULL BOXES, CONDUIT, ETC.

1.9 EQUIPMENT AND MATERIALS

A. ALL EQUIPMENT, DEVICES, AND MATERIALS USED ON THIS PROJECT SHALL BE NEW AND U.L. LISTED AND LABELED FOR THE APPLICATION.
B. PROVIDE MATERIAL AND LABOR WHICH IS NEITHER DRAWN NOR SPECIFIED, BUT WHICH IS OBVIOUSLY A COMPONENT PART OF, AND NECESSARY TO COMPLETE WORK AND WHICH IS CUSTOMARILY A PART OF WORK OF SIMILAR CHARACTER.
C. EQUIPMENT AND MATERIALS FOR THE CONSTRUCTION SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR AND SHALL BE PROTECTED BY SAME UNTIL FORMALLY ACCEPTED BY THE OWNER.
D. ALL MANUFACTURERS OF ELECTRICAL EQUIPMENT SHALL VERIFY TO THE SATISFACTION OF THE CONTRACTOR AND ENGINEER THAT THEIR EQUIPMENT WILL FUNCTION PROPERLY UNDER THE CONDITIONS OF USE, AS SHOWN ON THE DRAWINGS AND AS SPECIFIED HEREIN. DIMENSIONS, WEIGHTS, OPERATING CHARACTERISTICS AND ALL OTHER RELATED APPURTENANCES SHALL BE VERIFIED BEFORE SUBMITTAL OF SHOP DRAWINGS.

1.10 MATERIAL SUBSTITUTION

A. BIDS SHALL BE BASED UPON THE SPECIFIED PRODUCTS OR LISTED ALTERNATIVES. DRAWINGS AND SPECIFICATIONS ARE BASED ON THE PRODUCTS SPECIFIED BY TYPE, MODEL, AND SIZE AND THUS ESTABLISH MINIMUM QUALITIES, WHICH SUBSTITUTES MUST MEET TO QUALIFY FOR REVIEW.
B. SHOULD THE CONTRACTOR PROPOSE TO FURNISH MATERIALS, EQUIPMENT, AND DEVICES, OTHER THAN THOSE SPECIFIED AND LISTED, THE CONTRACTOR SHALL SUBMIT A WRITTEN REQUEST FOR SUBSTITUTIONS, TO THE ENGINEER AT LEAST TEN (10) BUSINESS DAYS PRIOR TO BID OPENING. THE REQUEST SHALL BE AN ALTERNATE TO THE ORIGINAL BID AND SHALL INCLUDE AND BE ACCOMPANIED WITH COMPLETE SPECIFICATIONS CUT SHEET SUBMITTAL AS OUTLINED IN THIS SPECIFICATION SECTION, COMPLETE WITH DESCRIPTIVE (MANUFACTURER, BRAND NAME, CATALOG NUMBER, ETC.) AND TECHNICAL DATA FOR ALL ITEMS. INDICATE ANY ADDITIONS OR DEDUCTIONS TO THE CONTRACT PRICE WITH THE SUBSTITUTION SUBMITTAL AND ON THE BID FORM.
C. ACCEPTANCE OR REJECTION OF THE PROPOSED SUBSTITUTIONS SHALL BE SUBJECT TO APPROVAL OF THE ARCHITECT AND ENGINEER. IF REQUESTED, THE CONTRACTOR SHALL SUBMIT INSPECTION SAMPLES OF BOTH THE SPECIFIED AND THE PROPOSED SUBSTITUTE ITEMS.
D. IF ANY SUBSTITUTIONS ARE APPROVED, AN ADDENDUM LISTING THE APPROVED ITEM(S) WILL BE ISSUED TO ALL BIDDING CONTRACTORS PRIOR TO THE BID DATE.
E. IN ALL CASES WHERE SUBSTITUTIONS ARE PERMITTED, THE CONTRACTOR SHALL BEAR ANY EXTRA COST OF EVALUATING THE EQUALITY OF THE MATERIAL AND EQUIPMENT TO BE INSTALLED.
F. WHERE ONLY ONE MAKE IS NAMED IN THE SPECIFICATIONS OR ON THE DRAWINGS, IT SHALL BE PROVIDED.
G. VERBAL REQUESTS OR APPROVALS SHALL NOT BE BINDING ON THE ENGINEER OR OWNER.

1.11 QUALITY ASSURANCE

A. ELECTRICAL COMPONENTS, DEVICES, AND ACCESSORIES: UNDERWRITER LABORATORIES, INC. (UL) LISTED AND LABELED AS DEFINED IN NFPA 70, ARTICLE 100, BY A TESTING AGENCY ACCEPTABLE TO AUTHORITIES HAVING JURISDICTION, AND MARKED FOR INTENDED USE.

1.12 SUBMITTALS

A. SHOP DRAWINGS AND PRODUCT DATA
   1. PREPARE SHOP DRAWINGS AND PRODUCT DATA FOR ELECTRICAL EQUIPMENT WITH ADEQUATE DETAILS AND SCALES AS NECESSARY TO CLEARLY SHOW CONSTRUCTION. INDICATE OPERATING CHARACTERISTICS FOR EACH REQUIRED ITEM AND DESIGN CONDITIONS FOR EACH. CLEARLY IDENTIFY EACH ITEM ON THE DRAWINGS AS TO MARK, LOCATION, AND USE.
   2. THIS CONTRACTOR SHALL REVIEW, STAMP WITH APPROVAL AND SUBMIT, WITH REASONABLE PROMPTNESS AND IN ORDERLY SEQUENCE SO AS TO CAUSE NO DELAY IN WORK OR IN THE WORK OF ANY OTHER CONTRACTOR, ALL SHOP DRAWINGS AND SAMPLES REQUIRED BY THE CONTRACT DOCUMENTS. SHOP DRAWINGS NOT STAMPED WITH CONTRACTOR APPROVAL WILL BE RETURNED FOR REPROCESSING. THE SUBMITTALS WILL BE REVIEWED ONLY FOR GENERAL COMPLIANCE AND NOT FOR DIMENSIONS, QUANTITIES, ETC.
   3. IF THE SUBMITTAL SHOWS VARIATIONS FROM THE REQUIREMENTS OF THE CONTRACT DOCUMENTS FOR ANY REASON, THE CONTRACTOR SHALL MAKE MENTION OF SUCH VARIATION IN A LETTER OF TRANSMITTAL. THE CONTRACTOR SHALL NOTE IN RED ON THE SUBMITTAL ANY CHANGE IN DESIGN OR DIMENSION ON THE ITEMS SUBMITTED INCLUDING CHANGES MADE BY THE MANUFACTURER WHICH MAY DIFFER FROM CATALOG INFORMATION.
   4. CONTRACTOR FURTHER AGREES THAT IF DEVIATIONS, DISCREPANCIES, OR CONFLICTS BETWEEN SHOP DRAWING SUBMITTALS AND THE CONTRACT DOCUMENTS IN THE FORM OF DESIGN DRAWING AND SPECIFICATIONS ARE DISCOVERED, EITHER PRIOR TO OR AFTER SHOP DRAWING SUBMITTALS ARE PROCESSED BY THE ENGINEER, THE DESIGN DRAWINGS, AND SPECIFICATIONS SHALL CONTROL AND SHALL BE FOLLOWED.
   5. THE SUBMITTALS THAT ARE RETURNED SHALL BE USED FOR PROCUREMENT. THE RESPONSIBILITY OF CORRECT PROCUREMENT REMAINS SOLELY WITH THE CONTRACTOR. THE SUBMITTAL REVIEW SHALL NOT RELIEVE THE CONTRACTOR OF RESPONSIBILITY FOR ERRORS OR OMISSIONS AND DEVIATIONS FROM THE CONTRACT REQUIREMENTS.
   6. IN CHECKING SHOP DRAWINGS, THE ARCHITECT AND ENGINEER WILL MAKE EVERY EFFORT TO DETECT AND CORRECT ERRORS, OMISSIONS, AND INACCURACIES IN SUCH DRAWINGS. HOWEVER, FAILURE TO DETECT ERRORS, OMISSIONS, AND INACCURACIES SHALL NOT RELIEVE THE CONTRACTOR OF RESPONSIBILITY FOR THE PROPER AND COMPLETE INSTALLATION IN ACCORDANCE WITH THE INTENT OF THE CONTRACT DOCUMENTS.
   7. CONTRACTOR AGREES THAT SHOP DRAWING SUBMITTALS PROCESSED BY THE ENGINEER ARE NOT CHANGE ORDERS; THAT THE PURPOSE OF SHOP DRAWING SUBMITTALS BY THE CONTRACTOR IS TO DEMONSTRATE TO THE ENGINEER THAT THE CONTRACTOR UNDERSTANDS THE DESIGN, THAT THEY DEMONSTRATE THEIR UNDERSTANDING BY INDICATING WHICH EQUIPMENT AND MATERIAL THEY INTEND TO FURNISH AND INSTALL, AND BY DETAILING THE FABRICATION AND INSTALLATION METHODS THEY INTEND TO USE.
   8. SHOP DRAWINGS SHALL CLEARLY INDICATE ALL DIMENSIONAL DATA FOR ALL PARTS OF THE ITEM, TYPES AND MATERIALS FOR ALL CONNECTIONS, FINISHES, THE EXACT RELATION OF THE ITEM TO ADJACENT MATERIALS AND EQUIPMENT IN THE COMPLETED STRUCTURE INCLUDING CLEARANCE, ANY NECESSARY ISOLATION AND FASTENING METHODS AND DEVICES AND MECHANICAL AND ELECTRICAL CONNECTIONS.
   9. SUBMIT NEWLY PREPARED INFORMATION, DRAWN TO ACCURATE SCALE. HIGHLIGHT, ENCIRCLE, OR OTHERWISE INDICATE DEVIATIONS FROM THE CONTRACT DOCUMENTS. DO NOT REPRODUCE CONTRACT DOCUMENTS OR COPY STANDARD INFORMATION PRINTED AS THE BASIS OF SHOP DRAWINGS. STANDARD INFORMATION PREPARED WITHOUT SPECIFIC REFERENCE TO THE PROJECT IS NOT CONSIDERED SHOP DRAWINGS AND WILL BE CAUSE FOR REJECTION.
   10. SHOP DRAWINGS AND PRODUCT DATA SHALL BE SUBMITTED AS FOLLOWS:
      a. CONFORM TO SUBMITTAL REQUIREMENTS OUTLINED IN DIVISION 1 OF THESE SPECIFICATIONS.
      b. WHERE CONTENTS OF SUBMITTAL LITERATURE INCLUDE DATA NOT PERTINENT TO THE SUBMITTAL, CLEARLY INDICATE WHICH PORTION OF CONTENT IS BEING SUBMITTED FOR REVIEW.
   11. SHOP DRAWINGS SHALL INCLUDE FABRICATION AND INSTALLATION DRAWINGS, SETTING DIAGRAMS, SCHEDULES, PATTERNS, TEMPLATES AND SIMILAR DRAWINGS. ALSO, INCLUDE THE FOLLOWING INFORMATION:
      a. DIMENSIONS.
      b. IDENTIFICATION OF PRODUCTS AND MATERIALS INCLUDED.
      c. COMPLIANCE WITH SPECIFIED STANDARDS.
      d. NOTATION OF COORDINATION REQUIREMENTS.
      e. NOTATION OF DIMENSIONS ESTABLISHED BY FIELD MEASUREMENT.
   12. WHERE ADDITIONAL INSTALLATION DRAWINGS, WIRING DIAGRAMS OR OTHER DRAWINGS ARE SPECIFIED AS A PART OF THE SUBMITTAL, THEY SHALL BE SUBMITTED AT THE SAME TIME WITH SHOP DRAWINGS AND PRODUCT DATA. PARTIAL SUBMITTALS ARE NOT ACCEPTABLE.
   13. SHOP DRAWINGS SHALL BE SUBMITTED FOR ALL MATERIAL ITEMS AS OUTLINED IN THESE SPECIFICATIONS. ANY DEVIATIONS FROM CONTRACT REQUIREMENTS MUST BE CLEARLY INDICATED ON SHOP DRAWINGS AND JUSTIFICATION FOR THEIR CONSIDERATION MUST BE INCLUDED.
   14. APPROVAL OF SUBMITTAL ITEMS SHALL NOT PRECLUDE REJECTION OF THOSE ITEMS UPON DISCOVERY OF DEFECTS IN THEM PRIOR TO FINAL ACCEPTANCE OF COMPLETE WORK.
   15. WHEN TWO OR MORE ITEMS OF THE SAME EQUIPMENT ARE REQUIRED (I.E. – LUMINAIRES, WIRING DEVICES, ETC.), EQUIPMENT ITEMS SHALL BE OF THE SAME MANUFACTURER.
   16. SUBMIT A MINIMUM OF SIX (6) COPIES OF SHOP DRAWINGS TO THE ARCHITECT. THE ARCHITECT AND ENGINEER SHALL EACH RETAIN ONE (1) COPY AND RETURN THE REMAINDER TO THE CONTRACTOR WHO SHALL DISTRIBUTE COPIES AS REQUIRED TO PROPERLY CONDUCT THE WORK, INCLUDING REQUIREMENTS OF THE OPERATING MANUAL.
   17. SHOP DRAWINGS AND PRODUCT DATA INCLUDES:
      a. POWER DISTRIBUTION EQUIPMENT
      b. WIRING DEVICES
      c. LIGHTING CONTROL DEVICES
      d. LUMINAIRES, LAMPS, DRIVERS, AND BALLASTS
      e. METER SOCKETS AND CT CABINETS
      f. FIRE ALARM DEVICES AND SYSTEM MODIFICATION DRAWINGS DESIGNED AND STAMPED BY A STATE CERTIFIED FIRE ALARM SYSTEM DESIGNER, INCLUDING BUT NOT LIMITED TO THE FOLLOWING:
         i. BATTERY CALCULATIONS
         ii. FLOOR PLANS INDICATING LOCATION OF ALL FIRE ALARM DEVICES.
         iii. VOLTAGE DROP CALCULATIONS FOR ALL WIRING AND WIRE TYPES.
         iv. MANUFACTURER MODEL NUMBER AND LISTING INFORMATION FOR ALL DEVICES AND WIRING BEING PROVIDED.
         v. LOCATION OF ALL REQUIRED SYNCHRONIZATION UNITS IF REQUIRED.
         vi. POWER EXTENDER PANELS, ADDRESSABLE MODULES, ETC.
         vii. IN–BUILDING TWO–WAY EMERGENCY RESPONDER COMMUNICATION ENHANCEMENT SYSTEM.

B. DESIGN DRAWINGS
   1. THE DESIGN DRAWINGS, AS SUBMITTED, ARE DIAGRAMMATIC AND ARE NOT INTENDED TO SHOW EXACT LOCATION OF EQUIPMENT, CONDUITS, ETC. UNLESS DIMENSIONS ARE GIVEN. DRAWINGS ARE NOT TO BE SCALED. EQUIPMENT, CONDUIT, ETC. TO BE INSTALLED ALONG THE GENERAL PLANS SHOWN ON THE DRAWINGS, BUT KEEPING IN MIND ACTUAL BUILDING CONDITIONS WHICH MUST BE CONFIRMED WITHIN THE ACTUAL WORK.
   2. IF THIS CONTRACTOR PROPOSES TO INSTALL EQUIPMENT REQUIRING SPACE CONDITIONS OTHER THAN THOSE AS SPECIFIED AND/OR SHOWN ON THE DESIGN DRAWINGS, OR TO REARRANGE THE EQUIPMENT, THEY SHALL ASSUME FULL RESPONSIBILITY FOR THE REARRANGEMENT OF THE SPACE AND SHALL OBTAIN THE FULL APPROVAL OF THE ARCHITECT BEFORE PROCEEDING WITH THE WORK.

C. COORDINATION DRAWINGS
   1. BEFORE BEGINNING CONSTRUCTION OF THE PROJECT, THE CONTRACTOR SHALL PROVIDE TO THE MECHANICAL CONTRACTOR MARKED UP PRINTS INDICATING ALL ELECTRICAL WORK WHICH AFFECTS LOCATION OF HEATING, VENTILATING, AIR CONDITIONING, PLUMBING PIPING, FIRE PROTECTION, AND DUCTWORK. REFER TO DIVISION 1 AND DIVISION 15 FOR RELATED WORK.
   2. COORDINATION DRAWINGS: REFLECTED CEILING PLANS DRAWN TO SCALE AND COORDINATING PENETRATIONS AND CEILING–MOUNTED ITEMS. SHOW THE FOLLOWING:
      a. CEILING SUSPENSION ASSEMBLY MEMBERS.
      b. METHOD OF ATTACHING HANGERS TO BUILDING STRUCTURE.
      c. CEILING–MOUNTED ITEMS INCLUDING LIGHTING FIXTURES, EXIT SIGNAGE, FIRE ALARM DEVICES, CCTV, SPEAKERS, ACCESS PANELS, ETC.

D. ELECTRICAL COMPONENTS, DEVICES, AND ACCESSORIES: UNDERWRITER LABORATORIES, INC. (UL) LISTED AND LABELED AS DEFINED IN NFPA 70, ARTICLE 100, BY A TESTING AGENCY ACCEPTABLE TO AUTHORITIES HAVING JURISDICTION, AND MARKED FOR INTENDED USE.
   1. EACH CONTRACTOR OR SUBCONTRACTOR FOR ELECTRICAL WORK SHALL KEEP ONE COMPLETE SET OF THE CONTRACT WORKING DRAWINGS ON THE PROJECT SITE ON WHICH THEY SHALL RECORD ANY DEVIATIONS OR CHANGES FROM SUCH CONTRACT DRAWINGS MADE DURING CONSTRUCTION. THESE CHANGES SHALL BE ACCURATELY RECORDED IN RED INK ON THE PRINTS. RECORD DRAWINGS SHALL SHOW CHANGES IN:
      a. SIZE, TYPE, CAPACITY, ETC. OF ANY MATERIAL, DEVICE, OR PIECE OF EQUIPMENT.
      b. LOCATION OF ANY DEVICE OR PIECE OF EQUIPMENT.
      c. LOCATION OF ANY OUTLET OR DEVICE AND ASSOCIATED WIRING.
      d. ROUTING OF FEEDER CONDUITS.
      e. BRANCH CIRCUIT NUMBER ASSIGNMENTS.
   2. RECORD DRAWINGS SHALL BE KEPT CLEAN AND UNDAMAGED, AND SHALL NOT BE USED FOR ANY PURPOSE OTHER THAN RECORDING DEVIATIONS FROM WORKING DRAWINGS.
   3. AFTER THE PROJECT IS COMPLETED, RECORD SETS OF DRAWINGS SHALL BE DELIVERED TO THE TENANT AND BUILDING MANAGEMENT IN GOOD CONDITION, AS A PERMANENT RECORD OF THE INSTALLATION AS CONSTRUCTED. ALL COSTS FOR PRODUCTION, PRINTING, ETC. SHALL BE BORNE BY THE CONTRACTOR AND SHALL BE INCLUDED IN THE BID.
   4. REFER TO DIVISION 1 SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS RELATED TO RECORD DRAWINGS.
   5. THIS CONTRACTOR SHALL RECORD ALL CHANGES FROM ORIGINAL DESIGN DRAWINGS WHICH WERE THE INSTALLATION OF THE WORK. THESE CHANGES SHALL BE RECORDED IN RED INK ON THE PRINTS. CHANGES SHALL BE ACCURATELY DIMENSIONED.
   6. THIS CONTRACTOR SHALL KEEP AN UPDATED SET OF PRINTS, INCLUDING CHANGES, ON THE JOB SITE AT ALL TIMES AND SHALL SUBMIT ONE (1) SET OF UPDATED AND LEGIBLE "AS–BUILT" PRINTS TO THE ARCHITECT WHEN THE WORK IS COMPLETE.
   7. PREPARE RECORD DOCUMENTS IN ACCORDANCE WITH THE REQUIREMENTS IN DIVISION 1 SECTION "PROJECT CLOSEOUT".
   8. IN ADDITION TO THE REQUIREMENTS SPECIFIED IN DIVISION 1, INDICATE THE FOLLOWING INSTALLED CONDITIONS (ACCURATELY DIMENSIONED):
      a. CONCEALED EQUIPMENT, UNITS, DEVICES, ETC., REQUIRING PERIODIC MAINTENANCE OR REPAIR.
   9. APPROVED SUBSTITUTIONS, CONTRACT MODIFICATIONS AND ACTUAL EQUIPMENT AND MATERIALS INSTALLED. EQUIPMENT LOCATIONS (EXPOSED AND CONCEALED), DIMENSIONED FROM PROMINENT BUILDING LINES.

1.13 OPERATING/MAINTENANCE MANUALS

A. PREPARE FOUR (4) COMPLETE BOUND SETS OF OPERATING/MAINTENANCE MANUALS IN ACCORDANCE WITH DIVISION 1 SECTION "PROJECT CLOSEOUT". CONTAINING OPERATING AND MAINTENANCE DATA, INSTRUCTIONS, AND MANUFACTURER START–UP REPORTS FOR ALL ELECTRICAL EQUIPMENT AND CONTROLS.
B. SERVICE MANUALS SHALL BE ASSEMBLED INTO ONE VINYL COVERED THREE RING "D" TYPE BINDER WITH HARD COVER AND WITH WRITTEN INSTRUCTIONS FOR EACH SYSTEM LISTED IN THE SPECIFICATIONS.
C. WRITTEN OPERATING INSTRUCTIONS, SUBMITTAL DRAWINGS, WIRING DIAGRAMS, EQUIPMENT CATALOG DATA SHEETS AND MANUFACTURER'S INSTRUCTIONS SHALL BE ACCOMMODATED INTO 8–1/2" X 11" AND/OR 11" X 17" SIZE. EACH SECTION SHALL BE TABULATED AND INDEXED AS FOLLOWS:
   1. FIRST PAGE ––– TITLE OF PROJECT, OWNER, ADDRESS, DATE OF SUBMITTAL, NAME OF CONTRACTOR, AND NAME OF ARCHITECT AND ENGINEER.
   2. SECOND PAGE ––– INDEX
   3. FIRST SECTION ––– WRITTEN DESCRIPTION OF SYSTEM CONTENTS WHERE ACTUALLY LOCATED IN BUILDING, HOW EACH PART FUNCTIONS INDIVIDUALLY, AND HOW SYSTEM WORKS AS A WHOLE. CONCLUDE WITH A LIST OF ITEMS REQUIRING SERVICE AND EITHER STATE THE SERVICE NEEDED OR REFER TO THE MANUFACTURER'S DATA IN THE BINDER THAT DESCRIBES THE PROPER SERVICE.
   4. SECOND SECTION ––– A COPY OF EACH SUBMITTAL DRAWING WITH AN INDEX AT THE BEGINNING OF THE SECTION.
   5. THIRD SECTION ––– A COPY OF EACH MANUFACTURER'S OPERATING INSTRUCTIONS WITH AN INDEX AT THE BEGINNING OF THE SECTION, AND A COPY OF EACH MANUFACTURER'S START UP REPORT FOR (FIRE ALARM, ETC.)
   6. FOURTH SECTION ––– A COPY OF ALL TEST RESULTS, IN CHART FORM, PERFORMED BY THE CONTRACTOR
   7. FIFTH SECTION ––– COPIES OF ALL WARRANTIES, APPROVALS, ETC.
D. SUBMIT ONE (1) COPY TO THE ENGINEER FOR APPROVAL. AFTER APPROVAL, SUBMIT THREE (3) COPIES TO THE ARCHITECT FOR DELIVERY TO THE OWNER.

1.14 PRODUCT DELIVERY, STORAGE, AND HANDLING

A. DELIVER PRODUCTS TO THE PROJECT PROPERLY IDENTIFIED WITH NAMES, MODEL NUMBERS, TYPES, COMPLIANCE LABELS AND SIMILAR INFORMATION NEEDED FOR IDENTIFICATION. MATERIALS MUST BE ADEQUATELY PACKAGED OR PROTECTED TO PREVENT DETERIORATION DURING SHIPMENT, STORAGE AND HANDLING.
B. THE CONTRACTOR SHALL MAKE PROVISIONS FOR THE DELIVERY AND SAFE STORAGE OF THEIR MATERIALS AND EQUIPMENT IN COORDINATION WITH THE WORK OF OTHERS. MATERIALS AND EQUIPMENT SHALL BE DELIVERED AT SUCH STAGES OF THE WORK AS WILL PERMIT THE PROPER WORK AS A WHOLE AND SHALL BE MARKED AND STORED IN SUCH A WAY AS TO BE EASILY CHECKED AND INSPECTED. THE ARRIVAL AND PLACING OF LARGE EQUIPMENT ITEMS SHALL BE SCHEDULED EARLY ENOUGH TO PERMIT ENTRY AND SETTING WHEN THERE IS NO RESTRICTION OR PROBLEM DUE TO SIZE AND WEIGHT.
C. MATERIALS SHALL BE STORED TO PROTECT THEM FROM INJURY PRIOR TO INSTALLATION. MATERIAL SHOULD NOT BE STORED DIRECTLY ON THE GROUND OR FLOOR AND SHALL BE KEPT AS CLEAN AND DRY AS POSSIBLE AND FREE FROM DAMAGE OR DETERIORATING ELEMENTS.
D. IN GENERAL, DO NOT DELIVER ITEMS OF ELECTRICAL EQUIPMENT TO THE PROJECT SUBSTANTIALLY BEFORE THE TIME OF INSTALLATION. LIMIT EACH SHIPMENT OF BULK AND MULTIPLE–USE MATERIALS TO THE QUANTITIES NEEDED FOR INSTALLATION WITHIN 3–WEEKS OF RECEIPT.

1

1. PANELBOARDS RATED 240V AC OR LESS SHALL HAVE SHORT–CIRCUIT RATINGS AS SHOWN ON THE DRAWINGS OR AS HEREIN SCHEDULED, BUT NOT LESS THAN 10,000 AMPERES RMS SYMMETRICAL.
2. PANELBOARDS RATED 480V AC OR LESS SHALL HAVE SHORT–CIRCUIT RATINGS AS SHOWN ON THE DRAWINGS OR AS HEREIN SCHEDULED, BUT NOT LESS THAN 14,000 AMPERES RMS SYMMETRICAL.
3. PANELBOARDS SHALL BE LABELED WITH A UL SHORT–CIRCUIT RATING. WHEN SERIES RATINGS ARE APPLIED WITH INTEGRAL OR REMOTE UPSTREAM DEVICES, A LABEL OR MANUAL SHALL BE PROVIDED. IT SHALL STATE THE CONDITIONS OF THE UL SERIES RATINGS INCLUDING:
   a. SIZE AND TYPE OF UPSTREAM DEVICE.
   b. UL RECOGNIZED BRANCH DEVICES THAT CAN BE USED.
   c. UL SERIES SHORT–CIRCUIT RATING.

C. CONSTRCTION
1. INTERIORS SHALL BE COMPLETELY FACTORY ASSEMBLED DEVICES. THEY SHALL BE DESIGNED SUCH THAT SWITCHING AND PROTECTIVE DEVICES CAN BE REPLACED WITHOUT DISTURBING ADJACENT UNITS AND WITHOUT REMOVING THE MAIN BUS CONNECTORS.
2. TRIMS FOR BRANCH CIRCUIT PANELBOARDS SHALL BE SUPPLIED WITH A HINGED DOOR OVER ALL CIRCUIT BREAKER HANDLES. DOORS IN PANELBOARD TRIMS SHALL NOT UNCOVER ANY LIVE PARTS. DOORS SHALL HAVE A SEMI FLUSH CYLINDER LOCK AND CATCH ASSEMBLY. DOORS OVER 48 INCHES IN HEIGHT SHALL HAVE AUXILIARY FASTENERS.
3. DISTRIBUTION PANELBOARD TRIMS SHALL COVER ALL LIVE PARTS. SWITCHING DEVICE HANDLES SHALL BE ACCESSIBLE.
4. SURFACE TRIMS SHALL BE SAME HEIGHT AND WIDTH AS BOX. FLUSH TRIMS SHALL OVERLAP THE BOX BY 3/4 OF AN INCH ON ALL SIDES.
5. A DIRECTORY CARD WITH A CLEAR PLASTIC COVER SHALL BE SUPPLIED AND MOUNTED ON THE INSIDE OF EACH DOOR.
6. ALL LOCKS SHALL BE KEYED ALIKE.

D. BUS
1. MAIN BUS BARS SHALL BE TIN–PLATED ALUMINUM SIZED IN ACCORDANCE WITH UL STANDARDS TO LIMIT TEMPERATURE RISE ON ANY CURRENT CARRYING PART TO A MAXIMUM OF 65 DEGREES C ABOVE AN AMBIENT OF 40 DEGREES C MAXIMUM.
2. A SYSTEM GROUND BUS SHALL BE INCLUDED IN ALL PANELS.
3. FULL–SIZE (100%–RATED) INSULATED NEUTRAL BARS SHALL BE INCLUDED FOR PANELBOARDS SHOWN WITH NEUTRAL BUS BAR TYPE FOR PANELS WITH SINGLE–POLE BRANCHES SHALL BE ARRANGED FOR SEQUENCE PHASING OF THE BRANCH CIRCUIT DEVICES. NEUTRAL BUSING SHALL HAVE A SUITABLE LUG FOR EACH OUTGOING FEEDER AND/OR BRANCH CIRCUIT REQUIRING A NEUTRAL CONNECTION.

E. DISTRIBUTION PANELBOARDS (PANELS RATED 600A OR MORE)
1. DISTRIBUTION PANELBOARDS SHALL HAVE MAIN DEVICES AND BRANCH BREAKERS AS INDICATED ON DRAWINGS.
2. DISTRIBUTION PANELBOARDS WITH BOLT–ON DEVICES CONTAINED THEREIN SHALL HAVE INTERRUPTING RATINGS AS SPECIFIED HEREIN OR AS INDICATED ON THE DRAWINGS. PANELBOARDS SHALL BE FULLY RATED.
3. MOLDED CASE CIRCUIT BREAKERS SHALL BE PROVIDED WITH INVERSE TIME AND INSTANTANEOUS TRIPPING CHARACTERISTICS. ADJUSTABLE MAGNETIC TRIP SETTINGS SHALL BE PROVIDED FOR CIRCUIT BREAKER FRAME SIZES 400 AMPERES AND LARGER.
4. CIRCUIT BREAKERS SHALL BE OPERATED BY A TOGGLE–TYPE HANDLE AND SHALL HAVE A QUICK–MAKE, QUICK–BREAK OVER–CENTER SWITCHING MECHANISM THAT IS MECHANICALLY TRIP–FREE. AUTOMATIC TRIPPING OF THE BREAKER SHALL BE CLEARLY INDICATED BY THE HANDLE POSITION. CONTACTS SHALL BE NONWELDING SILVER ALLOY AND ARC EXTINCTION SHALL BE ACCOMPLISHED BY MEANS OF DE–ION ARC CHUTES.
5. FOR 600A RATED PANELS, PANEL SHALL BE ABLE TO ACCEPT 225A BRANCH BREAKER AT A MINIMUM. FOR 800A AND ABOVE RATED PANELS, PANEL SHALL BE ABLE TO ACCEPT 400A BRANCH BREAKERS AT A MINIMUM.

F. DWELLING UNIT LOAD CENTERS
1. PROVIDE RESIDENTIAL BRANCH CIRCUIT BREAKER LOAD CENTERS IN ALL APARTMENTS EQUIPPED WITH CIRCUIT BREAKERS OF THE FRAME AND TRIP RATING AS SPECIFIED HEREIN AND AS SHOWN ON THE CONTRACT DRAWINGS. LOAD CENTER BUS STRUCTURE AND MAIN LUG/MAIN CIRCUIT BREAKER SHALL HAVE THE CURRENT RATING AS SHOWN ON THE CONTRACT DRAWINGS. BUS BAR CONNECTIONS SHALL ACCEPT PLUG–IN CIRCUIT BREAKERS. PANELBOARDS SHALL BE DEAD FRONT SAFETY TYPE. WHERE MULTI–POLE CIRCUIT BREAKERS ARE REQUIRED, THEY SHALL BE COMMON TRIP. CIRCUIT BREAKERS SHALL BE THERMAL MAGNETIC, MOLDED CASE PLUG–IN TYPE.
   a. PANELBOARDS AS A COMPLETE UNIT SHALL HAVE A MINIMUM SHORT CIRCUIT RATING OF 208(240)/120 – 10,000 AIC, UNLESS OTHERWISE NOTED.
   b. WHEN WORK IS COMPLETE, BALANCE THE CONTINUOUS LOAD ON EACH PHASE OF ALL PANELBOARDS.

G. BRANCH CIRCUIT PANELBOARDS
1. THE MINIMUM SHORT–CIRCUIT RATING FOR BRANCH CIRCUIT PANELBOARDS SHALL BE AS SPECIFIED HEREIN OR AS INDICATED ON THE DRAWINGS. PANELBOARDS SHALL BE FULLY RATED.
2. BOLT–ON TYPE, HEAVY–DUTY, QUICK–MAKE, QUICK–BREAK, SINGLE AND MULTI–POLE CIRCUIT BREAKERS OF THE TYPES SPECIFIED HEREIN, SHALL BE PROVIDED FOR EACH CIRCUIT WITH TOGGLE HANDLES THAT INDICATE WHEN UNIT HAS TRIPPED.
3. CIRCUIT BREAKERS SHALL BE THERMAL–MAGNETIC TYPE WITH COMMON TYPE HANDLE FOR ALL MULTIPLE POLE CIRCUIT BREAKERS. CIRCUIT BREAKERS SHALL BE MINIMUM 100–AMPERE FRAME AND THROUGH 100–AMPERE TRIP SIZES SHALL TAKE UP THE SAME POLE SPACING. CIRCUIT BREAKERS SHALL BE UL LISTED AS TYPE 'SWD' FOR LIGHTING CIRCUITS, AND UL LISTED AS TYPE 'HACR' FOR HEATING, AIR–CONDITIONING AND REFRIGERATING EQUIPMENT.
   a. CIRCUIT BREAKER HANDLE LOCKS SHALL BE PROVIDED FOR ALL CIRCUITS THAT SUPPLY EXIT SIGNS, EMERGENCY LIGHTS, ENERGY MANAGEMENT, AND CONTROL SYSTEM (EMCS) PANELS AND FIRE ALARM PANELS.
4. BRANCH CIRCUIT BREAKERS SHALL HAVE A MINIMUM INTERRUPTING RATING OF 10,000 AMPERES SYMMETRICAL AT 240 VOLTS AND 14,000 AMPERES SYMMETRICAL AT 480 VOLTS.

H. ENCLOSURE
1. ENCLOSURES SHALL BE AT LEAST 24 INCHES WIDE X 10 INCHES DEEP **(EXCEPT FOR DWELLING UNIT LOAD CENTERS)**, MADE FROM GALVANIZED STEEL PROVIDE MINIMUM GUTTER SPACE IN ACCORDANCE WITH THE NATIONAL ELECTRICAL CODE WHERE FEEDER CABLES SUPPLYING THE MAINS OF A PANEL ARE CARRIED THROUGH ITS BOX TO SUPPLY OTHER ELECTRICAL EQUIPMENT, THE BOX SHALL BE SIZED TO INCLUDE THE ADDITIONAL REQUIRED WIRING SPACE. AT LEAST FOUR INTERIOR MOUNTING STUDS WITH ADJUSTABLE NUTS SHALL BE PROVIDED.
2. ENCLOSURES SHALL BE PROVIDED WITH BLANK ENDS.
3. EXTERIOR MOUNTED PANELBOARDS SHALL BE NEMA 3R AND RATED FOR EXTERIOR APPLICATIONS.

I. SERVICE ENTRANCE LABEL
1. EQUIPMENT WITH MAIN SERVICE DISCONNECT SWITCH(ES) SHALL BE UL LABELED FOR USE AS SERVICE ENTRANCE EQUIPMENT.

J. FUTURE DEVICES
1. PANELBOARDS SHALL BE FULLY PROVISIONED WITH ALL NECESSARY MOUNTING BRACKETS, BUS CONNECTIONS AND APPURTENANCES REQUIRED FOR INSTALLATION OF FUTURE DEVICES.

K. ACCESSORY COMPONENTS AND FEATURES
1. PROVIDE ACCESSORY SET INCLUDING TOOLS AND MISCELLANEOUS ITEMS REQUIRED FOR OVERCURRENT PROTECTIVE DEVICE TEST, INSPECTION, MAINTENANCE AND OPERATION.

L. NAMEPLATES
1. PROVIDE AN ENGRAVED NAMEPLATE FOR EACH PANEL SECTION.

M. WARNING LABEL
1. PROVIDE ARC–FLASH HAZARD WARNING LABEL FOR EACH PANEL SECTION PER NATIONAL ELECTRIC CODE (N.E.C.) ARTICLE 110.16.

N. FINISH
1. SURFACES OF THE TRIM ASSEMBLY SHALL BE PROPERLY CLEANED, PRIMED, AND A FINISH COAT OF GRAY ANSI 61 PAINT APPLIED.

2.12 NOT USED.

2.13 FIRE ALARM SYSTEM

A. DESCRIPTION
1. PROVIDE ALL LABOR, MATERIAL, EQUIPMENT, TOOLS, AND SERVICES NECESSARY FOR, AND INCIDENTAL TO, THE PROPER INSTALLATION OF THE AUTOMATIC FIRE DETECTION AND ALARM SYSTEM HEREIN SPECIFIED, AND AS SHOWN ON THE CONSTRUCTION DOCUMENTS.
2. IN ORDER TO OBTAIN PROPER OPERATION, IF THE MANUFACTURER REQUIRES A GREATER NUMBER OF WIRES ANYWHERE IN THE SYSTEM THAN SCHEDULED ON CONSTRUCTION DOCUMENTS OR DESCRIBED HEREIN, THE MANUFACTURER SHALL INFORM CONTRACTORS DURING BIDDING PHASE, AND THE CONTRACTORS SHALL INCLUDE IN THEIR BIDS ALL COSTS FOR THE NECESSARY ADDITIONAL WIRING AND LARGER CONDUITS. SUCH ADDITIONAL WIRING REQUIREMENTS SHALL BE SPECIFICALLY NOTED ON SHOP DRAWINGS.

B. QUALITY ASSURANCE
1. THE FIRE ALARM SYSTEM SHALL CONFORM TO THE REQUIREMENTS OF THE NATIONAL ELECTRICAL CODE (NEC), NEMA, ANSI, IEEE, NFPA, AND SHALL BE LABELED WITH THE UNDERWRITERS LABORATORIES SEAL OF INSPECTION.
2. EQUIPMENT AND INSTALLATION SHALL COMPLY WITH ALL REQUIREMENTS OF THE REGULATORY AGENCIES AND SHALL CONFORM TO STATE CODE, LOCAL CODES AND ORDINANCES AND REQUIREMENTS OF THE LOCAL CITY FIRE DEPARTMENT.
3. INTEGRATED EQUIPMENT RATING TESTS SHALL BE FACTORY PERFORMED FOR PROPER SYSTEM OPERATION, ENCLOSURE STABILITY, AND DIELECTRIC STRENGTH.

C. FABRICATION AND MANUFACTURE
1. THIS CONTRACTOR SHALL PROVIDE A COMPLETE BUILDING GENERAL ALARM, ADDRESSABLE, SUPERVISED FIRE ALARM SYSTEM, WITH BATTERY BACKUP, AS COVERED BY THESE SPECIFICATIONS, TO BE WIRED, CONNECTED AND LEFT IN FIRST CLASS OPERATING CONDITION. THE CATALOG NUMBERS SPECIFIED ARE THOSE OF SILENT KNIGHT FARENHYT, AS FURNISHED BY LIFE SAFETY SYSTEMS LLC (216) 391–4840 AND CONSTITUTE THE TYPE AND QUALITY TO BE FURNISHED. EQUIVALENT SYSTEMS BY NOTIFIER, OR GAMEWELL–FCI WILL BE ACCEPTABLE, BUT MUST BE SO STATED AT TIME OF BIDDING.
2. THE FIRE ALARM STATIONS SHALL BE ADDRESSABLE SINGLE ACTION, PULL HANDLE, SILENT KNIGHT MODEL SD500–PS, SEMI–FLUSH MOUNTED. EACH STATION SHALL BE PROVIDED WITH A KEY OPERATED TEST SWITCH AND SHALL BE UL LISTED AND LABELED. THE FRONT OF THE STATION SHALL BE HINGED AND MUST BE OPERATED TO RESET THE STATION.
3. THE ALARM SIGNALS SHALL BE AUDIOVISUAL HORN DEVICE, GENTEX MODEL GEC324WR HORN VISIBLE APPLIANCE FOR FIRE ALARM SYSTEM NOTIFICATION. THE APPLIANCE SHALL BE MULTI CANDELA WITH POLAR DISTRIBUTION OF 75CD ILLUMINATION AS REQUIRED BY THE AMERICANS WITH DISABILITIES ACT (ADA). THE APPLIANCE SHALL BE UL LISTED TO STANDARD 1971. THE LIGHT UNIT SHALL BE OF ABS POLYCARBONATE AND THE LENS OF HIGH GRADE, OPTICAL QUALITY LEXAN. A SPECIAL COMPOUND REFLECTOR SHALL BE UTILIZED TO MAXIMIZE AND BEST DISTRIBUTE THE LIGHT PATTERN IN KEY AXIS DIRECTIONS.
   a. THE EFFECT OF THE ILLUMINATED VISIBLE APPLIANCE SHALL BE OBSERVABLE IN A CIRCUMPOLAR PATTERN. THE VISIBLE APPLIANCE SHALL BE LABELED WITH THE WORD "FIRE" IN A CONTRASTING COLOR.
   b. THE AUDIBLE PORTION OF THE APPLIANCE SHALL BE A SINGLE FREQUENCY ELECTRONIC OPTICALLY CONTROLLED VIBRATING HORN RATED AT 87DBA AT 10–FEET. FREQUENCY RESPONSE SHALL BE 2000HZ NOMINAL.
4. THE CEILING MOUNTED ANALOG SMOKE SENSORS SHALL BE SILENT KNIGHT MODEL SD505–APS, SENSORS SHALL BE OF THE SOLID STATE PHOTO–ELECTRIC TYPE, AND SHALL OPERATE ON LIGHT SCATTERING, CONDUCTIVE METAL–COATED PHOTOGRAPHIC FILM, AND SHALL BE LISTED BY UNDERWRITERS LABORATORIES AND APPROVED BY FACTORY MUTUAL UNDER THE CURRENT STANDARDS FOR PHOTO–ELECTRIC TYPE SMOKE SENSORS. FOR EASE OF MAINTENANCE AND INSTALLATION, SENSORS SHALL BE DESIGNED FOR TWIST LOCK MOUNTING TO A INDIVIDUAL BASE ASSEMBLY HAVING SCREW TERMINALS FOR EXTERNAL WIRING CONNECTIONS.
5. MODEL SD505–6AB ADDRESSABLE BASE ASSEMBLY SHALL BE PROVIDED WITH EACH SENSOR AND HEAT SENSOR UNLESS OTHERWISE NOTED ON THE PLANS. EACH SENSOR BASE ASSEMBLY SHALL BE EQUIPPED WITH AN OPTIONAL LOCKING MECHANISM TO PREVENT UNAUTHORIZED REMOVAL OF THE SENSOR AND SHALL MOUNT ON A STANDARD 3–1/2" ROUND, 4" SQUARE OR OCTAGONAL OUTLET BOX. BOX DEPTH SHALL BE SELECTED ON THE BASIS OF THE NUMBER OF CONDUCTORS CONNECTED IN ACCORDANCE WITH THE REQUIREMENTS OF THE NATIONAL ELECTRICAL CODE.
6. HEAT SENSORS, WHERE SHOWN ON THE CONSTRUCTION DOCUMENTS, SHALL BE SILENT KNIGHT MODEL SD505–AHS. MOUNT TO SD505–6AB SENSOR BASE.
7. THE DUCT SMOKE SHALL BE SILENT KNIGHT MODEL SD505–ADHR WITH APPROPRIATE SAMPLING TUBES. ALL DUCT SMOKE SENSORS SHALL HAVE A MODEL SD505–DTS REMOTE ALARM INDICATOR AND VISUAL INDICATOR AND TEST STATION.
8. VISUAL FLASHING LAMPS (XENON STROBE) SHALL BE GENTEX GES324WR ADA UL 1971 LISTED.

D. PRODUCTS AND SYSTEM OPERATION
1. THE FIRE ALARM SYSTEM SHALL BE AS DESCRIBED HEREIN AND AS SHOWN ON THE CONSTRUCTION DOCUMENTS. ACTUATION OF ANY ALARM INITIATING DEVICE SHALL CAUSE ALL AUDIBLE ALARM DEVICES TO SOUND CONTINUOUSLY. ALARM INITIATING DEVICES SHALL BE GROUPED IN ZONES. ANY ZONE INVOLVED IN AN ALARM SHALL BE INDICATED BY A LIGHTED RED LED ON THE ZONE MODULE INVOLVED. THE ACTUATION OF ANY ALARM INITIATING DEVICE SHALL ALSO SHUTDOWN CERTAIN AIR HANDLING UNITS AS MAY BE DESIGNATED, RETURN ELEVATORS TO THE FIRST OR GROUND FLOOR, AND ANNUNCIATE THE ALARM CONDITION ON THE ANNUNCIATOR(S) LOCATED AS SHOWN ON THE CONSTRUCTION DOCUMENTS.
2. THE GENERAL ALARM DEVICES MAY BE SILENCED BY AUTHORIZED PERSONNEL ONLY BY ENTERING A LOCKED CONTROL CABINET AND OPERATING THE PROPER SILENCING SWITCH, BUT ANY SUBSEQUENT ALARMS SHALL SOUND ALL SIGNALS. OPERATION OF THIS SWITCH SHALL BE INDICATED BY A TROUBLE LIGHT AND AUDIBLE SIGNAL. SYSTEM SHALL BE COMPLETELY SUPERVISED, AND ANY OPEN, SHORT, OR GROUNDED CONDITION SHALL CAUSE THE TROUBLE LIGHT TO BE ILLUMINATED AND THE TROUBLE SIGNAL TO SOUND. THIS SHALL REMAIN UNTIL TROUBLE CONDITION HAS BEEN CORRECTED.
3. THE MAIN FIRE ALARM CONTROL PANELS SHALL BE SILENT KNIGHT FARENHYT IFP 100 AND 1000 WITH A 40 CHARACTER LCD ANNUNCIATOR INSTALLED WHERE INDICATED ON THE CONSTRUCTION DOCUMENTS. ALL ZONE LIGHTS, TROUBLE LIGHTS, TROUBLE SIGNALS, SILENCING SWITCHES AND LIGHTS, AND RESET SWITCHES SHALL BE MOUNTED IN SURFACE CABINET AND CONTAIN THE FOLLOWING ITEMS:
   a. GREEN POWER ON LIGHT
   b. MANUAL EVACUATION SWITCH
   c. SYSTEM RESET SWITCH
   d. TROUBLE LIGHTS
   e. TROUBLE SILENCING SWITCH
   f. TONE ALERT TROUBLE SIGNAL
   g. AUXILIARY RELAYS WITH BYPASS SWITCHES
   h. PROVISION FOR REMOTE REPORT
4. ZONES SHALL BE AS INDICATED ON CONSTRUCTION DOCUMENTS. A SCHEDULE OF ANNUNCIATION ENGRAVING SHALL BE FURNISHED FOR ARCHITECT'S AND ENGINEER'S APPROVAL. MAIN CONTROL PANEL SHALL HAVE CAPACITY TO INDICATE ZONES IN THIS BUILDING PLUS REMOTE REPORTS FROM FUTURE BUILDINGS, AND FROM THE GENERATOR, FIRE PUMP AND TRANSFER SWITCHES.
5. CONTROL MODULES SHALL BE PROVIDED FOR THE OPERATING AND INDICATING FUNCTIONS OF THE FIRE ALARM SYSTEM. THESE FUNCTIONS SHALL INCLUDE SYSTEM RESET, ALARM LOCK–IN, ALARM RESOUND, ACKNOWLEDGE SWITCH, EARTH LED, SYSTEM TROUBLE LED, TONE ALERT, AND LED TEST. UPON HAVING DETECTED AN ALARM CONDITION, A ZONE SHALL REPORT ITS STATUS TO THE CONTROL MODULE. THIS ALARM CONDITION SHALL CAUSE THE ALARM TO BE DISPLAYED ON THE LCD. ONCE ACTUATED UPON SYSTEM OPERATION, IT SHALL ALSO CAUSE THE SOUNDING OF ALARM SIGNALS, THE ACTIVATION OF THE GAMEWELL BOX, OR THE OPERATION OF THE FAN CONTROL RELAYS. WHEN THE ACKNOWLEDGE SWITCH IS PRESSED, THE ZONE'S ALARM LED SHALL CEASE TO FLASH AND REMAIN ILLUMINATED. AT THIS TIME, THE SIGNALS SHALL ALSO BE SILENCED. THE ACTIVATION OF ANOTHER ZONE SHALL REPEAT THE ENTIRE ALARM PROCESS, THUS CAUSING THE SIGNALS TO "RESOUND". ALL ALARM AND TROUBLE EVENTS SHALL BE AUTOMATICALLY STORED INTO THE SYSTEM HISTORY LOG AND MAY BE REVIEWED BY OPERATOR WITH PROPER AUTHORIZATION CODE.
6. THE ZONE ALARM INITIATING CIRCUIT MODULES SHALL PROVIDE EIGHT ELECTRICALLY SUPERVISED NORMALLY OPEN CIRCUITS PER MODULE FOR CURRENT LIMITED SERIES MONITORING FOR ALARM, TROUBLE (SHORT OR OPEN) AND GROUND FAULT. EACH MODULE SHALL PROVIDE LCD ANNUNCIATION FOR ALARM, OPEN TROUBLE ANNUNCIATOR OUTPUTS, ALARM AND TROUBLE PRINTER OUTPUTS, AUXILIARY ALARM CONTACTS (N/O – 2 AMPERE RESISTIVE), AND ALARM AND TROUBLE RESOUND WITH TROUBLE ACKNOWLEDGE (FROM CONTROL). THE MODULE SHALL BE CAPABLE OF POWERING CURRENT LIMITED TWO–WIRE SMOKE SENSORS. THE MODULE SHALL BE COMPLETELY PLUGGABLE AND INHERENTLY POWER LIMITED (LIMITED ENERGY). PROVIDE THE NUMBER OF ZONES REQUIRED PLUS TEN PERCENT ACTIVE SPARES. THE COMMUNICATING DEVICE MODULE SHALL PROVIDE 127 ADDRESSES FOR MONITOR OR CONTROL POINTS PER LOOP.
7. THE NAC SIGNAL CIRCUITS SHALL BE CAPABLE OF SUPPLYING A MINIMUM OF (2.5) AMPERES OF SIGNAL CAPACITY TO EACH OF THE SIGNAL CIRCUITS. IT SHALL SUPPLY POWER VIA A TWO–WIRE SUPERVISED CIRCUIT AND INDICATE OPEN OR SHORTED FAULTS ON THE LCD ANNUNCIATOR. EACH SHALL BE PROTECTED BY A FUSE.
8. THE POWER SUPPLY MODULE SHALL SUPPLY 6 AMPERES OF CONTINUOUS FILTERED POWER OF THE PROPER VOLTAGE. THE POWER SUPPLY SHALL BE CAPABLE OF FURNISHING THE SYSTEM POWER AND POWER FOR DEVICES SUCH AS SMOKE SENSORS, AUXILIARY RELAYS, DOOR HOLDERS, ETC. IT SHALL CONTAIN A NORMAL, BATTERY TROUBLE, AND POWER SUPPLY TROUBLE, ANNUNCIATION ON THE LCD ANNUNCIATOR ALL VIEWABLE ON FRONT OF ENCLOSURE. A 250 VA TRANSFORMER SHALL BE PROVIDED IN THE CONTROL PANEL.
9. N AUTOMATIC DUAL–RATE BATTERY CHARGER MODULE SHALL BE INSTALLED IN THE CONTROL PANEL WHICH SHALL BE CAPABLE OF CHARGING EITHER A CEL OR WET CELL BATTERY. A CONSTANT TRICKLE CHARGE SHALL CONTINUOUSLY BE APPLIED TO KEEP THE BATTERY IN ORDER TO MAINTAIN IT AT FULL CHARGE STATE. A METHOD OF ADJUSTING THE TRICKLE CHARGE RATE SHALL ALSO BE PROVIDED IN ORDER TO SUPPLY THE SELECTED BATTERY WITH THE EXACT CHARGE RATE REQUIRED. IN THE EVENT OF A FAILURE OF THE CHARGER, THE CHARGER FAILURE LCD SHALL ILLUMINATE. SHOULD THE BATTERY CAPACITY DROP BELOW SPECIFIED LIMITS, THE CHARGER SHALL AUTOMATICALLY CHANGE TO HIGH RATE CONDITION, AND AN LCD SHALL ILLUMINATE. MAINTENANCE FREE BATTERIES SHALL BE PROVIDED TO DELIVER 24–HOURS OF STANDBY POWER.
10. A BATTERY MONITOR MODULE SHALL BE PROVIDED TO MONITOR THE CHARGER AND THE BATTERY MOUNTED WITHIN THE CONTROL PANEL. SHOULD THE BATTERY VOLTAGE BECOME EXCESSIVELY LOW, EXCESSIVELY HIGH OR NOT BE CONNECTED AT ALL, THE SYSTEM TROUBLE MODULE SHALL BE ACTIVATED AND THE BATTERY TROUBLE LCD SHALL ILLUMINATE.
11. THE AUXILIARY FLEXPUTS SHALL HAVE THE ABILITY TO BE CONFIGURED FOR NAC, INITIATING ZONE, OR RELAY OUTPUT RATED AT 3 AMPERES. THESE RELAYS SHALL BE USED FOR OPERATING EXTERNAL CIRCUITS INDEPENDENT OF THE FIRE ALARM SYSTEM (I.E., FAN CONTROLS, DOOR RELEASE, ETC.).
12. THE CONTROL CABINET SHALL BE SURFACE MOUNTED AND ACCOMMODATE ALL THE MODULES, RELAYS, TERMINAL CONNECTIONS, AND BATTERIES NECESSARY FOR THE PROPER OPERATION OF THE SYSTEM. THE ENCLOSURE SHALL BE EQUIPPED WITH AN OUTER DOOR AND FRAME ASSEMBLY WHICH SHALL BE FITTED WITH A LOCK AND THE TRANSPARENT DOOR PANEL, TO PERMIT FULL VIEWING OF THE VARIOUS MODULE LIGHTS AND SWITCHES.
13. REMOTE ANNUNCIATOR – THE SERIAL ANNUNCIATOR SHALL PROVIDE AN ALPHANUMERIC, 80 CHARACTER LIQUID CRYSTAL DISPLAY (LCD) THAT PROVIDES CLEAR LANGUAGE INFORMATION AS TO THE IFP PANEL POINT STATUS (ALARM, TROUBLE, ETC.), TYPE OF ALARM (SMOKE SENSOR, PULL STATION, ETC.), NUMBER OF ALARMS ON THE SYSTEM, AND A CUSTOM LOCATION LABEL. THE ANNUNCIATOR(S) SHALL HAVE A GREY OR RED FINISH AS DIRECTED BY THE ARCHITECT. THE ANNUNCIATOR SHALL COMMUNICATE TO THE CONTROL PANEL OVER 4 CONDUCTOR #16AWG WIRE AND OPERATING POWER SHALL BE 24 VDC AND BE FUSED AT THE CONTROL PANEL. POINT WIRED ANNUNCIATORS WILL NOT BE CONSIDERED AS EQUAL. STATUS INFORMATION OF EACH DEVICE MAY BE INDIVIDUALLY DISPLAYED TO INVESTIGATE SPECIFIC POINT DETAIL. REMOTE ANNUNCIATOR MAY NOT BE REQUIRED ON ALL PROJECTS; REFER TO FIRE ALARM RISER DIAGRAM. PROGRAMMABLE CONTROL SHALL BE PROVIDED FOR THE FOLLOWING FUNCTIONS:
   a. SYSTEM RESET
   b. ALARM SILENCE
   c. TROUBLE SILENCE
   d. MANUAL EVACUATION
   e. CITY DISCONNECT

E. INSTALLATION
1. EXAMINE AREA TO RECEIVE MAIN CONTROL PANEL TO ASSURE ADEQUATE SPACE AND CLEARANCES.
2. THE CONTRACTOR SHALL FURNISH AND INSTALL IN ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS ALL WIRING, CONDUIT, AND OUTLET BOXES FOR THE ERECTION OF A COMPLETE SYSTEM AS DESCRIBED HEREIN AND AS SHOWN ON THE CONSTRUCTION DOCUMENTS. ALL WIRING SHALL BE IN CONDUIT OF THE SAME APPROVED TYPE AS USED FOR ELECTRIC LIGHT AND POWER WIRING, AND SHALL BE THOSE SPECIFIED BY THE MANUFACTURER. BEFORE INSTALLATION, A COLOR CODING SYSTEM SHALL BE COORDINATED WITH THE MANUFACTURER AND ADHERED TO THROUGHOUT THE INSTALLATION. COLOR CODES SHALL BE USED AND ALL WIRES SHALL BE TAGGED AT ALL JUNCTION POINTS AND SHALL TEST FREE FROM GROUND OR CROSSES BETWEEN CONDUCTORS. FINAL CONNECTIONS SHALL BE MADE UNDER THE DIRECT SUPERVISION OF THE MANUFACTURER'S FACTORY TRAINED TECHNICAL REPRESENTATIVE.
3. ALL JUNCTION BOXES SHALL BE PROVIDED WITH NUMBERED TERMINAL STRIPS. NO OTHER TYPES OF SPLICES WILL BE PERMITTED.
4. AN AS–BUILT DRAWING OF THE FIRE ALARM SYSTEM SHALL BE PROVIDED UPON COMPLETION OF THE FIRE ALARM INSTALLATION SHOWING ALL DEVICES, NUMBER AND SIZE OF CONDUCTORS, COLOR CODE SYSTEM ETC., AND BUILDING FLOOR PLANS. ALL WIRING SHALL BE IN ACCORDANCE WITH THE FIRE ALARM MANUFACTURER'S RECOMMENDATIONS TO THE NATIONAL ELECTRICAL CODE.
5. A QUALIFIED ENGINEER, APPROVED BY THE ARCHITECT, SHALL BE PROVIDED BY THE MANUFACTURER TO SUPERVISE THE ENTIRE MECHANICAL AND ELECTRICAL INSTALLATION, AS WELL AS TO ALSO THOROUGHLY INSTRUCT THE RESPONSIBLE PERSONNEL OF THE OWNER IN THE PROPER OPERATION AND MAINTENANCE OF THE EQUIPMENT.
6. CONTRACTOR SHALL PROVIDE A CERTIFICATE OF INSPECTION AND OPERATION FROM THE SUPPLIER BEFORE THE REQUEST FOR FINAL PAYMENT IS MADE.

F. FIELD QUALITY CONTROL
1. WIRING SHALL BE CHECKED AND TESTED BY THE CONTRACTOR IN ACCORDANCE WITH THE INSTRUCTIONS PROVIDED BY THE MANUFACTURER TO INSURE THAT THE SYSTEM IS FREE FROM GROUNDS, SHORTS, OPENS, AND THAT THE INSULATION RESISTANCE BETWEEN CURRENT CARRYING CONDUCTORS IS 10 MEGOHMS OR GREATER.
2. THE SENSITIVITY OF EACH SENSOR SHALL BE CHECKED AND RECORDED IN MEMORY AFTER SENSORS HAVE BEEN INSTALLED. ANY SENSORS FOUND TO BE OUT OF TOLERANCE SHALL BE ADJUSTED OR REPLACED. THE RECORDED DATA SHALL BE INCLUDED IN THE OPERATING AND MAINTENANCE MANUAL.
3. THE SYSTEM SHALL BE TESTED BY AN ENGINEERING REPRESENTATIVE OF THE FIRE ALARM SYSTEM MANUFACTURER UPON THE COMPLETION OF THE INSTALLATION. EACH ZONE SHALL BE TESTED BY SCATTERING, TEMPORARILY, CANDLE SMOKE, DUCT SENSOR, AND MANUAL STATION FOR PROPER OPERATION OF ALARM, NON–INTERFERENCE OF THE SUCCESSIVE ALARMS, A PROPER SHUTDOWN OF THE VENTILATION EQUIPMENT, IF REQUIRED. THE RESULTS OF THE TESTS SHALL BE RECORDED AND SHALL BE INCLUDED IN THE OPERATING AND MAINTENANCE MANUAL.
4. THE FIRE PUMP AND ALL CONTROLS ARE EXISTING. THE CONTRACTOR UNDER THIS DIVISION SHALL PROVIDE FOR MONITORING AND REMOTE ANNUNCIATION OF SEVERAL OPERATIONAL POINTS ASSOCIATED WITH THE FIRE PUMP. THE POINTS SHALL BE ANNUNCIATED OF THE FIRE PUMPS OPERATIONAL FUNCTIONS SHALL INCLUDE BUT NOT BE LIMITED TO THE FOLLOWING:
   a. FIRE PUMP RUNNING
   b. LOW SUCTION PRESSURE
   c. POWER AVAILABLE
5. AN ALARM SIGNAL SHALL BE CAPABLE OF RETRANSMISSION DIRECTLY TO SECURITY MONITORING SERVICE THROUGH A 2–CONDUCTOR TWISTED SHIELD PAIR, NO. 18 CABLE IMMEDIATELY UPON RECEIPT AT THE FACP. IN ADDITION, ALL SIGNALS INCLUDING TROUBLE ALARMS SHALL BE INDICATED ON THE BUILDINGS LOCAL ANNUNCIATION PANELS AS SHOWN ON THE PLANS. THE SPECIFIC NAME AND LOCATION OF THE ALARM DEVICE SHALL BE INDICATED.
6. THE CONTRACTOR UNDER THIS DIVISION IS RESPONSIBLE FOR PROVIDING ALL SIGNALS INTERNAL OR EXTERNAL TO ALL LOCATIONS DESIGNATED OR REQUIRED BY LOCAL, STATE OR NATIONAL FIRE CODES GOVERNING BUILDINGS OF THE TYPE AND DESCRIPTION INVOLVED.

2.14 IN–BUILDING TWO–WAY EMERGENCY RESPONDER COMMUNICATION ENHANCEMENT SYSTEM (IF REQUIRED AFTER INITIAL SURVEY).

A. GENERAL
1. A. PROVIDE AN IN–BUILDING RADIO SIGNAL AMPLIFICATION SYSTEM TO PROVIDE COMPLETE COVERAGE IN THE BUILDING FOR THE PUBLIC SAFETY AGENCIES AS REQUIRED BY THE LOCAL AHJ (AUTHORITY HAVING JURISDICTION). THE SYSTEM LISTED IN THIS SPECIFICATION IS BASED ON PRODUCTS AND SERVICES PROVIDED BY LIFE SAFETY SYSTEMS, LLC.
   FOR INFORMATION AND PRICING PLEASE CONTACT:
   DAN KLAG – DKLAG@LIFESAFETYLLC.COM – (216) 347–0950
   JAKE KLAG – JKLAG@LIFESAFETYLLC.COM – (216) 633–6469
   SYSTEM USERS SHALL RECEIVE AND TRANSMIT RADIO SIGNALS FROM THEIR PORTABLE RADIO UNITS WITHIN THE BUILDING. THIS SHALL BE ACCOMPLISHED UTILIZING THE FOLLOWING COMPONENTS:
   a. BI DIRECTIONAL AMPLIFIERS (SIGNAL BOOSTERS) / FIBER DAS MASTER & REMOTES
   b. COAXIAL CABLE / FIBER CABLE
   c. ANTENNAS
   d. CABLE TAPS
   e. CONNECTORS
   f. POWER DIVIDERS
   g. OTHER COMPONENTS AND INTERCONNECTING CIRCUITRY AS REQUIRED
2. THE SYSTEM SHALL COMPLY WITH THE REQUIREMENTS OF UL2524 2ND EDITION ST

LUMINAIRE SCHEDULE

TYPE	LAMP	WATTAGE	VOLTAGE	DIMMING TYPE	DESCRIPTION	CATALOG NUMBER	EQUIVALENT MANUFACTURER/REMARKS
D-1			120		LED DOWNLIGHT USED IN CLUBROOM.		
D-1A			120		LED DOWNLIGHT USED IN FIRST FLOOR CORRIDOR.		
D-2			120		LED DOWNLIGHT USED THROUGHOUT THE BUILDINGS.		
D-4			120		LED DOWNLIGHT USED IN CLUBROOM.		
D-5			120		LED DOWNLIGHT USED IN OUTDOOR CORRIDOR.		
D-6			120		LED DOWNLIGHT USED IN CORRIDORS.		
ESC-2			120		LED EXTERIOR LIGHT.		
S-2			120		LED STRING LIGHT USED IN CLUBROOM.		
S-3			120		LED SQUARE LIGHT USED IN CORRIDOR.		
S-4			120		LED LIGHT USED IN PUBLIC BATHROOMS.		
P-1			120		LED SPHERE LIGHT USED IN CLUBROOM AND LOBBY AREA.		
P-6			120		LED STRIP LIGHT USED IN FITNESS AREA.		
P-7			120		LED STRIP LIGHT USED IN FITNESS AREA.		
P-8			120		LED STRING LIGHT USED ABOVE FITNESS AREA.		
SM-2			120		LED FIXTURE USED IN OFFICES.		
SM-3			120		LED FIXTURE USED IN PUBLIC BATHROOMS.		
U-1			120		LED STRIP LIGHT USED IN HOUSE AREAS.		
U-2			120		LED STRIP LIGHT USED IN STAIRWAYS, WITH ON-FIXTURE MOTION SENSOR.		
WS-1			120		LED SQUARE LIGHT USED IN CORRIDORS.		
X1							
X2							
X3							

LUMINAIRE SCHEDULE GENERAL NOTES:

1. VERIFY ALL LUMINAIRE COLORS, TRIMS, LENGTHS, ETC. WITH THE **ARCHITECT/OWNER/DEVELOPER/DESIGNER** PRIOR TO PLACING FINAL PURCHASE ORDERS. SUBMISSION OF SHOP DRAWINGS WILL BE INTERPRETED AS HAVING BEEN COORDINATED WITH THE AFOREMENTIONED PARTIES.

2. PROVIDE ALL LENGTHS, FEEDS, ACCESSORIES, CONNECTORS, WIRING, POWER SUPPLIES, DRIVERS, ETC. FOR A COMPLETE INSTALLATION. VERIFY THE COMPLETE BILL OF MATERIAL WITH MANUFACTURER'S REPRESENTATIVE AND ENSURE ALL EQUIPMENT IS INCLUDED IN BID PRICE. COORDINATE INSTALLATION WITH ARCHITECTURAL DETAILS.

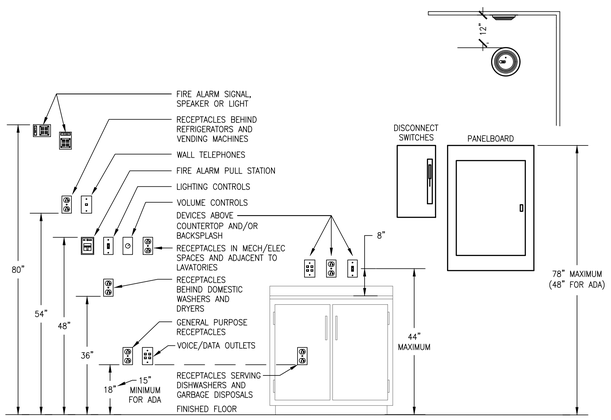
3. VERIFY FINAL LUMINAIRE LOCATIONS WITH OTHER CEILING MOUNTED EQUIPMENT SUCH AS DIFFUSERS, FIRE ALARM DEVICES, SPEAKERS, ETC., WITH ARCHITECTURAL REFLECTED CEILING PLANS.

4. VERIFY EXACT MOUNTING HEIGHT AND LOCATIONS OF ALL WALL MOUNTED AND PENDANT MOUNTED LUMINAIRES WITH ARCHITECTURAL PLANS AND ELEVATIONS PRIOR TO ROUGH-IN.

5. ANY PROPOSED ALTERNATE LUMINAIRES SHALL BE APPROVED BY THE **ARCHITECT/OWNER/DEVELOPER/DESIGNER** PRIOR TO FINAL BID PRICING.

6. SHOULD THE CONTRACTOR PROPOSE TO FURNISH MATERIALS, EQUIPMENT, AND DEVICES, OTHER THAN THOSE SPECIFIED AND LISTED, THE CONTRACTOR SHALL SUBMIT A WRITTEN REQUEST FOR SUBSTITUTIONS, TO THE ENGINEER AT LEAST TEN (10) BUSINESS DAYS PRIOR TO BID OPENING. THE REQUEST SHALL BE AN ALTERNATE TO THE ORIGINAL BID AND SHALL INCLUDE A COMPLETE SPECIFICATIONS CUT SHEET SUBMITTAL AS OUTLINED IN THE SPECIFICATIONS, COMPLETE WITH DESCRIPTIVE (MANUFACTURER, BRAND NAME, CATALOG NUMBER, ETC.) AND TECHNICAL DATA FOR ALL ITEMS. INDICATE ANY ADDITIONS OR DEDUCTIONS TO THE CONTRACT PRICE WITH THE SUBSTITUTION SUBMITTAL AND ON THE BID FORM.

7. ANY FINAL SELECTED/INSTALLED LIGHT FIXTURES WITHOUT A SPECIFICATION (OR TO BE SPECIFIED LATER BY **ARCHITECT/OWNER/DEVELOPER/DESIGNER**) LISTED IN THIS SCHEDULE SHALL NOT EXCEED THE WATTAGE LISTED, SHALL NOT HAVE A LOWERED DELIVERED LUMENS (IF LISTED), AND SHALL MATCH THE COLOR TEMPERATURE (IF LISTED).



TYPICAL DEVICE MOUNTING DETAIL

NO SCALE

TYPICAL DEVICE MOUNTING DETAIL GENERAL NOTES

A. ALL MOUNTING HEIGHTS SHALL BE MEASURED FROM FINISHED FLOOR TO THE CENTERLINE OF DEVICE EXCEPT FIRE ALARM A/V DEVICES.

B. NO WIRING DEVICES OR OUTLET BOXES SHALL BE MOUNTED BACK TO BACK.

C. ALL MOUNTING DEVICES SHALL BE INSTALLED AT MOUNTING HEIGHTS AS INDICATED ON THIS DETAIL UNLESS OTHERWISE NOTED.

D. FOR ALL ELEVATIONS (WHERE APPLICABLE), CASEWORK DETAILS, FIRE WALLS, SMOKE WALLS, LOCATION OF COUNTERTOP RECEPTACLES, LIGHTING FIXTURE SWITCHES, TELEPHONE OUTLETS, EQUIPMENT ROUGH-INS, HEADWALLS, ETC., SEE ARCH DRAWINGS. WHERE NO ARCHITECTURAL ELEVATIONS OR DETAILS OCCUR, THE ELECTRICAL CONTRACTOR SHALL USE MEANS AND METHODS AS WELL AS THEIR FIELD KNOWLEDGE TO SPOT DEVICES IN THE BEST LOCATIONS FOR THE PROJECT.

ELEVATOR INSTALLATION GENERAL CONSTRUCTION NOTES

1. INCLUDE EXTERNALLY OPERABLE SHUNT TRIP CIRCUIT BREAKERS CAPABLE OF BEING LOCKED IN THE "OFF" POSITION FOR ELEVATOR 3-PHASE POWER FEED (NEC 620.51). PROVIDE WIRING FROM THE FIRE ALARM SYSTEM TO THE CIRCUIT BREAKER FOR SHUNT TRIP ACTIVATION UPON OPERATION OF THE ELEVATOR CONTROL HEAT DETECTORS, UNLESS OTHERWISE DIRECTED BY LOCAL ELEVATOR INSPECTOR (ASME A17.1-102 AND NEC 620.51(B)). ROUTE 120V WIRING THROUGH FIRE ALARM CONTROL PANEL TO MONITOR LOSS OF SHUNT TRIP POWER (TROUBLE ONLY INDICATION). IF BATTERY POWERED LOWERING IS SPECIFIED BY THE ARCHITECT, PROVIDE A NORMALLY CLOSED CONTACT AND WIRING TO ELEVATOR CONTROLLER TO DISCONNECT ADDITIONAL POWER SOURCE (NEC 620.91(C)).

2. INCLUDE 4-PAIR UNSHIELDED TWISTED PAIR CABLE, MINIMUM-CATAGORY 3 FROM CONTROLLER TO NEAREST AVAILABLE TELEPHONE SERVICES FOR CAR EMERGENCY PHONE/INTERCOM (ANSI A-17.1 - 211). COORDINATE EXACT LOCATION OF TELEPHONE SERVICES. **TYPICAL OF EACH ELEVATOR.**

3. INCLUDE FUSIBLE DISCONNECT OR EXTERNALLY OPERABLE CIRCUIT BREAKER CAPABLE OF BEING LOCKED IN THE "OFF" POSITION FOR 120 VOLT SINGLE PHASE FEED (NEC 620.53). PROVIDE A SECOND SWITCH IF A BRANCH CIRCUIT FOR A HEATING OR AIR CONDITIONING POWER SUPPLY IS PROVIDED (NEC 620.54). **TYPICAL OF EACH ELEVATOR.**

4. INCLUDE DEDICATED 20 AMP, 120 VOLT BRANCH CIRCUIT ON FOR CAR LIGHTS, RECEPTACLES, AUXILIARY LIGHTING SOURCE AND VENTILATOR (NEC 620.22). PROVIDE A SECOND SWITCH IF A BRANCH CIRCUIT FOR A HEATING OR AIR CONDITIONING POWER SUPPLY IS PROVIDED (NEC 620.22). **TYPICAL OF EACH ELEVATOR.**

5. INCLUDE WEATHERPROOF DUPLEX RECEPTACLE - (FOR ELEVATOR SERVICING) AND SWITCH FOR LIGHTS. COORDINATE LOCATION IN FIELD WITH ELEVATOR INSTALLER.

6. INCLUDE SMOKE DETECTOR WITH AUXILIARY CONTACTS FOR ELEVATOR RECALL - LOCATE OUTSIDE EACH ELEVATOR LANDING, AT TOP AND BOTTOM OF THE HOISTWAY (2 FEET ABOVE FLOOR) AND IN THE MACHINE ROOM (ANSI A-17.1 - 211.3b).

7. INCLUDE FIXED TEMPERATURE HEAT DETECTOR AT SETTING LESS THAN SPRINKLER HEAD TO BE MOUNTED ADJACENT TO HEAD. HEAT DETECTORS SHALL INITIATE AN AUTOMATIC POWER SHUTDOWN FOR 3-PHASE POWER TO THE ELEVATOR CONTROLLER, UNLESS OTHERWISE DIRECTED BY THE LOCAL ELEVATOR INSPECTOR. LOCATE HEAT DETECTORS IN THE MACHINE ROOM, AT THE TOP OF THE SHAFT AND AT THE BOTTOM OF THE SHAFT (2 FEET ABOVE THE FLOOR).

GENERAL ELECTRICAL NOTES

1. ANY AND ALL "BUILDING STANDARDS" AND/OR "BUILDING SPECIFICATIONS" SHALL BE CONSIDERED AN INTEGRAL PART OF THESE DOCUMENTS AND THE CONTRACTOR SHALL BE RESPONSIBLE TO OBTAIN A COPY OF THESE REQUIREMENTS/THIS DOCUMENT AND COMPLY WITH ALL REQUIREMENTS AND STANDARDS CONTAINED WITHIN.

2. THESE DRAWINGS ARE GENERALLY DIAGRAMMATIC AND ARE INTENDED TO CONVEY THE SCOPE OF WORK AND INDICATE GENERAL ARRANGEMENT OF LIGHTING FIXTURES, DEVICES, CONTROLS, ELECTRICAL FIXTURES, MOTORS, PANELBOARDS, EQUIPMENT, ETC. THE LOCATIONS OF ALL ITEMS SHOWN ON THESE DRAWINGS OR CALLED FOR IN THE SPECIFICATIONS THAT ARE NOT DEFINITELY FIXED BY DIMENSIONS ARE APPROXIMATE. THE EXACT LOCATIONS NECESSARY TO SECURE THE BEST CONDITIONS AND RESULTS MUST BE DETERMINED AT THE PROJECT. ALL LOCATIONS OF WORK EXPOSED TO VIEW ARE SUBJECT TO APPROVAL OF THE ARCHITECT PRIOR TO INSTALLATION.

3. THE ELECTRICAL CONTRACTOR SHALL VERIFY EXISTING CONDITIONS TO ENSURE THAT ALL NEW WORK WILL FIT INTO THE EXISTING STRUCTURE IN THE MANNER INTENDED AND AS SHOWN ON THE DRAWINGS. THE CONTRACTOR SHALL NOTIFY THE ARCHITECT/OWNERS REPRESENTATIVE PRIOR TO ANY ROUGH-INS, FABRICATIONS, OR PERFORMING ANY WORK IN THE AREA INVOLVING DIFFERENCES. NOTIFICATION SHALL BE IN THE FORM OF A DRAWING OR SKETCH INDICATING FIELD MEASUREMENTS AND NOTES RELATED TO THE AREA.

4. ANY DISCREPANCIES BETWEEN DRAWINGS AND SPECIFICATIONS SHALL BE PROMPTLY BROUGHT TO THE ATTENTION OF THE ENGINEER FOR CLARIFICATION DURING THE BIDDING PERIOD. NO ALLOWANCE SHALL SUBSEQUENTLY BE MADE TO THE CONTRACTOR BY REASON OF THEIR FAILURE TO HAVE BROUGHT SAID DISCREPANCIES TO THE ATTENTION OF THE ENGINEER DURING THE BIDDING PERIOD OR OF ANY ERROR ON THE CONTRACTOR'S PART.

5. ALL EQUIPMENT SHALL BE INSTALLED IN A NEAT, PROFESSIONAL AND WORKMANLIKE MANNER, RECTILINEAR TO BUILDING STRUCTURE.

6. ALL COMPONENTS SHOWN ON THE RISER/ONE-LINE DIAGRAMS, BUT NOT ON THE PLAN OR VICE VERSA, SHALL BE INCLUDED AS IF SHOWN ON BOTH.

7. REVIEW ALL TRADES' CONTRACT DOCUMENTS TO DETERMINE SPECIFIC MOUNTING LOCATIONS FOR ELECTRICAL EQUIPMENT. COORDINATE EXACT MOUNTING LOCATIONS WITH THE ARCHITECT.

8. REFER TO ARCHITECTURAL ELEVATIONS TO DERIVE EXACT LOCATIONS OF ALL RECEPTACLES, OUTLETS/JACKS, SWITCHES, ETC. LUMINAIRES AND CEILING MOUNTED EQUIPMENT SHALL BE COORDINATED WITH THE ARCHITECTURAL REFLECTED CEILING PLANS.

9. EXACT LOCATION OF MECHANICAL EQUIPMENT THAT REQUIRE ELECTRICAL CONNECTION ARE SHOWN ON THE MECHANICAL DRAWINGS. FIELD VERIFY EXACT LOCATIONS WITH MECHANICAL CONTRACTOR PRIOR TO ANY ROUGH-INS.

10. ALL RACEWAY/CIRCUITING SHALL BE RUN CONCEALED UNLESS SPECIFIED OTHERWISE. CUT/FISH/PATCH WALLS AS REQUIRED. USE OF SURFACE RACEWAY MUST BE APPROVED BY ARCHITECT ALONG WITH PROPOSED ROUTING AND SURFACE RACEWAY TYPE.

11. ALL RACEWAYS RUNNING THROUGH BUILDING EXPANSION JOINTS SHALL BE EQUIPPED WITH EXPANSION FITTINGS.

12. CONDUIT HOME RUNS SHOWN ON THE DRAWING WITH MORE THAN (3) CURRENT CARRYING CONDUCTORS ARE SHOWN DIAGRAMATICALLY. THIS CONTRACTOR SHALL NOT INSTALL MORE THAN (3) CURRENT CARRYING CONDUCTORS IN A RACEWAY UNLESS NATIONAL ELECTRICAL CODE (N.E.C), ARTICLE 310.15 DERATING FACTORS ARE APPLIED.

13. ALL LIGHTING AND GENERAL POWER BRANCH CIRCUITS SHALL INCLUDE A SEPARATE NEUTRAL CONDUCTOR, UNLESS SPECIFICALLY NOTED OTHERWISE.

14. THE ELECTRICAL CONTRACTOR SHALL REFER TO THE ELECTRICAL SPECIFICATIONS FOR ACCEPTABLE CONDUIT TYPES/LOCATIONS. ALL CONDUIT SIZES ON THE DRAWINGS ARE BASED ON THE LATEST EDITION OF THE N.E.C. CONDUIT FILL TABLES FOR ELECTRICAL METALLIC TUBING (E.M.T). CONDUIT SIZES SHALL BE REVISED TO THE SIZE REQUIRED, RELATIVE TO THE ACTUAL CONDUIT TYPE TO BE INSTALLED.

15. IT IS NOT INTENDED THAT THE PLANS INDICATE ALL THE NECESSARY BENDS, OFFSETS, PULL BOXES AND OBSTRUCTIONS. IT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR TO INSTALL THEIR WORK TO CONFORM TO THE STRUCTURE, MAINTAIN HEADROOM AND KEEP OPENINGS AND PASSAGEWAYS CLEAR. REFER TO THE ARCHITECTURAL DRAWINGS FOR ALL DIMENSIONS AS REQUIRED.

16. IT IS NOT INTENDED THAT THE PLANS INDICATE ALL CONDUIT ROUTES, PULL BOXES, ETC. CONTRACTOR SHALL BE RESPONSIBLE FOR DETERMINING ACTUAL CONDUIT ROUTING, QUANTITY AND LOCATION OF PULL BOXES WITHIN ACCESSIBLE LOCATIONS.

17. PROVIDE SCREW-COVER PULL BOXES IN CONDUIT RUNS AS REQUIRED TO LIMIT THE NUMBER OF BENDS TO NO MORE THAN THREE (3) OR 270 DEGREES TOTAL. SIZE PULL BOXES IN ACCORDANCE WITH NEC, ARTICLE 314.28. DOCUMENT ON RECORD DRAWINGS, SIZE AND LOCATION OF PULL BOXES USED IN FEEDER CONDUIT RUNS.

18. ALL OUTLET BOXES IN WALLS SHALL HAVE A MINIMUM OF ONE (1) DEDICATED VERTICAL CONDUIT ENTERING AT THE TOP OF THE BOX. HORIZONTAL CONDUIT CONNECTIONS SHALL ONLY BE PERMITTED UNDER WINDOWS OR UNLESS OTHERWISE NOTED ON DRAWINGS.

19. WHERE MULTIPLE DEVICES ARE INDICATED IN A COMMON LOCATION, GANG INTO A SINGLE COVER PLATE.

20. NEW OR MODIFIED PANELS SHALL BE PROVIDED WITH ENGRAVED NAMEPLATES AS DESIGNATED ON PANEL SCHEDULES SECURED TO PANEL FACE AND DENOTING ORIGIN OF FEEDER FROM WHICH PANEL IS SERVED.

SYMBOL LEGEND

SYMBOL	DESCRIPTION
A o/EM	LUMINAIRE. "A" INDICATES TYPE; "o" INDICATES SWITCH CONTROL. "/EM" INDICATES LUMINAIRE HAS INTEGRAL 90-MINUTE EMERGENCY BATTERY DRIVER.
A/NL	LUMINAIRE WIRED FOR 24 HOUR OPERATION, INDICATED WITH HATCH AND A "/NL" DESIGNATION.
	SELF CONTAINED LED UNIVERSAL MOUNTED "EXIT" SIGN. PROVIDE QUANTITY OF FACES AND DIRECTIONAL ARROWS AS INDICATED ON FLOOR PLANS. REFER TO LUMINAIRE SCHEDULE FOR ADDITIONAL INFORMATION.
	EMERGENCY LIGHTING BATTERY UNIT WITH HEADS AS INDICATED. REFER TO LUMINAIRE SCHEDULE FOR ADDITIONAL INFORMATION.
x o $	LIGHT SWITCH. SINGLE POLE, 20 AMPERE, 120/277 VOLT, 1HP. "x" INDICATES SWITCH CONTROL. "x" INDICATES THE FOLLOWING: "o" CONTROL OF SPECIFIED LUMINAIRES. "3" THREE-WAY TYPE. "4" FOUR-WAY TYPE. "D" SWITCH TO BE WIRED AHEAD OF DOWNSTREAM LINE VOLTAGE WALLBOX TYPE DIMMING SWITCH, INDICATED BY THE SAME SWITCH CONTROL TAG. "2P" DOUBLE POLE TYPE. "TM" PROGRAMMABLE TIMER SWITCH. TYPICALLY HONEYWELL #RPLS730B. "OS" LINE VOLTAGE MULTI-TECHNOLOGY WALL SWITCH OCCUPANCY SENSOR WITH MANUAL ON/OFF SWITCH. REFER TO DIAGRAM ON DRAWING ####. "VS" LINE VOLTAGE MULTI-TECHNOLOGY WALL SWITCH VACANCY SENSOR WITH MANUAL ON/OFF SWITCH. REFER TO DIAGRAM ON DRAWING ####. "COS" LINE VOLTAGE PIR WALL SWITCH CORRIDOR-TYPE OCCUPANCY SENSOR WITHOUT CONTROL SWITCH. TYPICALLY SENSOR SWITCH #HWR13-SN SERIES. MOUNT AT **7'-0" A.F.F.** UNLESS OTHERWISE SHOWN. "CVS" LINE VOLTAGE PIR WALL SWITCH CORRIDOR-TYPE OCCUPANCY SENSOR WITH CONTROL SWITCH. TYPICALLY SENSOR SWITCH #HWR13 SERIES. MOUNT AT **7'-0" A.F.F.** UNLESS OTHERWISE SHOWN. "VDS" LINE VOLTAGE ON/OFF SWITCH WITH DIMMER FUNCTION AND VACANCY SENSOR. REFER TO DIAGRAM ON DRAWING ####. "LV" LOW VOLTAGE ON/OFF SWITCH. "LV2" DUAL ZONE LOW VOLTAGE ON/OFF SWITCH. "LV4" QUAD ZONE LOW VOLTAGE ON/OFF SWITCH. "LVD" LOW VOLTAGE ON/OFF SWITCH WITH DIMMER FUNCTION. "LVD2" DUAL ZONE LOW VOLTAGE ON/OFF SWITCH WITH DIMMER FUNCTION. "LVD4" QUAD ZONE LOW VOLTAGE ON/OFF SWITCH WITH DIMMER FUNCTION. "SCN" SCENE CONTROLLER CAPABLE OF A MINIMUM OF FOUR PRESETS.
	LINE VOLTAGE WALLBOX TYPE DIMMING SWITCH. COORDINATE DIMMING PROTOCOL OF THIS DIMMING SWITCH AND LIGHTING LOAD SERVED PRIOR TO PLACING PURCHASE ORDER. "o" INDICATES SWITCH CONTROL.
OS DR,x	120/277V RATED CEILING MOUNTED OCCUPANCY SENSOR. A "DR" DESIGNATIONS INDICATES A DUAL RELAY SENSOR FOR 3/4 WAY SWITCHING. "x" REPRESENTS CONTROLLED LIGHTING ZONE (IF LOCATED IN A ROOM/AREA WITH MULTIPLE LIGHTING ZONES). TYPICALLY SENSOR SWITCH #OMR SERIES.
OS LV,x	LOW VOLTAGE CEILING MOUNTED PHOTOSENSOR, "X" REPRESENTS CONTROLLED LIGHTING ZONE.
OS OV,x	LOW VOLTAGE MULTI-TECHNOLOGY CEILING MOUNTED OCCUPANCY SENSOR, "X" REPRESENTS CONTROLLED LIGHTING ZONE.
OS VV,x	LOW VOLTAGE MULTI-TECHNOLOGY CEILING MOUNTED VACANCY SENSOR, "X" REPRESENTS CONTROLLED LIGHTING ZONE.
	CIRCUITRY TURNING UP
	CIRCUITRY TURNING DOWN
	UNDERGROUND OR UNDERSLAB CONDUIT/RACEWAY AND/OR CIRCUITING.
	CONDUIT HOME RUN TO PANEL/CIRCUIT BREAKER AS INDICATED. ALL UNMARKED HOMERUNS TO CONTAIN 2-12 AWG & 1-12 AWG GND IN 3/4" CONDUIT (MINIMUM), UNLESS NOTED OTHERWISE. SIZE CONDUIT AND PROVIDE JUNCTION BOXES PER CODE UNLESS OTHERWISE NOTED.
	DWELLING UNITS: DWELLING UNIT SINGLE-PHASE LOAD CENTER.
	ALL OTHER AREAS: BRANCH CIRCUIT PANELBOARD.
	DISCONNECT SWITCH. REFER TO EQUIPMENT SCHEDULE OR PLAN NOTE FOR FUSING NECESSITY, SIZES, RATINGS, ETC.
	DUPLEX RECEPTACLE - 125 VOLT, 20 AMPERE, 2 POLE, 3 WIRE, GROUNDING TYPE MOUNTED AT 18" AFF UNLESS NOTED OTHERWISE
	DOUBLE DUPLEX RECEPTACLE - 125 VOLT, 20 AMPERE, 2 POLE, 3 WIRE, GROUNDING TYPE MOUNTED AT 18" AFF UNLESS NOTED OTHERWISE
	GFCI GROUND TYPE DUPLEX RECEPTACLE - 125 VOLT, 20 AMPERE, 2 POLE, 3 WIRE, GROUNDING TYPE MOUNTED AT 18" AFF UNLESS NOTED OTHERWISE
x o	DUPLEX RECEPTACLE - 125 VOLT, 20 AMPERE, 2 POLE, 3 WIRE, GROUNDING TYPE MOUNTED AT 18" AFF UNLESS NOTED OTHERWISE. RECEPTACLE SHALL BE SPLIT WIRED SUCH THAT TOP OUTLET TO BE WIRED FOR SWITCHING TO SWITCH INDICATED BY 'o'.
	TAMPER RESISTANT, POP-UP COUNTER RECEPTACLE - 125 VOLT, 20 AMPERE, 2 POLE, 3 WIRE, GROUNDING TYPE. TYPICALLY BRYANT #RCT220xx.
	FLOOR BOX WITH DUPLEX RECEPTACLE. PROVIDE APPROPRIATE BOX TYPE FOR CONCRETE OR WOOD FLOOR CONSTRUCTION.
	RECEPTACLE SYMBOL WITH ADDITIONAL SUBSCRIPT DESIGNATES TYPE OR SPECIFIC REQUIREMENTS AS FOLLOWS: "AC" INDICATES MOUNTED 8" ABOVE COUNTER UNLESS NOTED OTHERWISE ON ARCHITECTURAL ELEVATIONS. "CLG" FLUSH CEILING MOUNTED OR 12" ABOVE WINDOW. "48" RECEPTACLE MOUNTED AT HEIGHT INDICATED. "L" RECEPTACLE WITH ILLUMINATED NIGHT LIGHT. "VIF" VERIFY EXISTING MOUNTING HEIGHT IN FIELD, OR CONFIRM EXACT LOCATION IN OF NEW DEVICE IN FIELD. "WP" WEATHERPROOF "WHILE-IN-USE" (HUBBELL #WP26M) COVER. "U" DUPLEX RECEPTACLE WITH ONE USB TYPE "A" PORT AND ONE TYPE "C" PORT, LEVITON #T5633 SERIES OR EQUAL. "T" TAMPER RESISTANT (**ONLY INDICATED OUTSIDE OF DWELLING UNITS**). "C" RECEPTACLE PROTECTED BY UPSTREAM GFCI OR GFCI-TYPE BREAKER WITH "GFCI PROTECTED" LABEL ON THE FACE OF DEVICE. "CR" RECEPTACLE CONTROLLED BY AN AUTOMATIC CONTROL DEVICE FOR PLUG LOAD CONTROL, DURABLY AND PERMANENTLY MARKED WITH THE WORD "CONTROLLED" AND THE SYMBOL SHOWN IN NEC FIGURE **#406.3(F)**.
	SPECIAL PURPOSE RECEPTACLE. REFER TO EQUIPMENT SCHEDULE OR PLAN NOTE FOR RECEPTACLE TYPE AND RATINGS. COORDINATE EXACT NEMA CONFIGURATION WITH EQUIPMENT PRIOR TO ROUGH-IN.
	JUNCTION BOX - SIZE PER N.E.C. ARTICLE 314 REQUIREMENTS.
	ALL AREAS BESIDES IN DWELLING UNITS: COMBINATION VOICE/DATA OUTLET (4" x 2-1/8" DEEP JUNCTION BOX) COMPLETE WITH SINGLE GANG MUD RING (VERIFY DRYWALL THICKNESS) AND COVER PLATE (COLOR SELECTION BY ARCHITECT) MOUNTED AT 18" A.F.F. PROVIDE 1-0" CONDUIT WITH INSULATING BUSHING AND PULLSTRING STUBBED UP 12" INTO ACCESSIBLE CEILING SPACE. WITHIN DWELLING UNITS: REFER TO TYPICAL APARTMENT DATA FACEPLATE DETAILS ON DRAWING ####.
V	FLOOR BOX WITH DATA/VOICE OUTLET AS DESCRIBED ABOVE. PROVIDE APPROPRIATE BOX TYPE FOR CONCRETE OR WOOD FLOOR CONSTRUCTION.
F	FIRE ALARM SYSTEM MANUAL PULL STATION.
F H LF	FIRE ALARM COMBINATION AUDIBLE/VISUAL (HORN/STROBE) SIGNALING DEVICE. **LF DESIGNATION INDICATES LOW FREQUENCY ALARM DEVICE.**
V	FIRE ALARM VISUAL (STROBE) ONLY SIGNALING DEVICE.
S CO	ALL AREAS BESIDES IN DWELLING UNITS: FIRE ALARM SYSTEM SMOKE DETECTOR. WITHIN DWELLING UNITS: DWELLING UNIT SMOKE ALARM. REFER TO DETAIL(S) ON DRAWING ####. WITHIN DWELLING UNITS: BUILDING FIRE ALARM SYSTEM ADDRESSABLE SMOKE DETECTOR WITH 520HZ LOW FREQUENCY SOUNDER BASE. "CO" DESIGNATION INDICATES COMBINATION, "MULTI-CRITERIA" SMOKE/CARBON MONOXIDE CAPABILITY.
FACP	FIRE ALARM CONTROL PANEL. THE BRANCH CIRCUIT SERVING THIS PANEL SHALL ONLY SERVE FIRE ALARM EQUIPMENT. REFER TO PANEL SCHEDULE(S).
FAAP	FIRE ALARM ANNUNCIATOR PANEL. THE BRANCH CIRCUIT SERVING THIS PANEL SHALL ONLY SERVE FIRE ALARM EQUIPMENT. REFER TO PANEL SCHEDULE(S).
TS	TAMPER SWITCH. COORDINATE ALL REQUIREMENTS WITH SPRINKLER/FIRE SUPPRESSION DRAWINGS.
FS	FLOW SWITCH. COORDINATE ALL REQUIREMENTS WITH SPRINKLER/FIRE SUPPRESSION DRAWINGS.
H	MAGNETIC DOOR HOLD OPEN

WIRING SCHEDULE FOR 15/20A RECEPTACLE & LIGHTING CIRCUITS

DISTANCE	120V
1' - 100'	#12 (AWG)
100' - 150'	#10 (AWG)
150' - 200'	#10 (AWG)
200' - 300'	#8 (AWG)

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

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OSU AIRPORT WEST

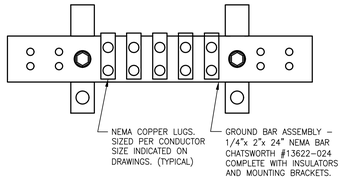
BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title

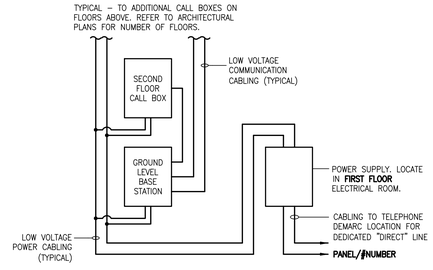
ELECTRICAL LEGEND AND DETAILS

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number
Checked by: KDS	E0.03A
Approved: GSS	



COMMUNICATION TERMINAL BOARD GROUNDING DETAIL

NO SCALE



TWO-WAY COMMUNICATION SYSTEM WIRING DIAGRAM

NO SCALE

TWO-WAY COMMUNICATION SYSTEM WIRING DIAGRAM GENERAL NOTES:

A. PROVIDE A COMPLETE, OPERATING TWO–WAY COMMUNICATION SYSTEM CONSISTING OF A **FIRST FLOOR** BASE STATION AND CALL BOX STATION ELEVATOR LANDING LOCATIONS ALL REMAINING FLOORS. SYSTEM SHALL COMPLY WITH ALL STATE CODES, LOCAL CODES AND ADA. MANUFACTURER SHALL BE RATH "AREA OF REFUGE" OR EQUAL BY CORNELL.

B. INSTALL ALL CABLING PER MANUFACTURER'S INSTRUCTIONS. WIRING SHALL BE INSTALLED IN 3/4" CONDUIT (MINIMUM).

C. COORDINATE LOCATION OF BASE STATION AND CALL BOX STATIONS WITH THE AHJ AND ARCHITECT PRIOR TO ROUGH–IN.

D. SUBMITTAL SHALL INCLUDE EQUIPMENT DATA SHEETS, OPERATING MANUALS AND DETAILED WIRING SCHEMATICS.

E. ALL RECESSED BOXES/BASE STATION SHALL BE FIRESTOPPED TO MAINTAIN FIRE RATING OF WALL.

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

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OSU AIRPORT WEST

BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title

ELECTRICAL SCHEDULES AND DETAILS

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number
Checked by: KDS	E0.04A
Approved: GSS	

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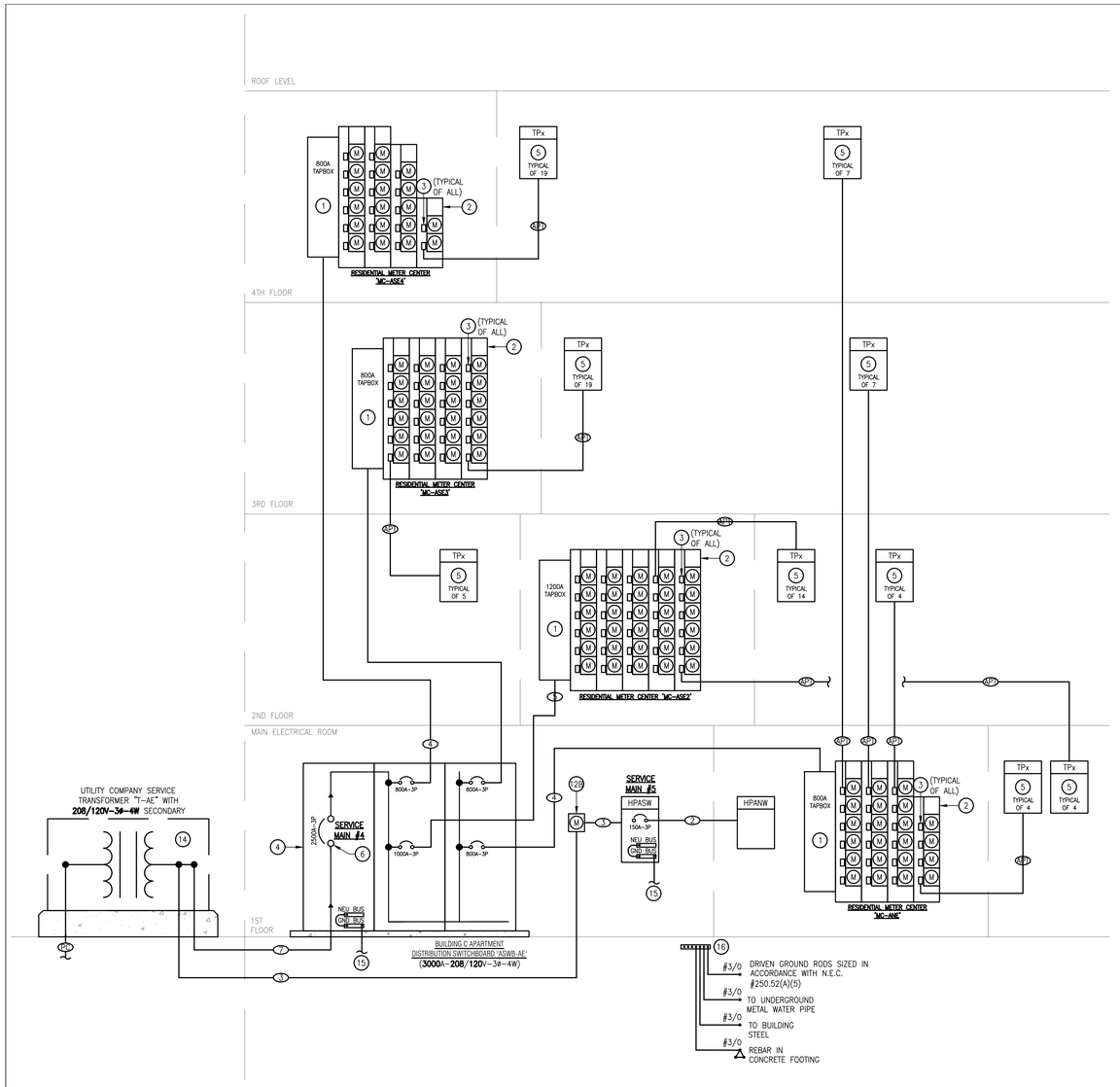

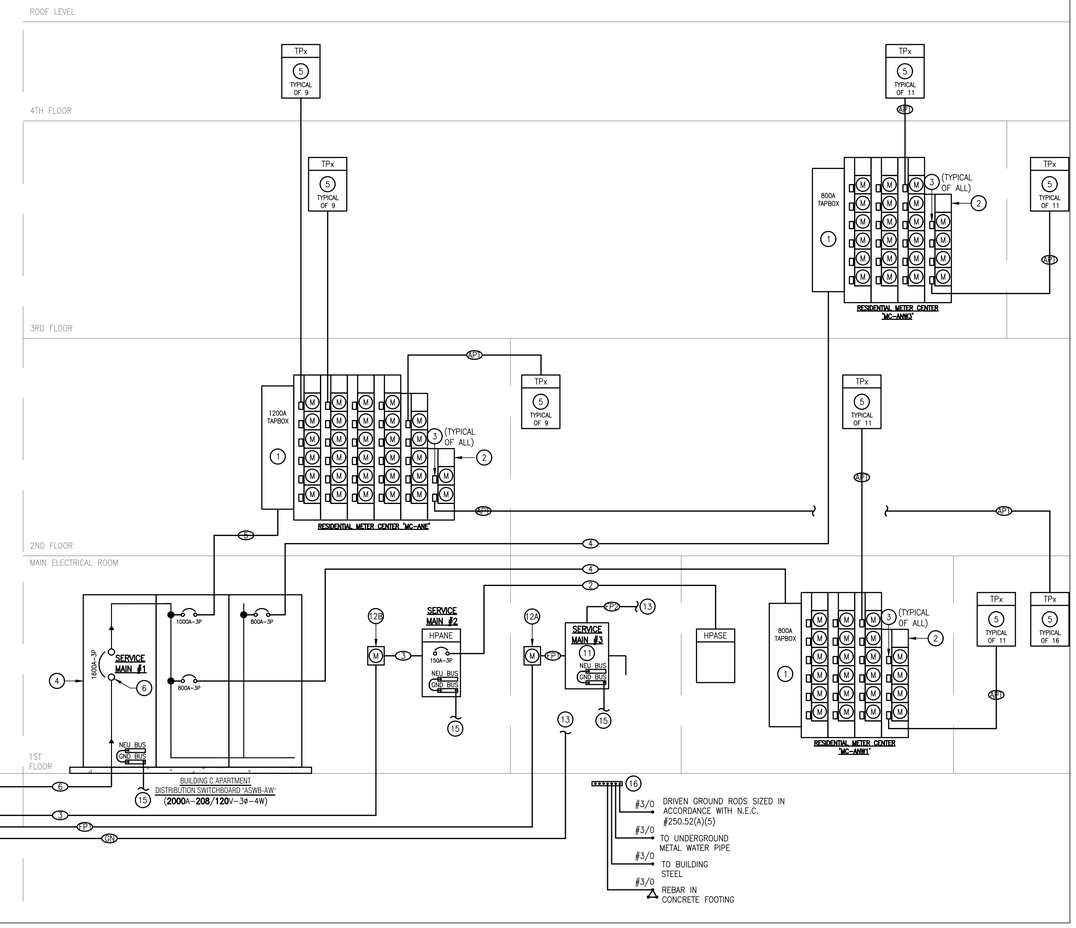

FEEDER SCHEDULE

ID	NOMINAL AMPACITY	CONDUCTOR & CONDUIT SIZE	CONDUCTOR MATERIAL	NOTES
1	20A	2#10, 1#10(G), 1-1/2"C.	COPPER	1,5
2	155A	4#3/0, 1#4(G), 2"C.	ALUMINUM	1,3
3	310A	4#500KCMIL, 3"C.	ALUMINUM	1,3
4	780A	2 SETS OF 4#750KCMIL, 1#3/0(G), 4"C.	ALUMINUM	1,3
5	1020A	3 SETS OF 4#600KCMIL, 1#4/0(G), 4"C.	ALUMINUM	1,3
6	1700A	5 SETS OF 4#600KCMIL, 4"C.	ALUMINUM	1,2
7	2695A	7 SETS OF 4#750KCMIL, 4"C.	ALUMINUM	1,2
FP1	85A	4#3, 1-1/2"C.	COPPER	1,2,5
FP2	85A	3#3, 1#8(G), 1-1/2"C.	COPPER	1,2,5
GN	85A	4#3, 1#8(G), 1-1/2"C.	COPPER	1,2,5
APT	120A (1&2 BED)	3#1/0, 1#4(G), 1-1/2"C. (1&2 BED)	ALUMINUM	1,3,4
	135A (3 BED)	3#2/0, 1#4(G), 2"C. (3 BED)	ALUMINUM	1,3,4
PC	–	TWO EMPTY **5"** CONDUITS.	–	6

FEEDER SCHEDULE NOTES:

1. ALL CONDUCTORS ARE BASED ON EITHER TYPE "TW"/"UF" (UNDER AND INCLUDING #1AWG) OR TYPE "THHW"/"THWN" (OVER #1AWG) INSULATION WITH A 60°C RATING (UNDER AND INCLUDING #1AWG) OR A 75°C RATING (OVER #1AWG) IN ADDITION TO 75°C TERMINATIONS. CONDUIT SIZES ARE ALSO INDICATED.
2. REFER TO RESPECTIVE TRENCH SECTION DETAIL ON SHEET #ES.02A FOR ADDITIONAL INFORMATION REGARDING THIS FEEDER.
3. COPPER CONDUCTORS MAY BE UTILIZED FOR THIS FEEDER. HOWEVER, THE E.C. IS RESPONSIBLE FOR PROVIDING THE APPROPRIATE CONDUCTOR SIZE IN ACCORDANCE WITH N.E.C. TABLE #310.15(B)(16).
4. TYPE SER CABLE FOR MAY BE UTILIZED FOR THIS FEEDER IN LIEU OF CONDUCTORS IN CONDUIT.
5. THIS FEEDER IS SIZED TO ACCOUNT FOR VOLTAGE DROP (ALL FEEDS WITH THIS NOTE) **AND SHARED CONDUIT DERATING FACTOR (FOR FEEDER #1).**
6. ROUTE TWO **5"** CONDUITS TO LOCATION DETERMINED BY LOCAL UTILITY COMPANY.

CODED NOTES

1. PROVIDE A U.L. LISTED, 1200A METER CENTER TAPBOX RATED AT **208/120V-3ø-4W**, TYPICALLY **SIEMENS #WTB SERIES**. ENTIRE ASSEMBLY SHALL BE RATED AT A MINIMUM **65,000 RMS** SYMMETRICAL SHORT CIRCUIT AMPERES. ACCEPTABLE ALTERNATIVES SHALL BE EATON, SQUARE D, OR GENERAL ELECTRIC, HOWEVER, THE E.C. SHALL BE RESPONSIBLE FOR ENSURING MODIFIED EQUIPMENT, SIZES, ETC. MEET N.E.C. CLEARANCE CRITERIA.
2. PROVIDE RESIDENTIAL METER STACK (**QUANTITY AS INDICATED**) WITH **THREE PHASE** HORIZONTAL BUS (**AMPACITY AS SHOWN**) AND BOLT-ON CIRCUIT BREAKERS, TYPICALLY SIEMENS #WMM SERIES. APPROXIMATE DIMENSIONS ARE **16.22"W X 43.31"H X 8.09"D (3 METER STACK) AND 16.22"W X 70.31"H X 8.09"D (6 METER STACK).** ACCEPTABLE ALTERNATIVES SHALL BE EATON, SQUARE D, OR GENERAL ELECTRIC, HOWEVER, THE E.C. SHALL BE RESPONSIBLE FOR ENSURING MODIFIED EQUIPMENT, SIZES, ETC. MEET N.E.C. CLEARANCE CRITERIA. METER STACK SHALL BE PROVIDED SUCH THAT NO CIRCUIT BREAKER SHALL EXCEED 6'-8" A.F.F. PROVIDE ALL REQUIRED HOUSEKEEPING PADS IN FRONT OF EQUIPMENT TO MEET THIS REQUIREMENT. COORDINATE ALL METER SOCKET REQUIREMENTS WITH LOCAL UTILITY COMPANY. ENTIRE ENCLOSURE SHALL BE RATED AT A MINIMUM **65,000 RMS** SYMMETRICAL SHORT CIRCUIT AMPERES.
3. LOCATION OF 125A-2P (ONE AND TWO BEDROOM APARTMENT UNITS) AND **150A-2P** (THREE BEDROOM APARTMENT UNITS) TENANT MAIN CIRCUIT BREAKER TO BE PROVIDED WITH METER STACK. REFER TO UNIT LOAD CALCULATIONS ON DRAWING **#E3.00C**. ALL MAIN CIRCUIT BREAKERS SHALL BE RATED AT A MINIMUM **100,000 RMS** SYMMETRICAL SHORT CIRCUIT AMPERES.
4. PROVIDE A U.L. LISTED "SERVICE ENTRANCE RATED" (RATINGS AS INDICATED ON RISER) DISTRIBUTION SWITCHBOARD, TYPICALLY **SIEMENS #SB2 SERIES.** ASSEMBLY SHALL BE RATED AT A MINIMUM 100,000 RMS SYMMETRICAL SHORT CIRCUIT AMPERES. SWITCHBOARD SHALL INCLUDE ANY SECTION/PROVISIONS FOR UTILITY COMPANY METERING COMPLYING WITH LOCAL UTILITY COMPANY REQUIREMENTS AND SHALL INCLUDE A SIEMENS #TPS4-06 SERIES SURGE PROTECTION DEVICE OR EQUAL BY OTHERS, AND A #PAC4220 METERING/MONITORING DEVICE. PROVIDE CONCRETE PAD IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS.
5. TYPICAL GUEST SUITE LOAD CENTER "GSx", TYPICALLY SIEMENS #ES SERIES. REFER TO TYPICAL PANEL SCHEDULES ON DRAWING **#E3.00C** FOR GUEST UNIT SERVICE CALCULATIONS, PANEL SIZES AND BREAKERS, AND OTHER ADDITIONAL INFORMATION. PROVIDE A TYPE 2 SURGE PROTECTIVE DEVICE (SPD) AND INSTALL IN EACH LOAD CENTER IN ACCORDANCE WITH NEC #215.18. TYPICALLY SIEMENS #QSPD-2-A-065 OR EQUAL BY OTHERS. THE KAIC RATING OF THE SPD SHALL NOT BE LESS THAN THE RATING OF THE LOAD CENTER AS SHOWN ON THE PANEL SCHEDULE.
6. PROVIDE ARC FAULT REDUCTION IN ACCORDANCE WITH N.E.C. ARTICLE #240.87.
7. CAM LOCK CONNECTION FOR REDUNDANT TEMPORARY MAINTENANCE GENERATOR LOCATED ON MANUAL TRANSFER SWITCH.
8. PRIOR TO ORDERING EQUIPMENT, PROVIDE SELECTIVE COORDINATION STUDY FOR EMERGENCY SYSTEM PER NEC #700.32. CIRCUIT BREAKERS IN PANEL SHALL BE COORDINATED WITH ALL UPSTREAM OVERCURRENT PROTECTION DEVICES.
9. THE EMERGENCY STANDBY GENERATOR MUST BE U.L. LISTED IN ACCORDANCE WITH U.L. #2200. PROVIDE A **20A-120V-1ø** BRANCH CIRCUIT FOR SERVICE TO BATTERY CHARGER AND A **20A-120V-1ø** BRANCH CIRCUIT FOR SERVICE TO GENERATOR BLOCK HEATER. CONNECT PER MANUFACTURER'S WIRING DIAGRAM.
10. PROVIDE ALL EQUIPMENT, ACCESSORIES, WIRING, ETC. IN COMPLIANCE WITH N.E.C. ARTICLE #700.3(F). THIS SHALL INCLUDE, BUT, NOT BE LIMITED TO THE PROVISION OF A PERMANENT SWITCHING MEANS TO CONNECT TO A PORTABLE/TEMPORARY GENERATOR, MECHANICAL/ELECTRICAL INTERLOCKING FOR THE PREVENTION OF INADVERTENT INTERCONNECTION OF POWER SOURCES, CONTACT POINTS THAT ANNUNCIATE AT A LOCATION REMOTE FROM THE GENERATOR, ETC. VERIFY LOCATION WITH ARCHITECT, OWNER, AND GENERATOR MANUFACTURER.
11. PROVIDE A 400A-3P, 240V RATED, SERVICE ENTRANCE RATED, FUSED DISCONNECT SWITCH, FUSED AT **400A**. DISCONNECT SHALL BE INSTALLED IN ACCORDANCE WITH NEC #695.4(3).
12A. PROVIDE A 100A, 240V, 4W SELF CONTAINED UTILITY METER SOCKET. COORDINATE PROVISION AND WORK REGARDING UTILITY METER WITH LOCAL UTILITY COMPANY.
12B. PROVIDE A 320A, 240V, 4W SELF CONTAINED UTILITY METER SOCKET. COORDINATE PROVISION AND WORK REGARDING UTILITY METER WITH LOCAL UTILITY COMPANY.
13. ROUTE TO THE FIRE CONTROL PANEL FOR THE 20HP FIRE PUMP WITH INTEGRAL AUTOMATIC TRANSFER SWITCHING CAPABILITY AND SOFT START METHOD TO BE SUPPLIED BY THE PLUMBING AND/OR FIRE SUPPRESSION CONTRACTOR. ALL WIRING TO BE PROVIDED BY THE E.C. INCLUDE ALL WIRING REQUIRED FOR A COMPLETE ELECTRICAL INSTALLATION IN ACCORDANCE WITH THE MANUFACTURER'S REQUIREMENTS, IN ADDITION TO STATE AND LOCAL CODES. THE INSTALLATION SHALL BE IN FULL COMPLIANCE WITH WITH N.E.C. ARTICLE #695.
14. PROVIDE CONCRETE PAD OR PRE-CAST CONCRETE VAULT IN ACCORDANCE WITH UTILITY COMPANY REQUIREMENTS. ADDITIONALLY, PROVIDE A TRANSFORMER GROUNDING MAT IN ACCORDANCE WITH UTILITY COMPANY REQUIREMENTS.
15. ROUTE A #3/0 COPPER GROUND CONDUCTOR TO THE GROUND BUS BAR DESCRIBED IN CODED NOTE #16.
16. PROVIDE A MASTER SERVICE ENTRANCE GROUND BUS TO WHICH ALL SERVICE ENTRANCE MAINS SHALL BE BONDED. TYPICAL EQUIPMENT SHALL BE A STORM COPPER #SCBG-6KT GROUND BAR KIT. BAR SHALL BE .25" THICK, 4" X 12" BAR MOUNTED ON (2) BRACKETS W/INSULATORS. PROVIDE A MINIMUM OF (6) .437" DIAMETER 2-HOLE AND (6) .281" DIAMETER MOUNTINGS OR AS OTHERWISE REQUIRED.

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

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OSU AIRPORT WEST

BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title

POWER RISER DIAGRAM

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number
Checked by: KDS	E0.05A
Approved: GSS	

METER CENTER "MC-ASW3" ELECTRIC SERVICE CALCULATION

DESCRIPTION	AREA	QUANTITY	LOAD	TOTAL WC	N.E.C. ARTICLE
1 BEDROOM UNITS					
1BED UNIT TYPE A1	713	2	3	4278	220.84(c)(1)
1BED UNIT TYPE AD2	897	4	3	10764	220.84(c)(1)
1BED UNIT TYPE AD2 PW	897	4	3	10764	220.84(c)(1)
1BED UNIT TYPE S1	704	6	3	12672	220.84(c)(1)
2 BEDROOM UNITS					
2BED UNIT TYPE B1.2	1035	2	3	6210	220.84(c)(1)
2BED UNIT TYPE BD3.1	1275	2	3	7650	220.84(c)(1)
3 BEDROOM UNITS					
3BED UNIT TYPE C3	1442	2	3	8652	220.84(c)(1)
TOTAL CONNECTED LIGHTING/RECEPTACLE LOAD:				60,990 W	
SMALL APPLIANCE BRANCH CIRCUITS	-	44	1500	66000	220.84(c)(2)
LAUNDRY CIRCUIT (WASHING MACHINE)	-	22	1500	33000	220.84(c)(2)
REFRIGERATOR	-	22	600	13200	220.84(c)(3)
ELECTRIC RANGE/OVEN	-	22	8000	176000	220.84(c)(3)
RANGE HOOD/MICROWAVE	-	22	1680	36960	220.84(c)(3)
DISHWASHER/DISPOSER	-	22	1200	26400	220.84(c)(3)
ELECTRIC DRYER	-	22	5000	110000	220.84(c)(3)
ELECTRIC WATER HEATER	-	22	4500	99000	220.84(c)(3)
1.5 TON HVAC	-	16	5180	82880	220.84(c)(5)
2 TON HVAC	-	4	6698	26792	220.84(c)(5)
2.5 TON HVAC	-	2	8871	17742	220.84(c)(5)
TOTAL CONNECTED EQUIPMENT LOAD:				687,974 W	
TOTAL CONNECTED LOAD:				748,964 W	
DEMAND FACTOR:				36%	TABLE 220.84
TOTAL DEMAND LOAD:				**269,627 W**	
AMPS AT 120/208V-3Ø				**748.4 A**	

METER CENTER "MC-ASE2" ELECTRIC SERVICE CALCULATION

DESCRIPTION	AREA	QUANTITY	LOAD	TOTAL WC	N.E.C. ARTICLE
1 BEDROOM UNITS					
1BED UNIT TYPE A1	713	8	3	17112	220.84(c)(1)
1BED UNIT TYPE AD2	897	4	3	10764	220.84(c)(1)
1BED UNIT TYPE AD2 PW	897	4	3	10764	220.84(c)(1)
1BED UNIT TYPE S1	704	6	3	12672	220.84(c)(1)
2 BEDROOM UNITS					
2BED UNIT TYPE B1	1035	3	3	9315	220.84(c)(1)
2BED UNIT TYPE B1.3	1062	1	3	3186	220.84(c)(1)
2BED UNIT TYPE BD3.1	1275	2	3	7650	220.84(c)(1)
3 BEDROOM UNITS					
3BED UNIT TYPE C3	1442	2	3	8652	220.84(c)(1)
TOTAL CONNECTED LIGHTING/RECEPTACLE LOAD:				80,115 W	
SMALL APPLIANCE BRANCH CIRCUITS	-	60	1500	90000	220.84(c)(2)
LAUNDRY CIRCUIT (WASHING MACHINE)	-	30	1500	45000	220.84(c)(2)
REFRIGERATOR	-	30	600	18000	220.84(c)(3)
ELECTRIC RANGE/OVEN	-	30	8000	240000	220.84(c)(3)
RANGE HOOD/MICROWAVE	-	30	1680	50400	220.84(c)(3)
DISHWASHER/DISPOSER	-	30	1200	36000	220.84(c)(3)
ELECTRIC DRYER	-	30	5000	150000	220.84(c)(3)
ELECTRIC WATER HEATER	-	30	4500	135000	220.84(c)(3)
1.5 TON HVAC	-	22	5180	113960	220.84(c)(5)
2 TON HVAC	-	6	6698	40188	220.84(c)(5)
2.5 TON HVAC	-	2	8871	17742	220.84(c)(5)
TOTAL CONNECTED EQUIPMENT LOAD:				936,290 W	
TOTAL CONNECTED LOAD:				1,016,405 W	
DEMAND FACTOR:				33%	TABLE 220.84
TOTAL DEMAND LOAD:				**335,414 W**	
AMPS AT 120/208V-3Ø				**931.0 A**	

APARTMENT SWITCHBOARD "ASWB-AW" ELECTRIC SERVICE CALCULATION

DESCRIPTION	AREA	QUANTITY	LOAD	TOTAL WC	N.E.C. ARTICLE
1 BEDROOM UNITS					
1BED UNIT TYPE A1	713	16	3	34224	220.84(c)(1)
1BED UNIT TYPE AD3	1025	1	3	3075	220.84(c)(1)
1BED UNIT TYPE AD2	897	8	3	21528	220.84(c)(1)
1BED UNIT TYPE AD2 PW	897	8	3	21528	220.84(c)(1)
1BED UNIT TYPE S1	704	12	3	25344	220.84(c)(1)
2 BEDROOM UNITS					
2BED UNIT TYPE B1	1035	7	3	21735	220.84(c)(1)
2BED UNIT TYPE B1.2	1035	4	3	12420	220.84(c)(1)
2BED UNIT TYPE BD3.2	1275	3	3	11475	220.84(c)(1)
2BED UNIT TYPE B1.1	1035	4	3	12420	220.84(c)(1)
2BED UNIT TYPE BD3.1	1275	4	3	15300	220.84(c)(1)
3 BEDROOM UNITS					
3BED UNIT TYPE C3	1062	8	3	25488	220.84(c)(1)
TOTAL CONNECTED LIGHTING/RECEPTACLE LOAD:				204,537 W	
SMALL APPLIANCE BRANCH CIRCUITS	-	150	1500	225000	220.84(c)(2)
LAUNDRY CIRCUIT (WASHING MACHINE)	-	75	1500	112500	220.84(c)(2)
REFRIGERATOR	-	75	600	45000	220.84(c)(3)
ELECTRIC RANGE/OVEN	-	75	8000	600000	220.84(c)(3)
RANGE HOOD/MICROWAVE	-	75	1680	126000	220.84(c)(3)
DISHWASHER/DISPOSER	-	75	1200	90000	220.84(c)(3)
ELECTRIC DRYER	-	75	5000	375000	220.84(c)(3)
ELECTRIC WATER HEATER	-	75	4500	337500	220.84(c)(3)
1.5 TON HVAC	-	45	5180	233100	220.84(c)(5)
2 TON HVAC	-	19	6698	127262	220.84(c)(5)
2.5 TON HVAC	-	8	8871	70968	220.84(c)(5)
TOTAL CONNECTED EQUIPMENT LOAD:				2,342,330 W	
TOTAL CONNECTED LOAD:				2,546,867 W	
DEMAND FACTOR:				23%	TABLE 220.84
TOTAL DEMAND LOAD:				**585,779 W**	
AMPS AT 120/208V-3Ø				**1,626.0 A**	

METER CENTER "MC-ASW1" ELECTRIC SERVICE CALCULATION

DESCRIPTION	AREA	QUANTITY	LOAD	TOTAL WC	N.E.C. ARTICLE
1 BEDROOM UNITS					
1BED UNIT TYPE A1	713	2	3	4278	220.84(c)(1)
1BED UNIT TYPE AD2	897	4	3	10764	220.84(c)(1)
1BED UNIT TYPE AD2 PW	897	4	3	10764	220.84(c)(1)
1BED UNIT TYPE S1	704	6	3	12672	220.84(c)(1)
2 BEDROOM UNITS					
2BED UNIT TYPE B1.2	1035	2	3	6210	220.84(c)(1)
2BED UNIT TYPE BD3.1	1275	2	3	7650	220.84(c)(1)
3 BEDROOM UNITS					
3BED UNIT TYPE C3	1442	2	3	8652	220.84(c)(1)
TOTAL CONNECTED LIGHTING/RECEPTACLE LOAD:				60,990 W	
SMALL APPLIANCE BRANCH CIRCUITS	-	44	1500	66000	220.84(c)(2)
LAUNDRY CIRCUIT (WASHING MACHINE)	-	22	1500	33000	220.84(c)(2)
REFRIGERATOR	-	22	600	13200	220.84(c)(3)
ELECTRIC RANGE/OVEN	-	22	8000	176000	220.84(c)(3)
RANGE HOOD/MICROWAVE	-	22	1680	36960	220.84(c)(3)
DISHWASHER/DISPOSER	-	22	1200	26400	220.84(c)(3)
ELECTRIC DRYER	-	22	5000	110000	220.84(c)(3)
ELECTRIC WATER HEATER	-	22	4500	99000	220.84(c)(3)
1.5 TON HVAC	-	16	5180	82880	220.84(c)(5)
2 TON HVAC	-	4	6698	26792	220.84(c)(5)
2.5 TON HVAC	-	2	8871	17742	220.84(c)(5)
TOTAL CONNECTED EQUIPMENT LOAD:				687,974 W	
TOTAL CONNECTED LOAD:				748,964 W	
DEMAND FACTOR:				36%	TABLE 220.84
TOTAL DEMAND LOAD:				**269,627 W**	
AMPS AT 120/208V-3Ø				**748.4 A**	

APARTMENT SWITCHBOARD "ASWB-AE" ELECTRIC SERVICE CALCULATION

DESCRIPTION	AREA	QUANTITY	LOAD	TOTAL WC	N.E.C. ARTICLE
1 BEDROOM UNITS					
1BED UNIT TYPE A1	713	27	3	57753	220.84(c)(1)
1BED UNIT TYPE A1.1	709	3	3	6381	220.84(c)(1)
1BED UNIT TYPE A1.2	709	3	3	6381	220.84(c)(1)
1BED UNIT TYPE AD2	897	8	3	21528	220.84(c)(1)
1BED UNIT TYPE AD2 PW	897	8	3	21528	220.84(c)(1)
1BED UNIT TYPE S1	704	12	3	25344	220.84(c)(1)
2 BEDROOM UNITS					
2BED UNIT TYPE B1	1035	13	3	40365	220.84(c)(1)
2BED UNIT TYPE B1.1	1035	3	3	9315	220.84(c)(1)
2BED UNIT TYPE B1.3	1062	1	3	3186	220.84(c)(1)
2BED UNIT TYPE BD3.1	1275	4	3	15300	220.84(c)(1)
2BED UNIT TYPE B1A	1035	1	3	3105	220.84(c)(1)
2BED UNIT TYPE BD3.2	1275	4	3	15300	220.84(c)(1)
3 BEDROOM UNITS					
3BED UNIT TYPE C3	1445	8	3	34680	220.84(c)(1)
TOTAL CONNECTED LIGHTING/RECEPTACLE LOAD:				260,166 W	
SMALL APPLIANCE BRANCH CIRCUITS	-	190	1500	285000	220.84(c)(2)
LAUNDRY CIRCUIT (WASHING MACHINE)	-	95	1500	142500	220.84(c)(2)
REFRIGERATOR	-	95	600	57000	220.84(c)(3)
ELECTRIC RANGE/OVEN	-	95	8000	760000	220.84(c)(3)
RANGE HOOD/MICROWAVE	-	95	1680	159600	220.84(c)(3)
DISHWASHER/DISPOSER	-	95	1200	114000	220.84(c)(3)
ELECTRIC DRYER	-	95	5000	475000	220.84(c)(3)
ELECTRIC WATER HEATER	-	95	4500	427500	220.84(c)(3)
1.5 TON HVAC	-	61	5180	315980	220.84(c)(5)
2 TON HVAC	-	21	6698	140658	220.84(c)(5)
2.5 TON HVAC	-	8	8871	70968	220.84(c)(5)
TOTAL CONNECTED EQUIPMENT LOAD:				2,948,206 W	
TOTAL CONNECTED LOAD:				3,208,372 W	
DEMAND FACTOR:				23%	TABLE 220.84
TOTAL DEMAND LOAD:				**737,926 W**	
AMPS AT 120/208V-3Ø				**2,048.3 A**	

METER CENTER "MC-ANE" TOTAL ELECTRIC SERVICE CALCULATION

DESCRIPTION	AREA	QUANTITY	LOAD	TOTAL WC	N.E.C. ARTICLE
1 BEDROOM UNITS					
1BED UNIT TYPE A1	713	5	3	10695	220.84(c)(1)
2 BEDROOM UNITS					
2BED UNIT TYPE B1	1035	5	3	15525	220.84(c)(1)
2BED UNIT TYPE B1.1	1035	3	3	9315	220.84(c)(1)
2BED UNIT TYPE B1A	1035	1	3	3105	220.84(c)(1)
2BED UNIT TYPE BD3.2	1275	4	3	15300	220.84(c)(1)
3 BEDROOM UNITS					
3BED UNIT TYPE C3	1445	4	3	17340	220.84(c)(1)
TOTAL CONNECTED LIGHTING/RECEPTACLE LOAD:				71,280 W	
SMALL APPLIANCE BRANCH CIRCUITS	-	44	1500	66000	220.84(c)(2)
LAUNDRY CIRCUIT (WASHING MACHINE)	-	22	1500	33000	220.84(c)(2)
REFRIGERATOR	-	22	600	13200	220.84(c)(3)
ELECTRIC RANGE/OVEN	-	22	8000	176000	220.84(c)(3)
RANGE HOOD/MICROWAVE	-	22	1680	36960	220.84(c)(3)
DISHWASHER/DISPOSER	-	22	1200	26400	220.84(c)(3)
ELECTRIC DRYER	-	22	5000	110000	220.84(c)(3)
ELECTRIC WATER HEATER	-	22	4500	99000	220.84(c)(3)
1.5 TON HVAC	-	5	5180	25900	220.84(c)(5)
2 TON HVAC	-	13	6698	87074	220.84(c)(5)
2.5 TON HVAC	-	4	8871	35484	220.84(c)(5)
TOTAL CONNECTED EQUIPMENT LOAD:				709,018 W	
TOTAL CONNECTED LOAD:				780,298 W	
DEMAND FACTOR:				36%	TABLE 220.84
TOTAL DEMAND LOAD:				**280,907 W**	
AMPS AT 120/208V-3Ø				**779.7 A**	

METER CENTER "MC-ASE4" ELECTRIC SERVICE CALCULATION

DESCRIPTION	AREA	QUANTITY	LOAD	TOTAL WC	N.E.C. ARTICLE
1 BEDROOM UNITS					
1BED UNIT TYPE A1	713	6	3	12834	220.84(c)(1)
1BED UNIT TYPE A1.1	709	1	3	2127	220.84(c)(1)
1BED UNIT TYPE A1.2	709	1	3	2127	220.84(c)(1)
1BED UNIT TYPE AD2	897	2	3	5382	220.84(c)(1)
1BED UNIT TYPE AD2 PW	897	2	3	5382	220.84(c)(1)
1BED UNIT TYPE S1	704	3	3	6336	220.84(c)(1)
2 BEDROOM UNITS					
2BED UNIT TYPE B1	1035	2	3	6210	220.84(c)(1)
2BED UNIT TYPE BD3.1	1275	1	3	3825	220.84(c)(1)
3 BEDROOM UNITS					
3BED UNIT TYPE C3	1492	1	3	4476	220.84(c)(1)
TOTAL CONNECTED LIGHTING/RECEPTACLE LOAD:				48,699 W	
SMALL APPLIANCE BRANCH CIRCUITS	-	38	1500	57000	220.84(c)(2)
LAUNDRY CIRCUIT (WASHING MACHINE)	-	19	1500	28500	220.84(c)(2)
REFRIGERATOR	-	19	600	11400	220.84(c)(3)
ELECTRIC RANGE/OVEN	-	19	8000	152000	220.84(c)(3)
RANGE HOOD/MICROWAVE	-	19	1680	31920	220.84(c)(3)
DISHWASHER/DISPOSER	-	19	1200	22800	220.84(c)(3)
ELECTRIC DRYER	-	19	5000	95000	220.84(c)(3)
ELECTRIC WATER HEATER	-	19	4500	85500	220.84(c)(3)
1.5 TON HVAC	-	15	5180	77700	220.84(c)(5)
2 TON HVAC	-	3	6698	20094	220.84(c)(5)
2.5 TON HVAC	-	1	8871	8871	220.84(c)(5)
TOTAL CONNECTED EQUIPMENT LOAD:				590,785 W	
TOTAL CONNECTED LOAD:				639,484 W	
DEMAND FACTOR:				38%	TABLE 220.84
TOTAL DEMAND LOAD:				**243,004 W**	
AMPS AT 120/208V-3Ø				**674.5 A**	

METER CENTER "MC-ANW" ELECTRIC SERVICE CALCULATION

DESCRIPTION	AREA	QUANTITY	LOAD	TOTAL WC	N.E.C. ARTICLE
1 BEDROOM UNITS					
1BED UNIT TYPE A1	713	12	3	25668	220.84(c)(1)
1BED UNIT TYPE AD3	1025	1	3	3075	220.84(c)(1)
2 BEDROOM UNITS					
2BED UNIT TYPE B1	1035	7	3	21735	220.84(c)(1)
2BED UNIT TYPE B1.1	1035	4	3	12420	220.84(c)(1)
2BED UNIT TYPE BD3.2	1275	3	3	11475	220.84(c)(1)
3 BEDROOM UNITS					
3BED UNIT TYPE C3	1442	4	3	17304	220.84(c)(1)
TOTAL CONNECTED LIGHTING/RECEPTACLE LOAD:				91,677 W	
SMALL APPLIANCE BRANCH CIRCUITS	-	62	1500	93000	220.84(c)(2)
LAUNDRY CIRCUIT (WASHING MACHINE)	-	31	1500	46500	220.84(c)(2)
REFRIGERATOR	-	31	600	18600	220.84(c)(3)
ELECTRIC RANGE/OVEN	-	31	8000	248000	220.84(c)(3)
RANGE HOOD/MICROWAVE	-	31	1680	52080	220.84(c)(3)
DISHWASHER/DISPOSER	-	31	1200	37200	220.84(c)(3)
ELECTRIC DRYER	-	31	5000	155000	220.84(c)(3)
ELECTRIC WATER HEATER	-	31	4500	139500	220.84(c)(3)
1.5 TON HVAC	-	13	5180	67340	220.84(c)(5)
2 TON HVAC	-	14	6698	93772	220.84(c)(5)
2.5 TON HVAC	-	4	8871	35484	220.84(c)(5)
TOTAL CONNECTED EQUIPMENT LOAD:				986,476 W	
TOTAL CONNECTED LOAD:				1,078,153 W	
DEMAND FACTOR:				32%	TABLE 220.84
TOTAL DEMAND LOAD:				**345,009 W**	
AMPS AT 120/208V-3Ø				**957.7 A**	

METER CENTER "MC-ASE3" ELECTRIC SERVICE CALCULATION

DESCRIPTION	AREA	QUANTITY	LOAD	TOTAL WC	N.E.C. ARTICLE
1 BEDROOM UNITS					
1BED UNIT TYPE A1	713	8	3	17112	220.84(c)(1)
1BED UNIT TYPE A1.1	709	2	3	4254	220.84(c)(1)
1BED UNIT TYPE A1.2	709	2	3	4254	220.84(c)(1)
1BED UNIT TYPE AD2	897	2	3	5382	220.84(c)(1)
1BED UNIT TYPE AD2 PW	897	2	3	5382	220.84(c)(1)
1BED UNIT TYPE S1	704	3	3	6336	220.84(c)(1)
2 BEDROOM UNITS					
2BED UNIT TYPE B1	1035	3	3	9315	220.84(c)(1)
2BED UNIT TYPE BD3.1	1275	1	3	3825	220.84(c)(1)
3 BEDROOM UNITS					
3BED UNIT TYPE C3	1442	1	3	4326	220.84(c)(1)
TOTAL CONNECTED LIGHTING/RECEPTACLE LOAD:				60,186 W	
SMALL APPLIANCE BRANCH CIRCUITS	-	48	1500	72000	220.84(c)(2)
LAUNDRY CIRCUIT (WASHING MACHINE)	-	24	1500	36000	220.84(c)(2)
REFRIGERATOR	-	24	600	14400	220.84(c)(3)
ELECTRIC RANGE/OVEN	-	24	8000	192000	220.84(c)(3)
RANGE HOOD/MICROWAVE	-	24	1680	40320	220.84(c)(3)
DISHWASHER/DISPOSER	-	24	1200	28800	220.84(c)(3)
ELECTRIC DRYER	-	24	5000	120000	220.84(c)(3)
ELECTRIC WATER HEATER	-	24	4500	108000	220.84(c)(3)
1.5 TON HVAC	-	19	5180	98420	220.84(c)(5)
2 TON HVAC	-	4	6698	26792	220.84(c)(5)
2.5 TON HVAC	-	1	8871	8871	220.84(c)(5)
TOTAL CONNECTED EQUIPMENT LOAD:				745,603 W	
TOTAL CONNECTED LOAD:				805,789 W	
DEMAND FACTOR:				35%	TABLE 220.84
TOTAL DEMAND LOAD:				**282,026 W**	
AMPS AT 120/208V-3Ø				**782.8 A**	

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

whs engineering
2012 West 25th Street, Suite 900
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Phone: 216 227 8505

OSU AIRPORT WEST

BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title

PANEL SCEHDULES AND CALCULATIONS

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number: E0.06A
Checked by: KDS	
Approved: GSS	

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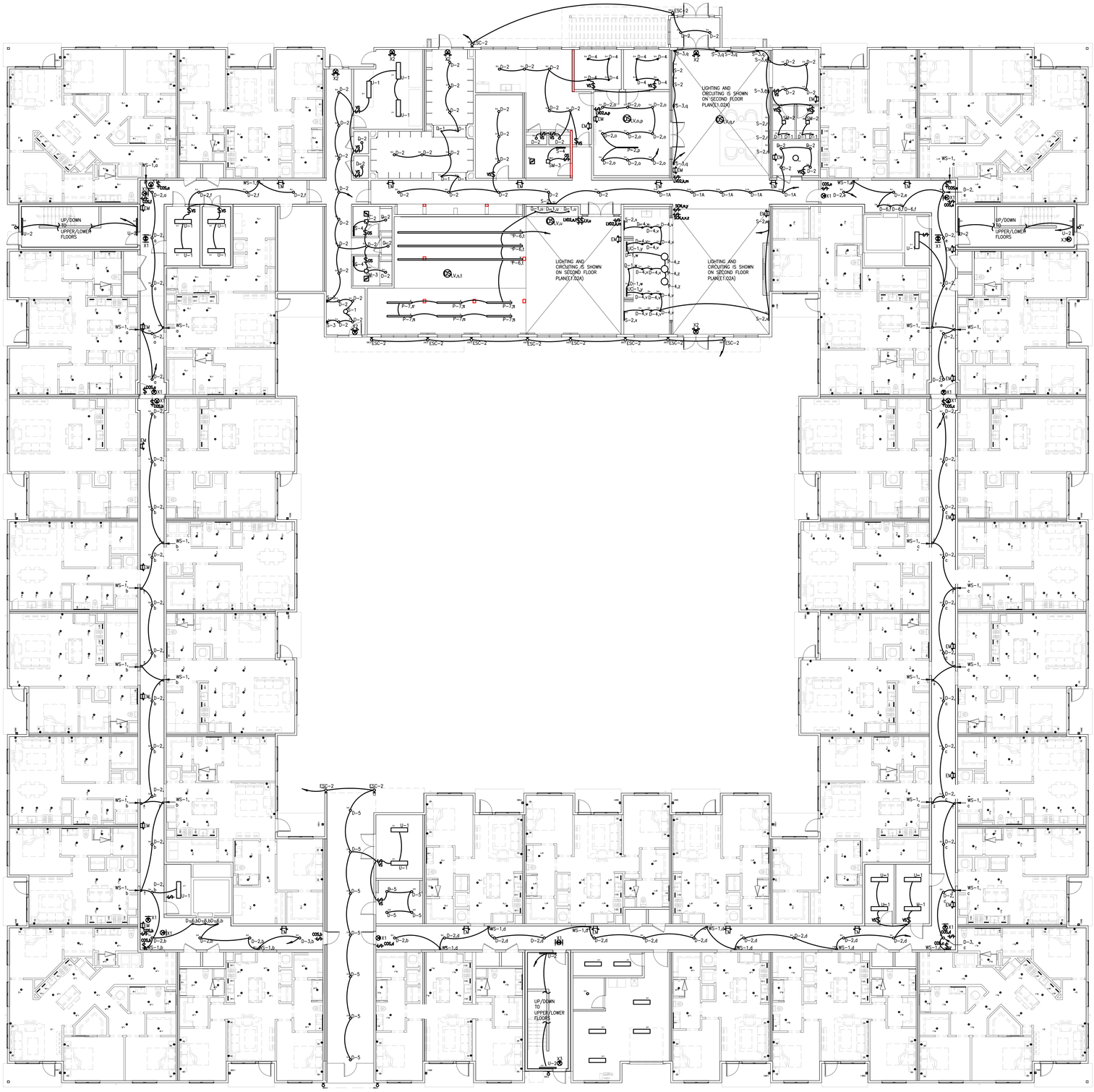

1 E1.01A BUILDING A
FIRST FLOOR LIGHTING PLAN
SCALE: 3/32" = 1'-0"

GENERAL LIGHTING PLAN NOTES

- ALL EXIT SIGNS, **EMERGENCY LIGHTING BATTERY PACKS, EMERGENCY LUMINAIRES AND NIGHT LIGHTS (DENOTED "NL")** SHALL BE CONNECTED TO **THE LOCAL LIGHTING CIRCUIT** AHEAD OF ANY CONTROLS SUCH AS: SWITCHES (DEVICE), OCCUPANCY SENSORS AND/OR RELAY CONTROLS.
- EXACT LOCATION OF ALL LUMINAIRES, AND EXACT MOUNTING HEIGHT OF ALL PENDANT MOUNTED LUMINAIRES SHALL BE COORDINATED WITH ARCHITECTURAL DRAWINGS PRIOR TO ANY ROUGH-INS.
- REFER TO WIRING SCHEDULE ON SHEET **#E0.03A/B/C** FOR 15A/20A LIGHTING BRANCH CIRCUITS FOR ALL CONDUCTOR SIZES TO ACCOUNT FOR VOLTAGE DROP.
- ALL WIRING SHALL BE IDENTIFIED BY PANELBOARD AND CIRCUIT NUMBER(S) IN ALL CABINETS, JUNCTION BOXES, WIRING TROUGHS, ENCLOSURES, SPLICE OR TERMINATION POINTS, ETC.
- A NEW TYPED PANELBOARD DIRECTORY CARD SHALL BE PROVIDED FOR ALL PANELS INSTALLED OR MODIFIED UNDER THIS CONTRACT. NEW DIRECTORY CARDS SHALL BE LOCATED ON THE INSIDE DOOR OF ASSOCIATED PANELS.

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Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

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OSU AIRPORT WEST
BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title
FIRST FLOOR LIGHTING PLAN

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number
Checked by: KDS	E1.01A
Approved: GSS	

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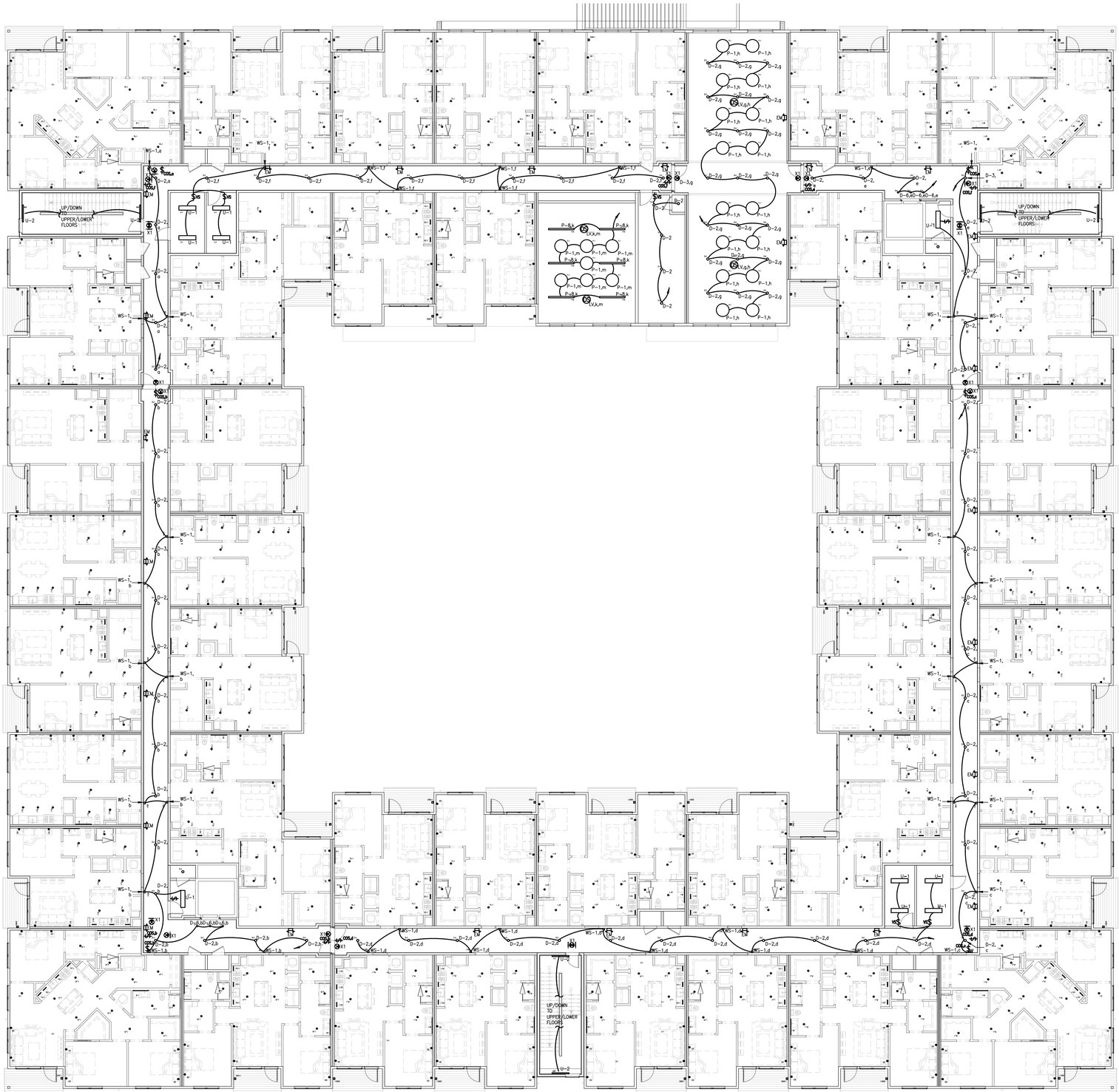

1 E1.02A BUILDING A
SECOND FLOOR LIGHTING PLAN
SCALE: 3/32" = 1'-0"

GENERAL LIGHTING PLAN NOTES

- ALL EXIT SIGNS, **EMERGENCY LIGHTING BATTERY PACKS, EMERGENCY LUMINAIRES AND NIGHT LIGHTS (DENOTED "NL")** SHALL BE CONNECTED TO **THE LOCAL LIGHTING CIRCUIT** AHEAD OF ANY CONTROLS SUCH AS: SWITCHES (DEVICE), OCCUPANCY SENSORS AND/OR RELAY CONTROLS.
- EXACT LOCATION OF ALL LUMINAIRES, AND EXACT MOUNTING HEIGHT OF ALL PENDANT MOUNTED LUMINAIRES SHALL BE COORDINATED WITH ARCHITECTURAL DRAWINGS PRIOR TO ANY ROUGH-INS.
- REFER TO WIRING SCHEDULE ON SHEET **#E0.03A/B/C** FOR 15A/20A LIGHTING BRANCH CIRCUITS FOR ALL CONDUCTOR SIZES TO ACCOUNT FOR VOLTAGE DROP.
- ALL WIRING SHALL BE IDENTIFIED BY PANELBOARD AND CIRCUIT NUMBER(S) IN ALL CABINETS, JUNCTION BOXES, WIRING TROUGHS, ENCLOSURES, SPLICE OR TERMINATION POINTS, ETC.
- A NEW TYPED PANELBOARD DIRECTORY CARD SHALL BE PROVIDED FOR ALL PANELS INSTALLED OR MODIFIED UNDER THIS CONTRACT. NEW DIRECTORY CARDS SHALL BE LOCATED ON THE INSIDE DOOR OF ASSOCIATED PANELS.

Seal

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Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

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OSU AIRPORT WEST
BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title
SECOND FLOOR LIGHTING PLAN

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number: E1.02A
Designed by: KDS	
Approved: GSS	

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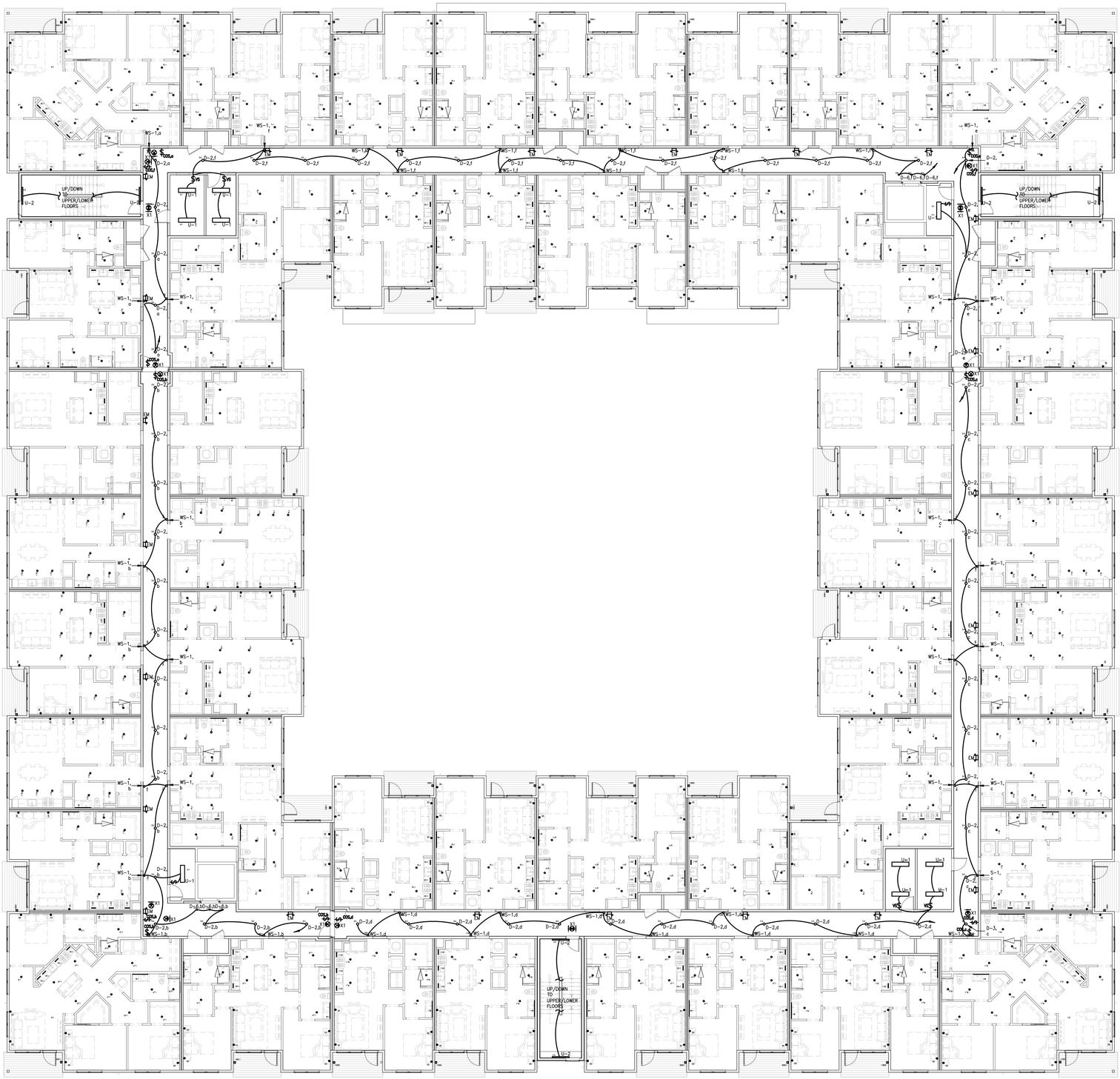

1 E1.03A BUILDING A
THIRD FLOOR LIGHTING PLAN
SCALE: 3/32" = 1'-0"

GENERAL LIGHTING PLAN NOTES

- ALL EXIT SIGNS, **EMERGENCY LIGHTING BATTERY PACKS, EMERGENCY LUMINAIRES AND NIGHT LIGHTS (DENOTED "NL")** SHALL BE CONNECTED TO **THE LOCAL LIGHTING CIRCUIT** AHEAD OF ANY CONTROLS SUCH AS: SWITCHES (DEVICE), OCCUPANCY SENSORS AND/OR RELAY CONTROLS.
- EXACT LOCATION OF ALL LUMINAIRES, AND EXACT MOUNTING HEIGHT OF ALL PENDANT MOUNTED LUMINAIRES SHALL BE COORDINATED WITH ARCHITECTURAL DRAWINGS PRIOR TO ANY ROUGH-INS.
- REFER TO WIRING SCHEDULE ON SHEET **#E0.03A/B/C** FOR 15A/20A LIGHTING BRANCH CIRCUITS FOR ALL CONDUCTOR SIZES TO ACCOUNT FOR VOLTAGE DROP.
- ALL WIRING SHALL BE IDENTIFIED BY PANELBOARD AND CIRCUIT NUMBER(S) IN ALL CABINETS, JUNCTION BOXES, WIRING TROUGHS, ENCLOSURES, SPLICE OR TERMINATION POINTS, ETC.
- A NEW TYPED PANELBOARD DIRECTORY CARD SHALL BE PROVIDED FOR ALL PANELS INSTALLED OR MODIFIED UNDER THIS CONTRACT. NEW DIRECTORY CARDS SHALL BE LOCATED ON THE INSIDE DOOR OF ASSOCIATED PANELS.

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

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OSU AIRPORT WEST

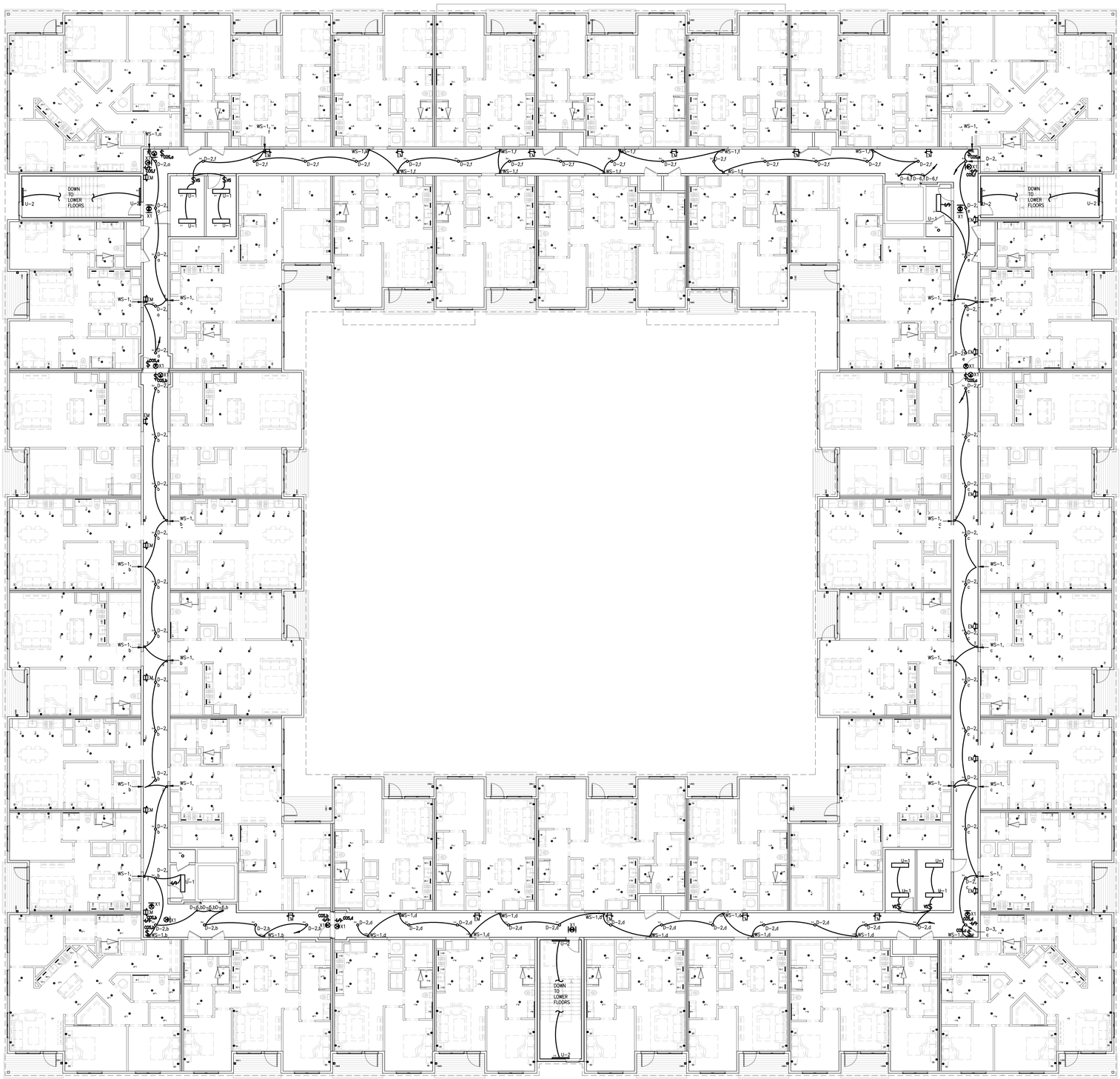
BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title

THIRD FLOOR LIGHTING PLAN

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number
Reviewed by: KDS	E1.03A
Approved: GSS	



1 / E1.04A — BUILDING A
FOURTH FLOOR LIGHTING PLAN
SCALE: 3/32" = 1'-0"

GENERAL LIGHTING PLAN NOTES

- ALL EXIT SIGNS, **EMERGENCY LIGHTING BATTERY PACKS, EMERGENCY LUMINAIRES AND NIGHT LIGHTS (DENOTED "NL")** SHALL BE CONNECTED TO **THE LOCAL LIGHTING CIRCUIT** AHEAD OF ANY CONTROLS SUCH AS: SWITCHES (DEVICE), OCCUPANCY SENSORS AND/OR RELAY CONTROLS.
- EXACT LOCATION OF ALL LUMINAIRES, AND EXACT MOUNTING HEIGHT OF ALL PENDANT MOUNTED LUMINAIRES SHALL BE COORDINATED WITH ARCHITECTURAL DRAWINGS PRIOR TO ANY ROUGH-INS.
- REFER TO WIRING SCHEDULE ON SHEET **#E0.03A/B/C** FOR 15A/20A LIGHTING BRANCH CIRCUITS FOR ALL CONDUCTOR SIZES TO ACCOUNT FOR VOLTAGE DROP.
- ALL WIRING SHALL BE IDENTIFIED BY PANELBOARD AND CIRCUIT NUMBER(S) IN ALL CABINETS, JUNCTION BOXES, WIRING TROUGHS, ENCLOSURES, SPLICE OR TERMINATION POINTS, ETC.
- A NEW TYPED PANELBOARD DIRECTORY CARD SHALL BE PROVIDED FOR ALL PANELS INSTALLED OR MODIFIED UNDER THIS CONTRACT. NEW DIRECTORY CARDS SHALL BE LOCATED ON THE INSIDE DOOR OF ASSOCIATED PANELS.

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

whs engineering
2012 West 25th Street, Suite 900
Cleveland, OH 44113
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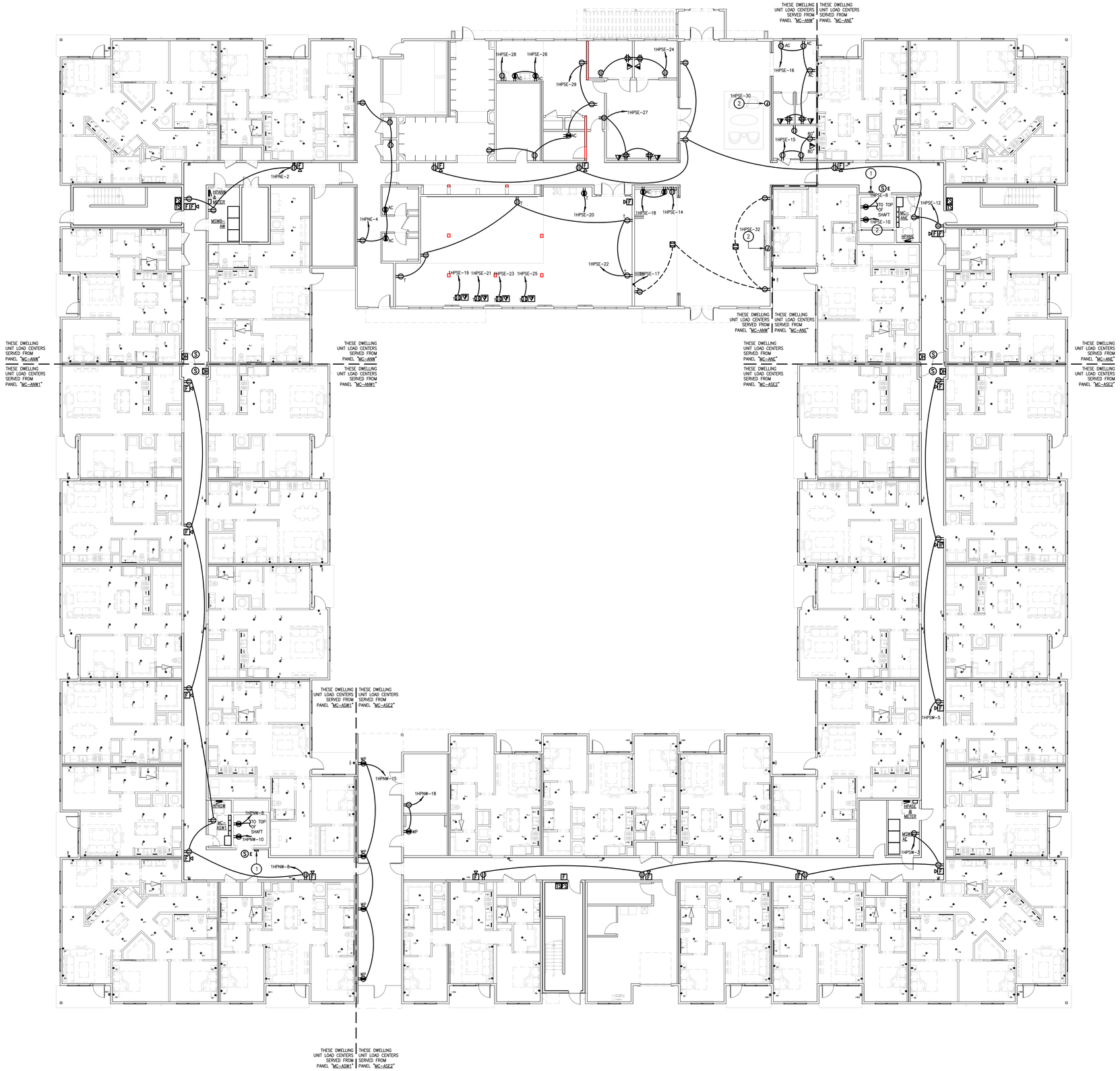
OSU AIRPORT WEST
BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title
FOURTH FLOOR LIGHTING PLAN

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number
Checked by: KDS	E1.04A
Approved: GSS	



1 / E2.01A BUILDING A
FIRST FLOOR POWER PLAN
SCALE: 3/32" = 1'-0"

GENERAL POWER AND COMMUNICATION PLAN NOTES

- EXACT LOCATION OF MECHANICAL, PLUMBING, KITCHEN, FURNITURE SYSTEMS, OWNER FURNISHED EQUIPMENT, ETC. THAT REQUIRE ELECTRICAL CONNECTIONS ARE SHOWN ON THE MECHANICAL, PLUMBING, AND/OR ARCHITECTURAL DRAWINGS. COORDINATE EXACT LOCATIONS WITH RESPECTIVE CONTRACTORS AND/OR VENDORS PRIOR TO ANY ROUGH-INS.
- PRIOR TO ROUGH-IN, REVIEW AND COORDINATE WITH ALL OTHER TRADES' CONTRACT DOCUMENTS TO DETERMINE SPECIFIC MOUNTING LOCATIONS AND ELECTRICAL REQUIREMENTS (VOLTAGE, PHASE, KW, HP, CONNECTION TYPE, ETC.) FOR HVAC AND PLUMBING EQUIPMENT. COORDINATE EXACT MOUNTING LOCATIONS WITH THE SPECIFIC TRADE AND ARCHITECT.
- REFER TO WIRING SCHEDULE ON SHEET **#E0.03A/B/C** FOR 15A/20A RECEPTACLE BRANCH CIRCUITS FOR ALL CONDUCTOR SIZES TO ACCOUNT FOR VOLTAGE DROP.
- ALL WIRING SHALL BE IDENTIFIED BY PANELBOARD AND CIRCUIT NUMBER(S) IN ALL CABINETS, JUNCTION BOXES, WIRING TROUGHS, ENCLOSURES, SPLICE OR TERMINATION POINTS, ETC.
- A NEW TYPED PANELBOARD DIRECTORY CARD SHALL BE PROVIDED FOR ALL PANELS INSTALLED OR MODIFIED UNDER THIS CONTRACT. NEW DIRECTORY CARDS SHALL BE LOCATED ON THE INSIDE DOOR OF ASSOCIATED PANELS.

CODED NOTES

1. LOCATION OF AREA OF RESCUE COMMUNICATION BASE STATION. REFER TO DIAGRAM ON DRAWING #E0-02.
2. REFER TO ELEVATOR GENERAL INSTALLATION NOTES ON DRAWING #E0-03 FOR ALL ELEVATOR ROOM AND ELEVATOR PIT ELECTRICAL REQUIREMENTS.
3. PROVIDE A 4' x 4' x 3/4" THICK PLYWOOD BACKBOARD PAINTED WITH FIRE RETARDANT PAINT FOR MAIN COMMUNICATION TERMINAL BOARD (TELEPHONE, TELEVISION, ETC.). INCLUDE ALL WORK REQUIRED BY RESPECTIVE UTILITIES. PROVIDE A COPPER GROUNDING BUS AND #6 AWG GROUNDING CONDUCTOR TO SERVICE GROUNDING POINT(S) (REFER TO DETAIL ON DRAWING **#E0-02**). INCLUDE (2) DOUBLE DUPLEX RECEPTACLES INSTALLED AT THIS TERMINAL BOARD SERVED FROM (2) 20A-120V-1ø DEDICATED CIRCUITS. REFER TO COMMUNICATION RISER DIAGRAM ON DRAWING **#E0-02** FOR ADDITIONAL INFORMATION.
4. ROUTE BRANCH CIRCUITING FOR RECEPTACLES AT THIS LOCATION VERTICALLY AT EACH FLOOR. PROVIDE HOMERUN AS SHOWN ON FOURTH FLOOR.

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

whs engineering
2012 West 25th Street, Suite 900
Cleveland, OH 44113
Phone: 216 227 8505

OSU AIRPORT WEST
BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title
FIRST FLOOR POWER PLAN

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number
Checked by: KDS	E2.01A
Approved: GSS	

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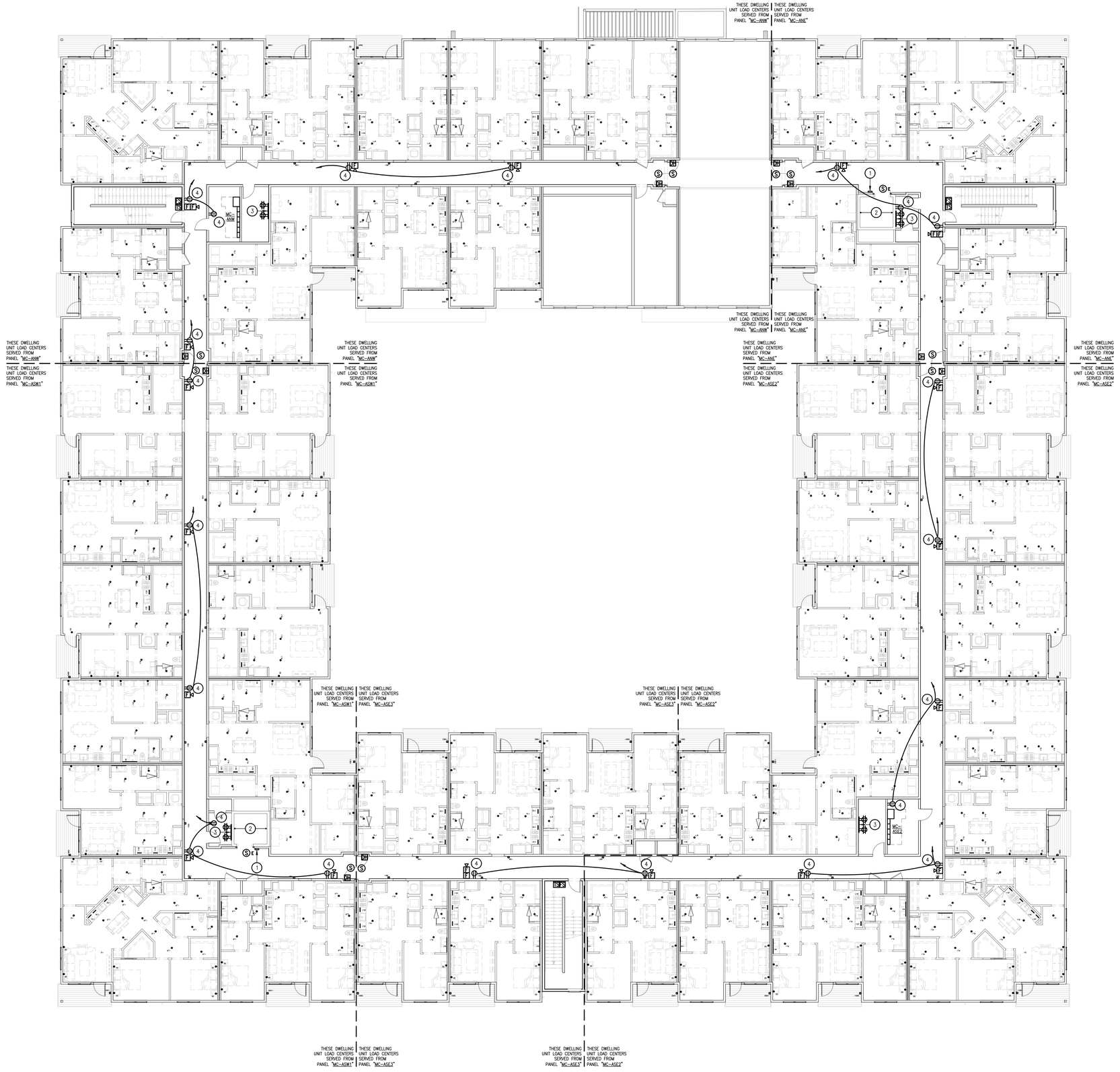

GENERAL POWER AND COMMUNICATION PLAN NOTES

- EXACT LOCATION OF MECHANICAL, PLUMBING, KITCHEN, FURNITURE SYSTEMS, OWNER FURNISHED EQUIPMENT, ETC. THAT REQUIRE ELECTRICAL CONNECTIONS ARE SHOWN ON THE MECHANICAL, PLUMBING, AND/OR ARCHITECTURAL DRAWINGS. COORDINATE EXACT LOCATIONS WITH RESPECTIVE CONTRACTORS AND/OR VENDORS PRIOR TO ANY ROUGH-INS.
- PRIOR TO ROUGH-IN, REVIEW AND COORDINATE WITH ALL OTHER TRADES' CONTRACT DOCUMENTS TO DETERMINE SPECIFIC MOUNTING LOCATIONS AND ELECTRICAL REQUIREMENTS (VOLTAGE, PHASE, KW, HP, CONNECTION TYPE, ETC.) FOR HVAC AND PLUMBING EQUIPMENT. COORDINATE EXACT MOUNTING LOCATIONS WITH THE SPECIFIC TRADE AND ARCHITECT.
- REFER TO WIRING SCHEDULE ON SHEET #**E0.03A/B/C** FOR 15A/20A RECEPTACLE BRANCH CIRCUITS FOR ALL CONDUCTOR SIZES TO ACCOUNT FOR VOLTAGE DROP.
- ALL WIRING SHALL BE IDENTIFIED BY PANELBOARD AND CIRCUIT NUMBER(S) IN ALL CABINETS, JUNCTION BOXES, WIRING TROUGHS, ENCLOSURES, SPLICE OR TERMINATION POINTS, ETC.
- A NEW TYPED PANELBOARD DIRECTORY CARD SHALL BE PROVIDED FOR ALL PANELS INSTALLED OR MODIFIED UNDER THIS CONTRACT. NEW DIRECTORY CARDS SHALL BE LOCATED ON THE INSIDE DOOR OF ASSOCIATED PANELS.

CODED NOTES

1. LOCATION OF AREA OF RESCUE COMMUNICATION BASE STATION. REFER TO DIAGRAM ON DRAWING #E0-02.
2. REFER TO ELEVATOR GENERAL INSTALLATION NOTES ON DRAWING #E0-03 FOR ALL ELEVATOR ROOM AND ELEVATOR PIT ELECTRICAL REQUIREMENTS.
3. PROVIDE A 4' x 4' x 3/4" THICK PLYWOOD BACKBOARD PAINTED WITH FIRE RETARDANT PAINT FOR MAIN COMMUNICATION TERMINAL BOARD (TELEPHONE, TELEVISION, ETC.). INCLUDE ALL WORK REQUIRED BY RESPECTIVE UTILITIES. PROVIDE A COPPER GROUNDING BUS AND #6 AWG GROUNDING CONDUCTOR TO SERVICE GROUNDING POINT(S) (REFER TO DETAIL ON DRAWING **#E0-02**). INCLUDE (2) DOUBLE DUPLEX RECEPTACLES INSTALLED AT THIS TERMINAL BOARD SERVED FROM (2) 20A-120V-1ø DEDICATED CIRCUITS. REFER TO COMMUNICATION RISER DIAGRAM ON DRAWING **#E0-02** FOR ADDITIONAL INFORMATION.
4. ROUTE BRANCH CIRCUITING FOR RECEPTACLES AT THIS LOCATION VERTICALLY AT EACH FLOOR. PROVIDE HOMERUN AS SHOWN ON FOURTH FLOOR.

1 / E2.02A — BUILDING A
SECOND FLOOR POWER PLAN
SCALE: 3/32" = 1'-0"

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

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OSU AIRPORT WEST
BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title
SECOND FLOOR POWER PLAN

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number: E2.02A
Designed by: KDS	
Approved: GSS	

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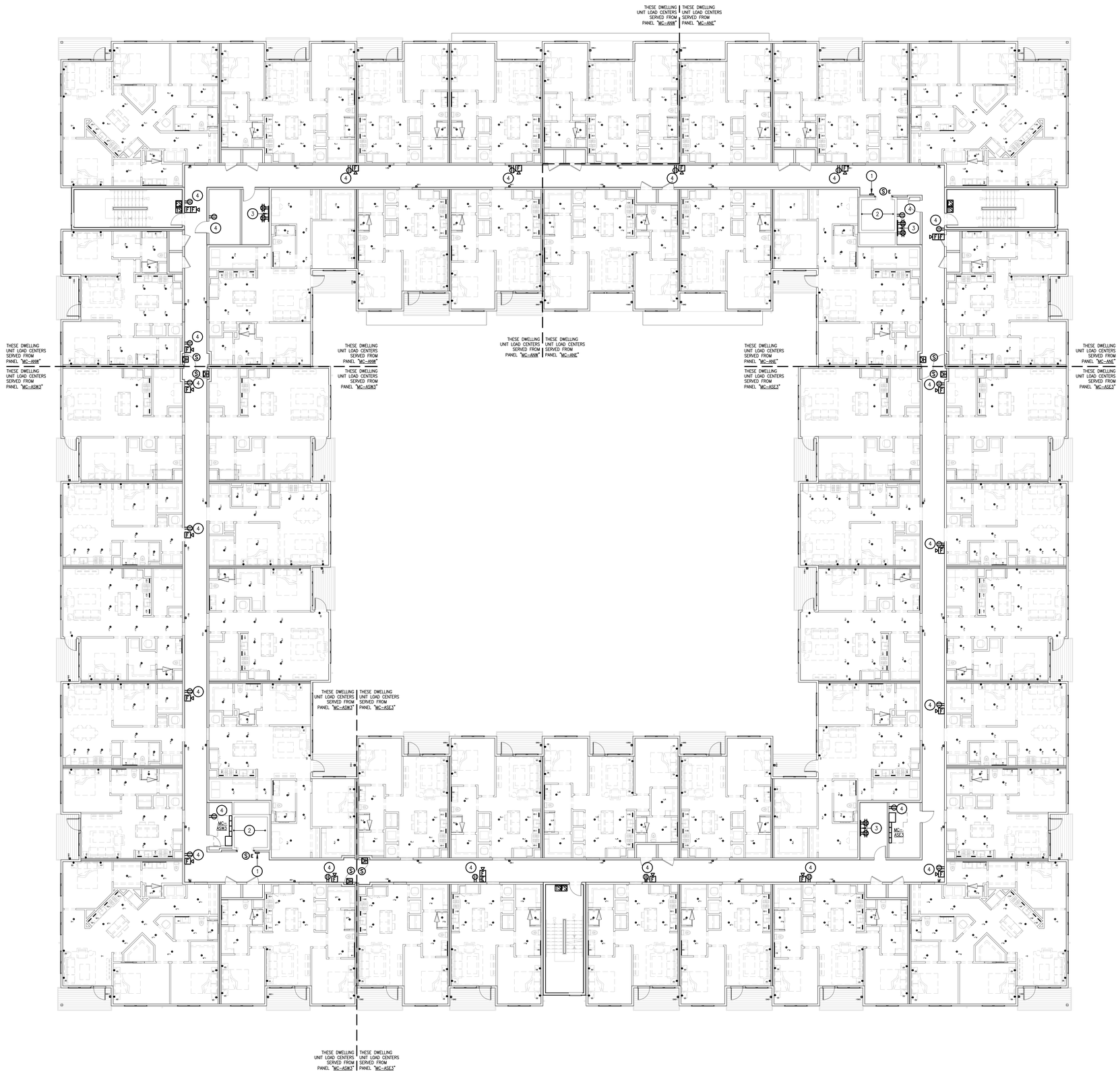

1 E2.03A BUILDING A
THIRD FLOOR POWER PLAN
SCALE: 3/32" = 1'-0"

GENERAL POWER AND COMMUNICATION PLAN NOTES

- EXACT LOCATION OF MECHANICAL, PLUMBING, KITCHEN, FURNITURE SYSTEMS, OWNER FURNISHED EQUIPMENT, ETC. THAT REQUIRE ELECTRICAL CONNECTIONS ARE SHOWN ON THE MECHANICAL, PLUMBING, AND/OR ARCHITECTURAL DRAWINGS. COORDINATE EXACT LOCATIONS WITH RESPECTIVE CONTRACTORS AND/OR VENDORS PRIOR TO ANY ROUGH-INS.
- PRIOR TO ROUGH-IN, REVIEW AND COORDINATE WITH ALL OTHER TRADES' CONTRACT DOCUMENTS TO DETERMINE SPECIFIC MOUNTING LOCATIONS AND ELECTRICAL REQUIREMENTS (VOLTAGE, PHASE, KW, HP, CONNECTION TYPE, ETC.) FOR HVAC AND PLUMBING EQUIPMENT. COORDINATE EXACT MOUNTING LOCATIONS WITH THE SPECIFIC TRADE AND ARCHITECT.
- REFER TO WIRING SCHEDULE ON SHEET **#E0.03A/B/C** FOR 15A/20A RECEPTACLE BRANCH CIRCUITS FOR ALL CONDUCTOR SIZES TO ACCOUNT FOR VOLTAGE DROP.
- ALL WIRING SHALL BE IDENTIFIED BY PANELBOARD AND CIRCUIT NUMBER(S) IN ALL CABINETS, JUNCTION BOXES, WIRING TROUGHS, ENCLOSURES, SPLICE OR TERMINATION POINTS, ETC.
- A NEW TYPED PANELBOARD DIRECTORY CARD SHALL BE PROVIDED FOR ALL PANELS INSTALLED OR MODIFIED UNDER THIS CONTRACT. NEW DIRECTORY CARDS SHALL BE LOCATED ON THE INSIDE DOOR OF ASSOCIATED PANELS.

CODED NOTES

1. LOCATION OF AREA OF RESCUE COMMUNICATION BASE STATION. REFER TO DIAGRAM ON DRAWING #E0-02.
2. REFER TO ELEVATOR GENERAL INSTALLATION NOTES ON DRAWING #E0-03 FOR ALL ELEVATOR ROOM AND ELEVATOR PIT ELECTRICAL REQUIREMENTS.
3. PROVIDE A 4' x 4' x 3/4" THICK PLYWOOD BACKBOARD PAINTED WITH FIRE RETARDANT PAINT FOR MAIN COMMUNICATION TERMINAL BOARD (TELEPHONE, TELEVISION, ETC.). INCLUDE ALL WORK REQUIRED BY RESPECTIVE UTILITIES. PROVIDE A COPPER GROUNDING BUS AND #6 AWG GROUNDING CONDUCTOR TO SERVICE GROUNDING POINT(S) (REFER TO DETAIL ON DRAWING **#E0-02**). INCLUDE (2) DOUBLE DUPLEX RECEPTACLES INSTALLED AT THIS TERMINAL BOARD SERVED FROM (2) 20A-120V-1ø DEDICATED CIRCUITS. REFER TO COMMUNICATION RISER DIAGRAM ON DRAWING **#E0-02** FOR ADDITIONAL INFORMATION.
4. ROUTE BRANCH CIRCUITING FOR RECEPTACLES AT THIS LOCATION VERTICALLY AT EACH FLOOR. PROVIDE HOMERUN AS SHOWN ON FOURTH FLOOR.

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OSU AIRPORT WEST
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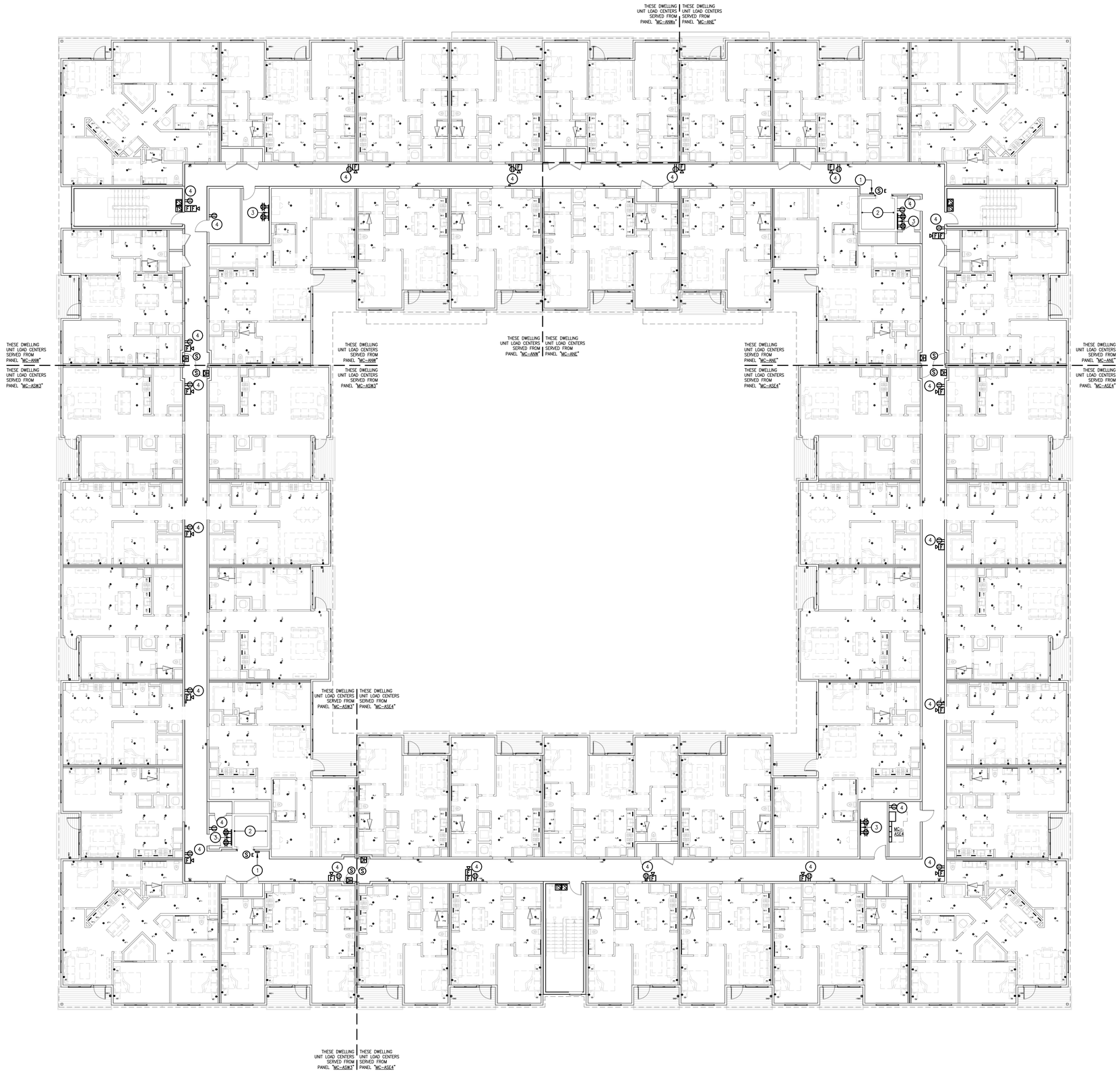
Drawing Title
THIRD FLOOR POWER PLAN

Date: 03.05.2026	Project Number: 24-065
Drawn by: DJS	Drawing Number: E2.03A
Reviewed by: KDS	
Approved: GSS	



1 E2.04A BUILDING A
FOURTH FLOOR POWER PLAN
SCALE: 3/32" = 1'-0"

GENERAL POWER AND COMMUNICATION PLAN NOTES

- EXACT LOCATION OF MECHANICAL, PLUMBING, KITCHEN, FURNITURE SYSTEMS, OWNER FURNISHED EQUIPMENT, ETC. THAT REQUIRE ELECTRICAL CONNECTIONS ARE SHOWN ON THE MECHANICAL, PLUMBING, AND/OR ARCHITECTURAL DRAWINGS. COORDINATE EXACT LOCATIONS WITH RESPECTIVE CONTRACTORS AND/OR VENDORS PRIOR TO ANY ROUGH-INS.
- PRIOR TO ROUGH-IN, REVIEW AND COORDINATE WITH ALL OTHER TRADES' CONTRACT DOCUMENTS TO DETERMINE SPECIFIC MOUNTING LOCATIONS AND ELECTRICAL REQUIREMENTS (VOLTAGE, PHASE, KW, HP, CONNECTION TYPE, ETC.) FOR HVAC AND PLUMBING EQUIPMENT. COORDINATE EXACT MOUNTING LOCATIONS WITH THE SPECIFIC TRADE AND ARCHITECT.
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- ALL WIRING SHALL BE IDENTIFIED BY PANELBOARD AND CIRCUIT NUMBER(S) IN ALL CABINETS, JUNCTION BOXES, WIRING TROUGHS, ENCLOSURES, SPLICE OR TERMINATION POINTS, ETC.
- A NEW TYPED PANELBOARD DIRECTORY CARD SHALL BE PROVIDED FOR ALL PANELS INSTALLED OR MODIFIED UNDER THIS CONTRACT. NEW DIRECTORY CARDS SHALL BE LOCATED ON THE INSIDE DOOR OF ASSOCIATED PANELS.

CODED NOTES

1. LOCATION OF AREA OF RESCUE COMMUNICATION BASE STATION. REFER TO DIAGRAM ON DRAWING #E0-02.
2. REFER TO ELEVATOR GENERAL INSTALLATION NOTES ON DRAWING #E0-03 FOR ALL ELEVATOR ROOM AND ELEVATOR PIT ELECTRICAL REQUIREMENTS.
3. PROVIDE A 4' x 4' x 3/4" THICK PLYWOOD BACKBOARD PAINTED WITH FIRE RETARDANT PAINT FOR MAIN COMMUNICATION TERMINAL BOARD (TELEPHONE, TELEVISION, ETC.). INCLUDE ALL WORK REQUIRED BY RESPECTIVE UTILITIES. PROVIDE A COPPER GROUNDING BUS AND #6 AWG GROUNDING CONDUCTOR TO SERVICE GROUNDING POINT(S) (REFER TO DETAIL ON DRAWING **#E0-02**). INCLUDE (2) DOUBLE DUPLEX RECEPTACLES INSTALLED AT THIS TERMINAL BOARD SERVED FROM (2) 20A-120V-1ϕ DEDICATED CIRCUITS. REFER TO COMMUNICATION RISER DIAGRAM ON DRAWING **#E0-02** FOR ADDITIONAL INFORMATION.
4. ROUTE BRANCH CIRCUITING FOR RECEPTACLES AT THIS LOCATION VERTICALLY AT EACH FLOOR. PROVIDE HOMERUN AS SHOWN ON FOURTH FLOOR.

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BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title
FOURTH FLOOR POWER PLAN

Date: 03.05.2026	Project Number: 24-065
Drawn by: DJS	Drawing Number: E2.04A
Checked By: KDS	
Approved: GSS	

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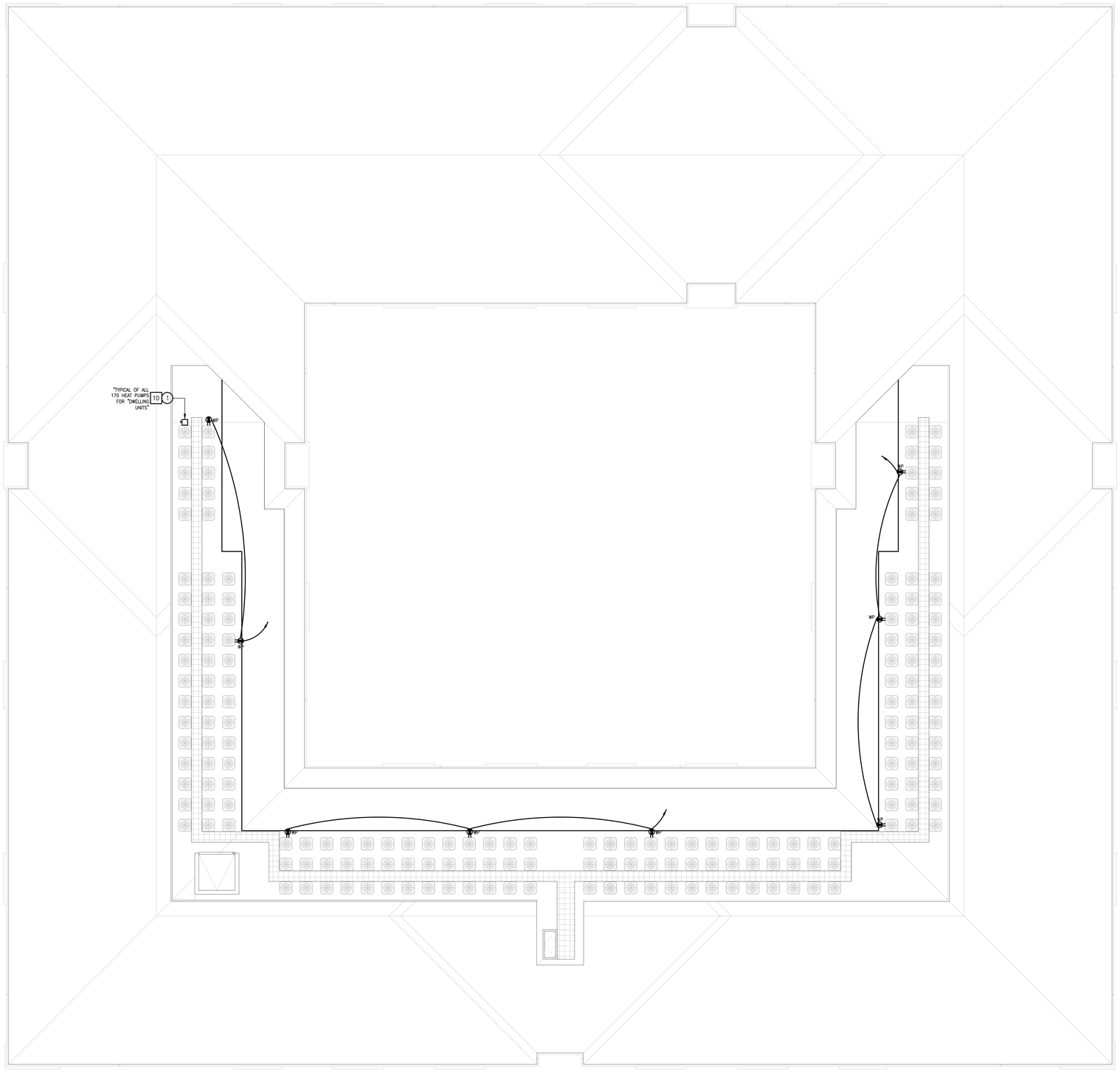

1 E2.05A BUILDING A
ROOF LEVEL POWER PLAN
SCALE: 3/32" = 1'-0"

GENERAL POWER AND COMMUNICATION PLAN NOTES

- EXACT LOCATION OF MECHANICAL, PLUMBING, KITCHEN, FURNITURE SYSTEMS, OWNER FURNISHED EQUIPMENT, ETC. THAT REQUIRE ELECTRICAL CONNECTIONS ARE SHOWN ON THE MECHANICAL, PLUMBING, AND/OR ARCHITECTURAL DRAWINGS. COORDINATE EXACT LOCATIONS WITH RESPECTIVE CONTRACTORS AND/OR VENDORS PRIOR TO ANY ROUGH-INS.
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- REFER TO WIRING SCHEDULE ON SHEET #**E0.03A/B/C** FOR 15A/20A RECEPTACLE BRANCH CIRCUITS FOR ALL CONDUCTOR SIZES TO ACCOUNT FOR VOLTAGE DROP.
- ALL WIRING SHALL BE IDENTIFIED BY PANELBOARD AND CIRCUIT NUMBER(S) IN ALL CABINETS, JUNCTION BOXES, WIRING TROUGHS, ENCLOSURES, SPLICE OR TERMINATION POINTS, ETC.
- A NEW TYPED PANELBOARD DIRECTORY CARD SHALL BE PROVIDED FOR ALL PANELS INSTALLED OR MODIFIED UNDER THIS CONTRACT. NEW DIRECTORY CARDS SHALL BE LOCATED ON THE INSIDE DOOR OF ASSOCIATED PANELS.

CODED NOTES

1. DWELLING UNIT HEAT PUMPS TO BE POWERED FROM RESPECTIVE TENANT PANEL. REFER TO TYPICAL DWELLING UNIT APPLIANCE CONNECTION SCHEDULE ON SHEET #E3-00 FOR LOAD INFORMATION, CIRCUIT BREAKER, DISCONNECT, AND CONDUCTOR SIZING.

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

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OSU AIRPORT WEST

BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title

ROOF LEVEL POWER PLAN

Date: 03.05.2026	Project Number: 24-065
Drawn by: DJS	Drawing Number: E2.05A
Reviewed by: KDS	
Approved: GSS	

GENERAL "DWELLING UNIT" ELECTRICAL CONSTRUCTION NOTES

A. EQUIPMENT AND MATERIALS USED ON THIS PROJECT SHALL BE NEW AND U.L. LABELED FOR THE APPLICATION.

B. WORK SHALL BE INSTALLED IN ACCORDANCE WITH ALL APPLICABLE PROVISIONS OF LOCAL AND STATE CODES AS WELL AS THE CURRENT EDITION OF THE NATIONAL ELECTRICAL CODE (NEC), AS INTERPRETED BY THE LOCAL AUTHORITY HAVING JURISDICTION.

C. MATERIAL OR LABOR WHICH IS NOT INDICATED ON THE DRAWINGS OR SPECIFICATION BUT WHICH IS OBVIOUSLY NECESSARY TO COMPLETE THE WORK SHALL BE PROVIDED. DRAWINGS AND SPECIFICATIONS ARE TO BE CONSIDERED AS SUPPLEMENTING EACH OTHER.

D. VISIT THE SITE OF THE WORK AND BECOME FAMILIAR WITH CONDITIONS AFFECTING THE INSTALLATION. SUBMISSION OF A PROPOSAL SHALL PRESUPPOSE KNOWLEDGE OF SUCH CONDITIONS AND NO ADDITIONAL COMPENSATION SHALL BE ALLOWED WHERE EXTRA LABOR OR MATERIALS ARE REQUIRED BECAUSE OF IGNORANCE OF THESE CONDITIONS.

E. WHERE LIGHT SWITCHES ARE SHOWN ADJACENT TO ONE ANOTHER, THEY SHALL BE GROUPED TOGETHER INTO A 2, 3, OR 4 GANG DEVICE BOX WITH COMMON COVERPLATE.

F. BACK BOXES IN DEMISING WALLS SHALL NOT BE ROUGHED–IN BACK TO BACK IN THE SAME STUD SPACE. PROVIDE FIRE–STOPPING AS REQUIRED TO MAINTAIN FIRE RATING.

G. ELECTRICAL BOXES, ELECTRICAL EQUIPMENT, ETC. MUST BE FIRESTOPPED UNDER THE FOLLOWING CONDITIONS: BOXES LARGER THAN 16 SQUARE INCHES, IF HORIZONTAL SPACING BETWEEN BOXES IS LESS THAN 24 INCHES. WHEN MULTIPLE BOXES ARE LOCATED IN ONE STUD CAVITY OR IF THE AGGREGATE OF ALL BOXES EXCEEDS 100 SQUARE INCHES PER 100 SQUARE FEET, REFER TO LISTED SYSTEM DETAILS AND APPLICABLE LOCAL BUILDING CODE REQUIREMENTS.

H. WHERE ELECTRICAL EQUIPMENT, OUTLET BOXES, ETC. ARE INSTALLED IN A FIRE RATED ASSEMBLY AND AS REQUIRED UNDER THE PARAMETERS LISTED ABOVE, FIRE (PUTTY) PADS SHALL BE PROVIDED TO MAINTAIN THE INTEGRITY OF THE FIRE RATED ASSEMBLY.

I. WHEN ELECTRICAL EQUIPMENT, OUTLET BOXES, ETC. ARE INSTALLED IN A STUD PARTITION WHERE ADJACENT ROOMS ARE SIMILAR, THE OUTLETS SHALL BE STAGGERED AND FIRE RATED (PUTTY) PADS SHALL ALSO BE INSTALLED.

J. FIRE RATED (PUTTY) PADS SHALL BE 3M "FIRE BARRIER MOLDABLE PUTTY PADS MPP+", FIRE RATED PRODUCTS SPECIALTIES CORPORATION TYPE "FRPS" ENCLOSURES, OR APPROVED EQUAL PRIOR TO PLACING BASE BID PRICE.

K. THE ELECTRICAL CONTRACTOR IS RESPONSIBLE FOR MAINTAINING THE FIRE RATING OF ALL CEILINGS, STRUCTURES, ETC. IN COMPLIANCE WITH ALL STANDARDS. WHERE BACK BOXES ARE REQUIRED TO BE INSTALLED TO SERVE LIGHT FIXTURES IN FIRE RATED CEILINGS, FIRE (PUTTY) PADS SHALL BE PROVIDED.

L. REFER TO ARCHITECTURAL ELEVATIONS FOR MOUNTING LOCATION AND HEIGHT OF ALL WIRING DEVICES.

M. COORDINATE THE EXACT LOCATION OF ALL RECEPTACLES AT CABINET LOCATIONS WITH G.C. CUT AND PATCH AS DIRECTED BY THE G.C.

N. ALL DEVICES SHALL BE RESIDENTIAL GRADE, 15/20 AMP DEVICES. RECEPTACLES SHALL BE TAMPER RESISTANT TYPE IN COMPLIANCE WITH N.E.C. #406.12.

O. SMOKE DETECTORS MUST BE INSTALLED FREE OF DUST OR ANY OTHER CONTAMINATION. LOCATE A MINIMUM OF 3'–0" FROM ALL SUPPLY AIR DIFFUSERS. LOCATE A MINIMUM OF 1'–0" FROM ANY PART OF ANY LIGHTING FIXTURES. DO NOT LOCATE ABOVE PENDANT MOUNTED OR INDIRECT LIGHTING FIXTURES. DO NOT LOCATE ABOVE CEILING FANS.

P. IN ALL SINGLE, OR, MULTI–FAMILY DWELLING UNITS, RECEPTACLES SHALL BE LOCATED AND INSTALLED IN ACCORDANCE WITH THE CURRENT EDITION OF THE N.E.C. ELECTRICAL EQUIPMENT, DEVICES, ETC. SHALL BE INSTALLED AND SPACED AS DIRECTED BY THE N.E.C. WITH SPECIFIC ATTENTION TO ARTICLE #210.52(A) THROUGH #210.52(I).

Q. ALL PENETRATIONS IN FIRE RATED FLOORS, SHAFTS AND WALLS SHALL BE FIRE STOPPED. THE INSTALLATION OF FIRE STOPPING MATERIALS SHALL CONFORM TO THE LATEST FIRE RESISTANCE DIRECTORY AS PUBLISHED BY U.L.

U. ALL RECEPTACLES SERVING KITCHEN COUNTER TOP SURFACES SHALL BE G.F.C.I. PROTECTED IN ACCORDANCE WITH N.E.C. ARTICLE #210.8(A)(6).

V. ALL RECEPTACLES SHALL LOCATED WITHIN 6'–0" OF THE OUTSIDE OF ALL SINKS, SHOWER STALLS, AND, BATH TUBS SHALL BE G.F.C.I. PROTECTED IN ACCORDANCE WITH N.E.C. ARTICLE #210.8(A)(7).

W. WHERE G.F.C.I. DEVICES ARE INSTALLED TO SERVE LARGE APPLIANCES, OR, APPLIANCES BELOW COUNTERTOP SURFACES, THE G.F.C.I DEVICE SHALL BE READILY ACCESSIBLE AS REQUIRED BY N.E.C. ARTICLE #422.5 AND AS DEFINED BY N.E.C. ARTICLE #100. PROVIDE G.F.C.I. TYPE CIRCUIT BREAKERS WHERE APPLICABLE. PROVIDE PROPER IDENTIFICATION AT RELATED DEVICE LOCATIONS INDICATING "G.F.C.I. PROTECTED".

X. PROVIDE ARC–FAULT PROTECTION FOR APPLICABLE BRANCH CIRCUITS IN ACCORDANCE WITH N.E.C. ARTICLE #210.12(A).

TYPICAL UNIT APPLIANCE/EQUIPMENT CONNECTION SCHEDULE

·	APPLIANCE/EQUIPMENT DESCRIPTION	VOLTAGE/ PHASE	KW.	CONNECTION TYPE			WIRE/CONDUIT SIZE	NOTES
				NEMA TYPE	DISCONNECT/SWITCH	HEIGHT		
1	REFRIGERATOR	120V–1ø	0.600	5–20R	–	54"	2#12, 1#12(G), 3/4"C.	1
2	ELECTRIC RANGE/OVEN	120/**208**V–1ø	8.000	14–50R	–	VERIFY	3#6, 1#10(G), 1"C.	1
3A	RANGE HOOD (ADA UNITS)	120V–1ø	0.480	–	DIRECT	VERIFY	2#12, 1#12(G), 3/4"C.	1,4
3B	RANGE HOOD/MICROWAVE	120V–1ø	1.680	5–20R	–	VERIFY	2#12, 1#12(G), 3/4"C.	1
4	DISHWASHER	120V–1ø	1.200	5–20R	–	SEE NOTE 2	2#12, 1#12(G), 3/4"C.	1,2
5	DISPOSER	120V–1ø						
6	WASHING MACHINE/LAUNDRY	120V–1ø	1.500	5–20R	–	42"	2#12, 1#12(G), 3/4"C.	1
7	ELECTRIC DRYER	120/**208**V–1ø	5.000	14–30R	–	42"	3#10, 1#10(G), 3/4"C.	1
8	ELECTRIC WATER HEATER	**208**V–1ø	4.500	–	30A–2P, NEMA 1	AT UNIT	2#10, 1#10(G), 3/4"C.	1,6
9A	1.5 TON AIR HANDLER	**208**V–1ø	4.784	–	30A–2P, NEMA 1	AT UNIT	2#10, 1#10(G), 3/4"C.	1,6
9B	2 TON AIR HANDLER	**208**V–1ø	5.824	–	30A–2P, NEMA 1	AT UNIT	2#10, 1#10(G), 3/4"C.	1,6
9C	2.5 TON AIR HANDLER	**208**V–1ø	8.528	–	60A–2P, NEMA 1	AT UNIT	2#6, 1#10(G), 1"C.	1,6
10A	1.5 TON ROOFTOP HEAT PUMP	**208**V–1ø	2.080	–	30A–2P, NEMA 3R	AT UNIT	2#10, 1#10(G), 3/4"C.	1,5
10B	2 TON ROOFTOP HEAT PUMP	**208**V–1ø	2.704	–	30A–2P, NEMA 3R	AT UNIT	2#10, 1#10(G), 3/4"C.	1,5
10C	2.5 TON ROOFTOP HEAT PUMP	**208**V–1ø	3.328	–	30A–2P, NEMA 3R	AT UNIT	2#8, 1#10(G), 3/4"C.	1,5
11	EXHAUST FAN	120V–1ø	0.055	–	INTEGRAL	CEILING	2#12, 1#12(G), 3/4"C.	1,3

APPLIANCE/EQUIPMENT CONNECTION SCHEDULE NOTES:

1. VERIFY ALL EQUIPMENT SPECIFICATIONS (VOLTAGE, CONNECTION, ETC.) WITH OWNER AND/OR ARCHITECT PRIOR TO PLACING PURCHASE ORDER FOR DEVICES AND ROUGH–IN. VERIFY THE EXACT MOUNTING HEIGHT WITH ARCHITECTURAL ELEVATIONS, MILLWORK, ETC. PRIOR TO ROUGH–IN.
2. PROVIDE A DUPLEX RECEPTACLE TO SERVE DISHWASHER AND DISPOSER PROTECTED BY AN UPSTREAM GFCI TYPE CIRCUIT BREAKER. COORDINATE LOCATION OF DEVICE IN MILLWORK WITH ARCHITECT/G.C. AT ISLAND LOCATIONS, INSTALL AN AIR PUSH BUTTON TYPE BUTTON (PROVIDED BY P.C.) FOR CONTROL OF DISHWASHER. INCLUDE ALL COSTS OF CORE DRILLING MILLWORK/SINK TO INSTALL PUSH BUTTON. ENSURE INSTALLED BUTTON ALLOWS FOR DISHWASHER TO BE SERVED BY THE SAME RECEPTACLE. AT NON–ISLAND KITCHEN COUNTER LOCATIONS, PROVIDE A SINGLE POLE SWITCH AT THE KITCHEN BACKSPLASH (FINAL LOCATION BY ARCHITECT). SPLIT–WIRE THE RECEPTACLE AND WIRE SUCH THE SWITCH ONLY CONTROLS THE TOP OUTLET. **ALL FINAL DEVICE LOCATIONS SHALL BE A.D.A. COMPLIANT IN ADA COMPLIANT UNITS.**
3. CONNECT TO INTEGRAL DISCONNECT SWITCH AND CONNECT TO APARTMENT UNIT LIGHTING CIRCUIT. VERIFY EXACT REQUIREMENTS WITH EQUIPMENT. WIRE SUCH THAT EXHAUST FAN RECEIVES CONSTANT LINE VOLTAGE POWER AHEAD OF ALL SWITCHING AND CONTROL. **REFER TO DETAIL ON THIS SHEET.**
4. PROVIDE A REDUNDANT (3–WAY) SWITCH IN ADA UNITS FOR CONTROL OF RANGE HOOD. SWITCH SHALL BE LOCATED IN ADA–COMPLIANT LOCATION. MODIFY RANGE HOOD WIRING AS REQUIRED (REFER TO HOOD UNIT WIRING DIAGRAM) FOR ADDITIONAL CONTROL LOCATION.
5. BRANCH CIRCUITING SIZED TO ACCOUNT FOR VOLTAGE DROP.
6. INSTALL DISCONNECT IN CLOSET SUCH THAT NEC #110.26 WORKING CLEARANCE IS MAINTAINED.

DWELLING UNIT LUMINAIRE SCHEDULE

TYPE	LAMP	WATTAGE	VOLTAGE	DIMMING TYPE	DESCRIPTION	CATALOG NUMBER	EQUIVALENT MANUFACTURER/REMARKS
M–1	–	–	120	–	LED MIRROR LIGHT USED IN BATHROOMS.	–	–
PL–1	–	–	120	–	LED PENDANT LIGHT USED THROUGHOUT THE UNITS.	–	–
PL–2	–	–	120	–	LED PENDANT USED OVER BATHROOM SHOWERS.	–	–
UC–1	–	–	120	–	LED UNDERCOUNTER LIGHT USED IN KITCHEN AREA.	–	–

LUMINAIRE SCHEDULE GENERAL NOTES:

1. VERIFY ALL LUMINAIRE COLORS, TRIMS, LENGTHS, ETC. WITH THE **ARCHITECT/OWNER/DEVELOPER/DESIGNER** PRIOR TO PLACING FINAL PURCHASE ORDERS. SUBMISSION OF SHOP DRAWINGS WILL BE INTERPRETED AS HAVING BEEN COORDINATED WITH THE AFOREMENTIONED PARTIES.

2. PROVIDE ALL LENGTHS, FEEDS, ACCESSORIES, CONNECTORS, WIRING, POWER SUPPLIES, DRIVERS, ETC. FOR A COMPLETE INSTALLATION. VERIFY THE COMPLETE BILL OF MATERIAL WITH MANUFACTURER'S REPRESENTATIVE AND ENSURE ALL EQUIPMENT IS INCLUDED IN BID PRICE. COORDINATE INSTALLATION WITH ARCHITECTURAL DETAILS.

3. VERIFY FINAL LUMINAIRE LOCATIONS WITH OTHER CEILING MOUNTED EQUIPMENT SUCH AS DIFFUSERS, FIRE ALARM DEVICES, SPEAKERS, ETC., WITH ARCHITECTURAL REFLECTED CEILING PLANS.

4. VERIFY EXACT MOUNTING HEIGHT AND LOCATIONS OF ALL WALL MOUNTED AND PENDANT MOUNTED LUMINAIRES WITH ARCHITECTURAL PLANS AND ELEVATIONS PRIOR TO ROUGH–IN.

5. ANY PROPOSED ALTERNATE LUMINAIRES SHALL BE APPROVED BY THE **ARCHITECT/OWNER/DEVELOPER/DESIGNER** PRIOR TO FINAL BID PRICING.

6. SHOULD THE CONTRACTOR PROPOSE TO FURNISH MATERIALS, EQUIPMENT, AND DEVICES, OTHER THAN THOSE SPECIFIED AND LISTED, THE CONTRACTOR SHALL SUBMIT A WRITTEN REQUEST FOR SUBSTITUTIONS, TO THE ENGINEER AT LEAST TEN (10) BUSINESS DAYS PRIOR TO BID OPENING. THE REQUEST SHALL BE AN ALTERNATE TO THE ORIGINAL BID AND SHALL INCLUDE A COMPLETE SPECIFICATIONS CUT SHEET SUBMITTAL AS OUTLINED IN THE SPECIFICATIONS, COMPLETE WITH DESCRIPTIVE (MANUFACTURER, BRAND NAME, CATALOG NUMBER, ETC.) AND TECHNICAL DATA FOR ALL ITEMS. INDICATE ANY ADDITIONS OR DEDUCTIONS TO THE CONTRACT PRICE WITH THE SUBSTITUTION SUBMITTAL AND ON THE BID FORM.

7. ANY FINAL SELECTED/INSTALLED LIGHT FIXTURES WITHOUT A SPECIFICATION (OR TO BE SPECIFIED LATER BY **ARCHITECT/OWNER/DEVELOPER/DESIGNER**) LISTED IN THIS SCHEDULE SHALL NOT EXCEED THE WATTAGE LISTED, SHALL NOT HAVE A LOWERED DELIVERED LUMENS (IF LISTED), AND SHALL MATCH THE COLOR TEMPERATURE (IF LISTED).

TYPICAL 1 BED & 2 BED DWELLING UNIT ELECTRIC SERVICE CALC.

DESCRIPTION	AREA	LOAD	TOTAL WC	N.E.C. ARTICLE
RECEPTACLES/LIGHTING	1200	3	3600	220.82(B)(1)
SMALL APPLIANCE CIRCUITS (2 x 1500)	-	3000	3000	220.82(B)(2)
LAUNDRY CIRCUIT (WASHING MACHINE)	-	1500	1500	220.82(B)(2)
REFRIGERATOR	-	600	600	220.82(B)(3)
ELECTRIC RANGE/OVEN	-	8000	8000	220.82(B)(3)
RANGE HOOD/MICROWAVE	-	1680	1680	220.82(B)(3)
DISHWASHER/DISPOSER	-	1200	1200	220.82(B)(3)
ELECTRIC DRYER	-	5000	5000	220.82(B)(3)
ELECTRIC WATER HEATER	-	4500	4500	220.82(B)(3)
TOTAL CONNECTED GENERAL LOAD	-	-	29080	-
100% OF FIRST 10kW + 40% OF REMAINING	-	-	17632	220.82(B)
INTERIOR AIR HANDLER	-	5824	5824	-
ROOFTOP HEAT PUMP	-	2704	2704	-
TOTAL CONNECTED HVAC LOAD	-	-	8528	-
65% OF AIR HAND(S). + 100% OF HEAT PUMP(S)	-	-	6490	220.82(C)(3)
TOTAL DEMAND LOAD	**-**	**-**	**24,122 W**	-
AMPS AT 120/208V-1Ø-3W			**116.0 A**	

TYPICAL 3 BED DWELLING UNIT ELECTRIC SERVICE CALC.

DESCRIPTION	AREA	LOAD	TOTAL WC	N.E.C. ARTICLE
RECEPTACLES/LIGHTING	1450	3	4350	220.82(B)(1)
SMALL APPLIANCE CIRCUITS (2 x 1500)	-	3000	3000	220.82(B)(2)
LAUNDRY CIRCUIT (WASHING MACHINE)	-	1500	1500	220.82(B)(2)
REFRIGERATOR	-	600	600	220.82(B)(3)
ELECTRIC RANGE/OVEN	-	8000	8000	220.82(B)(3)
RANGE HOOD/MICROWAVE	-	1680	1680	220.82(B)(3)
DISHWASHER/DISPOSER	-	1200	1200	220.82(B)(3)
ELECTRIC DRYER	-	5000	5000	220.82(B)(3)
ELECTRIC WATER HEATER	-	4500	4500	220.82(B)(3)
TOTAL CONNECTED GENERAL LOAD	-	-	29830	-
100% OF FIRST 10kW + 40% OF REMAINING	-	-	17932	220.82(B)
INTERIOR AIR HANDLER	-	8528	8528	-
ROOFTOP HEAT PUMP	-	3328	3328	-
TOTAL CONNECTED HVAC LOAD	-	-	11856	-
65% OF AIR HAND(S). + 100% OF HEAT PUMP(S)	-	-	8871	220.82(C)(3)
TOTAL DEMAND LOAD	**-**	**-**	**26,803 W**	-
AMPS AT 120/208V-1Ø-3W			**128.9 A**	

TYPICAL DWELLING UNIT PANEL SCHEDULE

NAME: TP1 | VOLTAGE: 120/**208**–1ø–3W
MINIMUM KAIC: 10 | TYPE: FLUSH MOUNTED
MCB: 125A MLO | KWD: SEE ADJACENT CALC.

	CKT.	DESCRIPTION	C/B	ø	C/B	DESCRIPTION	CKT.	
A,G	1	SMALL APPLIANCE REC.	20/1	A	20/1	REFRIGERATOR	2	A,G
A,G	3	SMALL APPLIANCE REC.	20/1	B	50/2	ELECTRIC RANGE	4	G2
A	5	LIVING REC.	20/1	A			6	
G	7	BATHROOM REC.	20/1	B	20/1	RANGE HOOD/MICRO.	8	A,G
A	9	LIGHTING/S.D.	20/1	A	20/1	DISHWASHER/DISPOSER	10	A,G
A	11	BEDROOM REC.	20/1	B	20/1	WASHING MACHINE	12	A,G
	13	SPACE	–	A	30/2	ELECTRIC DRYER	14	G2
	15	SPACE	–	B			16	
	17	SURGE PROTECTION	**–**	A	30/2	ELECTRIC WATER HEATER	18	
	19			B			20	
	21	AIR HANDLER	**25**/2	A	**15**/2	ROOFTOP HEAT PUMP	22	
	23			B			24	

TYPICAL DWELLING UNIT PANEL SCHEDULE NOTES:

A: PROVIDE AN ARC–FAULT CIRCUIT INTERRUPTER (AFCI) TYPE CIRCUIT BREAKER FOR THIS BRANCH CIRCUIT.

G: PROVIDE GROUND–FAULT CIRCUIT INTERRUPTER (GFCI) PROTECTION FOR THIS BRANCH CIRCUIT. PROVIDE A GFCI–TYPE CIRCUIT BREAKER IF GFCI–TYPE RECEPTACLES/DEVICES ARE NOT PROVIDED DOWNSTREAM ON THIS BRANCH CIRCUIT.

G2: PROVIDE A GROUND–FAULT CIRCUIT INTERRUPTER (GFCI) TYPE CIRCUIT BREAKER FOR THIS BRANCH CIRCUIT.

TYPICAL DWELLING UNIT PANEL SCHEDULE

NAME: TP2 | VOLTAGE: 120/**208**–1ø–3W
MINIMUM KAIC: 10 | TYPE: FLUSH MOUNTED
MCB: 125A MLO | KWD: SEE ADJACENT CALC.

	CKT.	DESCRIPTION	C/B	ø	C/B	DESCRIPTION	CKT.	
A,G	1	SMALL APPLIANCE REC.	20/1	A	20/1	REFRIGERATOR	2	A,G
A,G	3	SMALL APPLIANCE REC.	20/1	B	50/2	ELECTRIC RANGE	4	G2
A	5	LIVING REC.	20/1	A			6	
G	7	BATHROOM REC.	20/1	B	20/1	RANGE HOOD/MICRO.	8	A,G
A	9	LIGHTING/S.D.	20/1	A	20/1	DISHWASHER/DISPOSER	10	A,G
A	11	BEDROOM REC.	20/1	B	20/1	WASHING MACHINE	12	A,G
A	13	BEDROOM REC.	20/1	A	30/2	ELECTRIC DRYER	14	G2
G	15	BATHROOM REC.	20/1	B			16	
	17	SURGE PROTECTION	**–**	A	30/2	ELECTRIC WATER HEATER	18	
	19			B			20	
	21	AIR HANDLER	**30**/2	A	**20**/2	ROOFTOP HEAT PUMP	22	
	23			B			24	

TYPICAL DWELLING UNIT PANEL SCHEDULE NOTES:

A: PROVIDE AN ARC–FAULT CIRCUIT INTERRUPTER (AFCI) TYPE CIRCUIT BREAKER FOR THIS BRANCH CIRCUIT.

G: PROVIDE GROUND–FAULT CIRCUIT INTERRUPTER (GFCI) PROTECTION FOR THIS BRANCH CIRCUIT. PROVIDE A GFCI–TYPE CIRCUIT BREAKER IF GFCI–TYPE RECEPTACLES/DEVICES ARE NOT PROVIDED DOWNSTREAM ON THIS BRANCH CIRCUIT.

G2: PROVIDE A GROUND–FAULT CIRCUIT INTERRUPTER (GFCI) TYPE CIRCUIT BREAKER FOR THIS BRANCH CIRCUIT.

TYPICAL DWELLING UNIT PANEL SCHEDULE

NAME: TP3 | VOLTAGE: 120/**208**–1ø–3W
MINIMUM KAIC: 10 | TYPE: FLUSH MOUNTED
MCB: 150A MLO | KWD: SEE ADJACENT CALC.

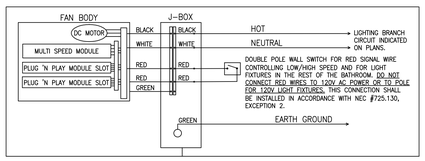
	CKT.	DESCRIPTION	C/B	ø	C/B	DESCRIPTION	CKT.	
A,G	1	SMALL APPLIANCE REC.	20/1	A	20/1	REFRIGERATOR	2	A,G
A,G	3	SMALL APPLIANCE REC.	20/1	B	50/2	ELECTRIC RANGE	4	G2
A	5	LIVING REC.	20/1	A			6	
G	7	BATHROOM REC.	20/1	B	20/1	RANGE HOOD/MICRO.	8	A,G
A	9	LIGHTING/S.D.	20/1	A	20/1	DISHWASHER/DISPOSER	10	A,G
A	11	BEDROOM REC.	20/1	B	20/1	WASHING MACHINE	12	A,G
A	13	BEDROOM REC.	20/1	A	30/2	ELECTRIC DRYER	14	G2
G	15	BATHROOM REC.	20/1	B			16	
	17	SURGE PROTECTION	**–**	A	30/2	ELECTRIC WATER HEATER	18	
	19			B			20	
	21	INTERIOR AIR HANDLER	**45**/2	A	**25**/2	ROOFTOP HEAT PUMP	22	
	23			B			24	
A	25	BEDROOM REC.	20/1	A	–	SPACE	26	
	27	SPACE	–	B	–	SPACE	28	

TYPICAL DWELLING UNIT PANEL SCHEDULE NOTES:

A: PROVIDE AN ARC–FAULT CIRCUIT INTERRUPTER (AFCI) TYPE CIRCUIT BREAKER FOR THIS BRANCH CIRCUIT.

G: PROVIDE GROUND–FAULT CIRCUIT INTERRUPTER (GFCI) PROTECTION FOR THIS BRANCH CIRCUIT. PROVIDE A GFCI–TYPE CIRCUIT BREAKER IF GFCI–TYPE RECEPTACLES/DEVICES ARE NOT PROVIDED DOWNSTREAM ON THIS BRANCH CIRCUIT.

G2: PROVIDE A GROUND–FAULT CIRCUIT INTERRUPTER (GFCI) TYPE CIRCUIT BREAKER FOR THIS BRANCH CIRCUIT.



DWELLING UNIT BATHROOM EXHAUST FAN WIRING DIAGRAM
NO SCALE

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

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2012 West 25th Street, Suite 900
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Phone: 216 227 8505

OSU AIRPORT WEST

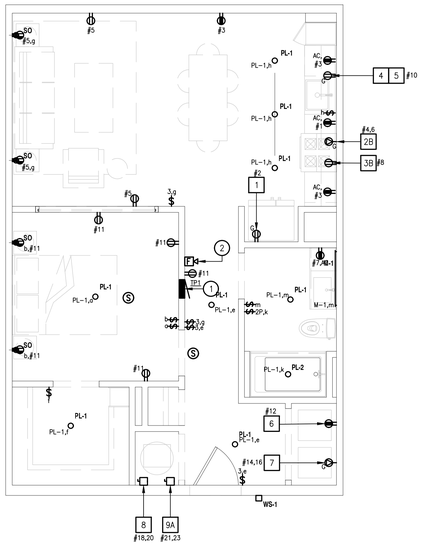
BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

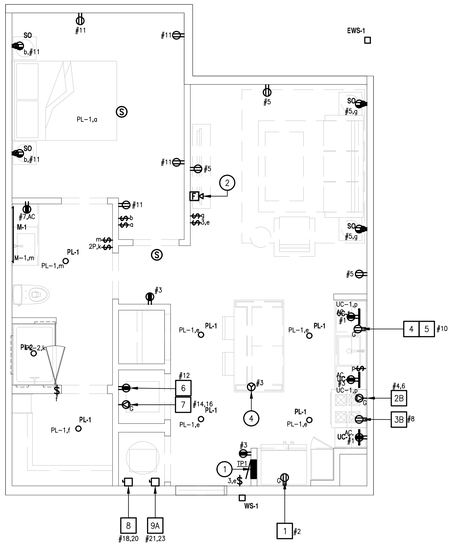
Drawing Title

TYPICAL DWELLING UNIT ELECTRICAL DETAILS

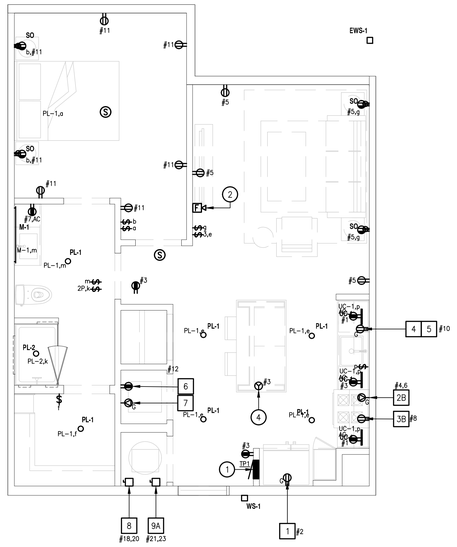
Date: 03.05.2026	Project Number: 24-065
Drawn by: DJS	Drawing Number: E3.00A
Checked by: KDS	
Approved: GSS	



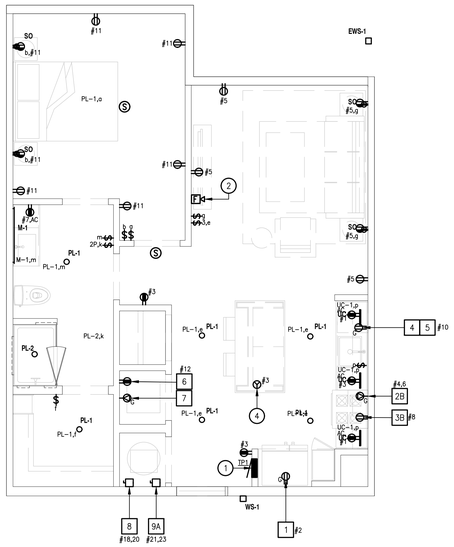
1 A3.01A BUILDING 1
UNIT TYPE S1 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"



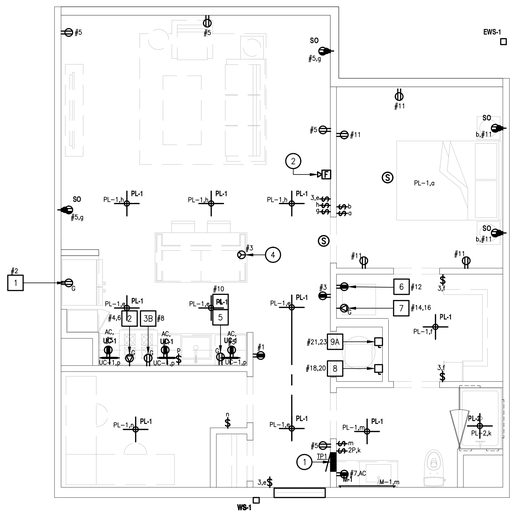
2 A3.01A BUILDING 1
UNIT TYPE A1 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"



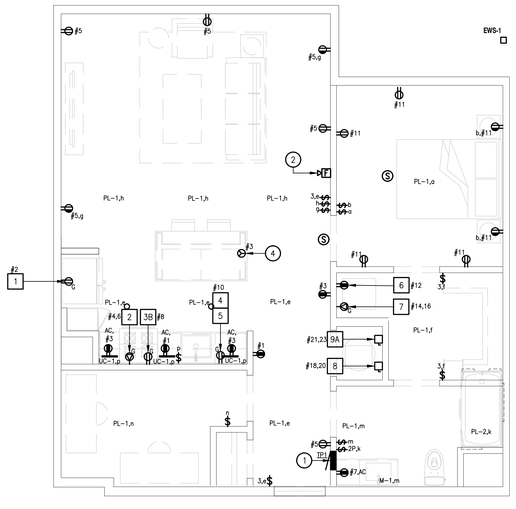
3 A3.01A BUILDING 1
UNIT TYPE A1.1 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"



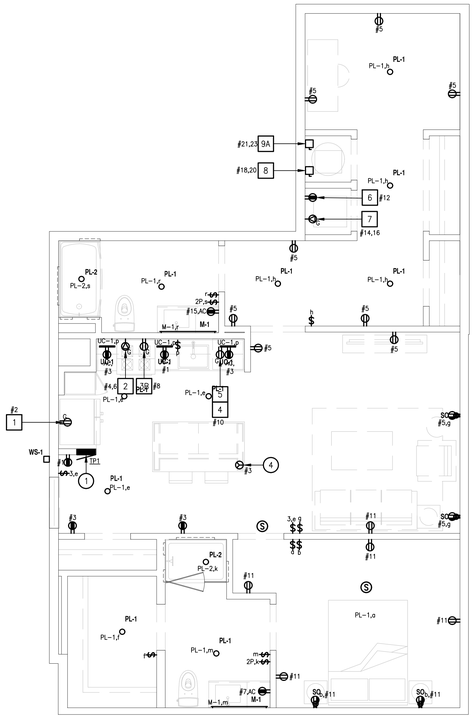
4 A3.01A BUILDING 1
UNIT TYPE A1.2 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"



5 A3.01A BUILDING 1
UNIT TYPE AD2 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"



6 A3.01A BUILDING 1
UNIT TYPE AD2 PW ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"



7 A3.01A BUILDING 1
UNIT TYPE AD3 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"

GENERAL DWELLING UNIT ELECTRICAL PLAN NOTES

- ALL LIGHT FIXTURES, CEILING FANS, AND BATHROOM/KITCHEN EXHAUST FANS SHALL BE CONNECTED TO CIRCUIT **#9** IN RESPECTIVE UNIT LOAD CENTER UNLESS SHOWN OTHERWISE.
- REFER TO SHEET **#E3-00** FOR ALL UNIT DETAILS, INCLUDING APPLIANCE SCHEDULES.
- FLOOR RECEPTACLES MAY BE INSTALLED IN LIEU OF WALL RECEPTACLES SHOULD WINDOWS/GLAZING OR SLIDING DOOR RECESSED POCKETS INTERFERE WITH WALL MOUNTING AT ANY LOCATION WITHIN THE DWELLING UNIT. FLOOR RECEPTACLES SHALL BE INSTALLED WITHIN 18" OF WINDOWS/WALLS OR PER NEC #210.52(A)(3).

CODED NOTES

1. LOCATION OF NEW TENANT CIRCUIT BREAKER LOAD CENTER. REFER TO TYPICAL UNIT CIRCUIT BREAKER PANEL SCHEDULE ON SHEET **#E3-00** FOR ADDITIONAL INFORMATION. ALL LIVING AREA BRANCH CIRCUITS SHALL BE PROVIDED WITH ARC-FAULT CIRCUIT INTERRUPTER PROTECTION IN ACCORDANCE WITH N.E.C. ARTICLE #210.12(A).
2. THIS FIRE ALARM DEVICE SHALL BE CONNECTED TO THE BASE BUILDING FIRE ALARM SYSTEM. VERIFY FINAL LOCATION WITH A.H.J. AND FIRE MARSHAL.
3. LOCATION OF STRUCTURED MEDIA ENCLOSURE "SME". REFER TO DETAIL ON DRAWING **#E3-00** FOR ADDITIONAL INFORMATION. PROVIDE POWER FROM NEAREST LIVING AREA BRANCH CIRCUIT.
4. COORDINATE FINAL LOCATION OF JUNCTION BOX FOR FUTURE KITCHEN COUNTER RECEPTACLE WITH G.C. AND OWNERSHIP. ROUTE AN EMPTY 1"C. WITH PULLWIRE TO NEAREST SMALL APPLIANCE BRANCH CIRCUIT (#1 OR #3) FOR FUTURE WIRING. THE FINAL INSTALLATION SHALL BE IN COMPLIANCE WITH NEC SECTION 210.52(C)(3).

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

whs engineering
2012 West 25th Street, Suite 900
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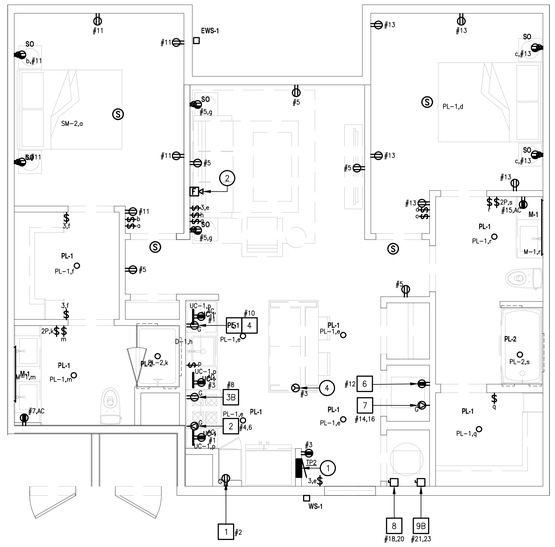
Phone: 216 227 8505

OSU AIRPORT WEST
BUILDING A

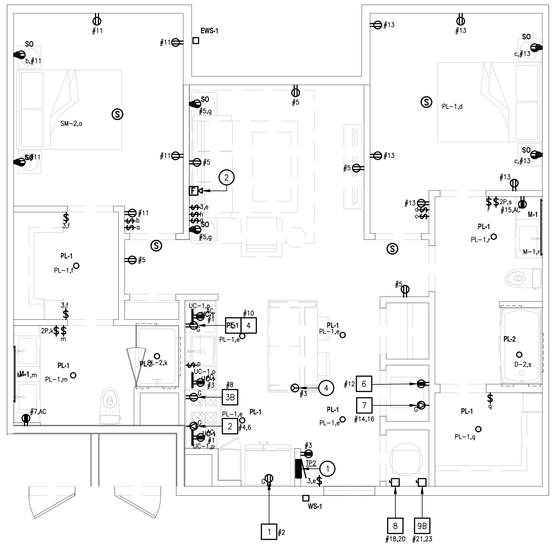
3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title
TYPICAL DWELLING UNIT ELECTRICAL PLANS

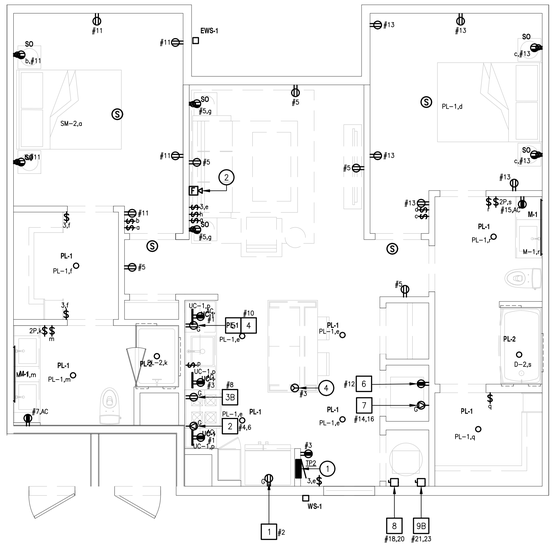
Date 03.05.2026	Project Number 24-065
Drawn by DJS	Drawing Number E3.01A
Reviewed by KDS	
Approved GSS	



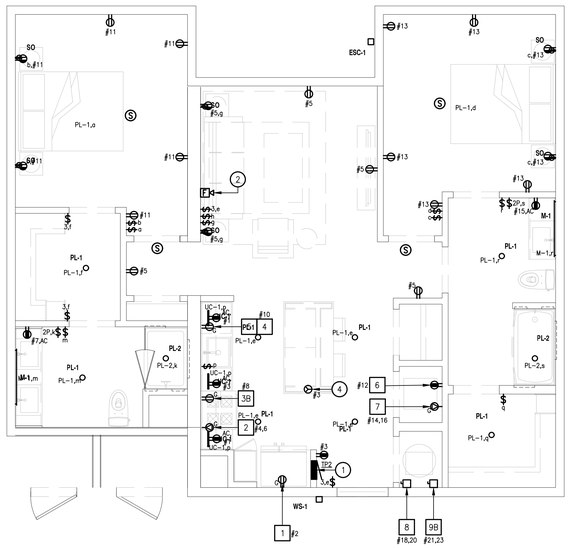
1 A3.02A BUILDING 1
UNIT TYPE B1 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"



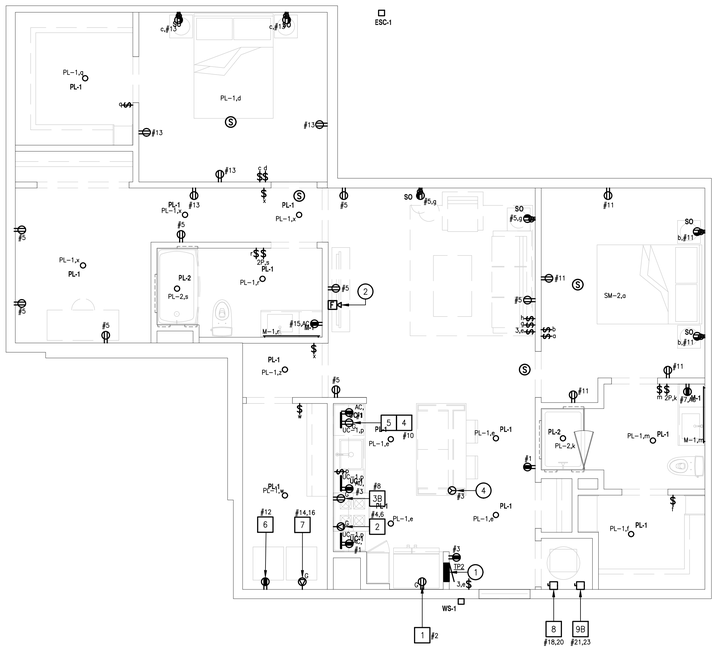
2 A3.02A BUILDING 1
UNIT TYPE B1.1 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"



3 A3.02A BUILDING 1
UNIT TYPE B1.2 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"



4 A3.02A BUILDING 1
UNIT TYPE B1.3 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"



5 A3.02A BUILDING 1
UNIT TYPE BD3.1 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"

GENERAL DWELLING UNIT ELECTRICAL PLAN NOTES

- ALL LIGHT FIXTURES, CEILING FANS, AND BATHROOM/KITCHEN EXHAUST FANS SHALL BE CONNECTED TO CIRCUIT #9 IN RESPECTIVE UNIT LOAD CENTER UNLESS SHOWN OTHERWISE.
- REFER TO SHEET #E3-00 FOR ALL UNIT DETAILS, INCLUDING APPLIANCE SCHEDULES.
- FLOOR RECEPTACLES MAY BE INSTALLED IN LIEU OF WALL RECEPTACLES SHOULD WINDOWS/GLAZING OR SLIDING DOOR RECESSED POCKETS INTERFERE WITH WALL MOUNTING AT ANY LOCATION WITHIN THE DWELLING UNIT. FLOOR RECEPTACLES SHALL BE INSTALLED WITHIN 18" OF WINDOWS/WALLS OR PER NEC #210.52(A)(3).

CODED NOTES

1. LOCATION OF NEW TENANT CIRCUIT BREAKER LOAD CENTER. REFER TO TYPICAL UNIT CIRCUIT BREAKER PANEL SCHEDULE ON SHEET #E3-00 FOR ADDITIONAL INFORMATION. ALL LIVING AREA BRANCH CIRCUITS SHALL BE PROVIDED WITH ARC-FAULT CIRCUIT INTERRUPTER PROTECTION IN ACCORDANCE WITH N.E.C. ARTICLE #210.12(A).
2. THIS FIRE ALARM DEVICE SHALL BE CONNECTED TO THE BASE BUILDING FIRE ALARM SYSTEM. VERIFY FINAL LOCATION WITH A.H.J. AND FIRE MARSHAL.
3. LOCATION OF STRUCTURED MEDIA ENCLOSURE "SME". REFER TO DETAIL ON DRAWING #E3-00 FOR ADDITIONAL INFORMATION. PROVIDE POWER FROM NEAREST LIVING AREA BRANCH CIRCUIT.
4. COORDINATE FINAL LOCATION OF JUNCTION BOX FOR FUTURE KITCHEN COUNTER RECEPTACLE WITH G.C. AND OWNERSHIP. ROUTE AN EMPTY 1"C. WITH PULLWIRE TO NEAREST SMALL APPLIANCE BRANCH CIRCUIT (#1 OR #3) FOR FUTURE WIRING. THE FINAL INSTALLATION SHALL BE IN COMPLIANCE WITH NEC SECTION 210.52(C)(3).

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

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Cleveland, OH 44113
Phone: 216 227 8505

OSU AIRPORT WEST
BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title
TYPICAL DWELLING UNIT ELECTRICAL PLANS

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number
Checked by: KDS	E3.02A
Approved: GSS	

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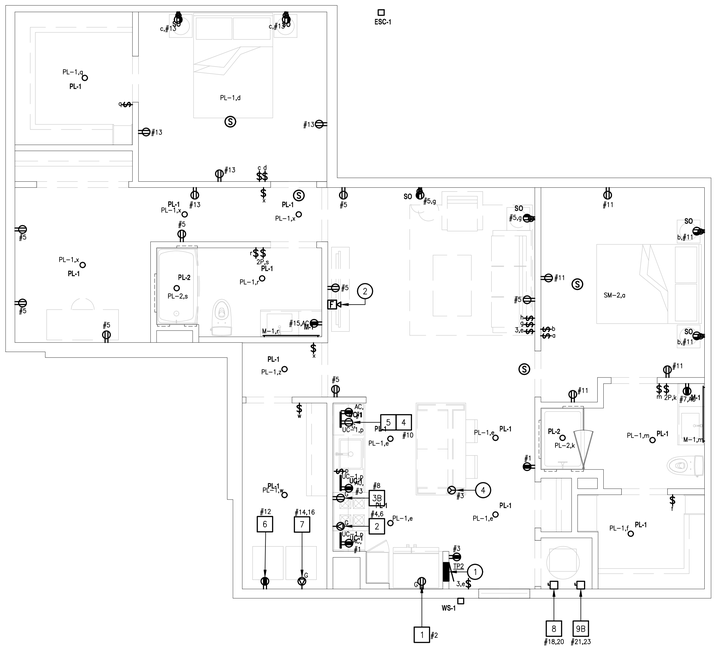
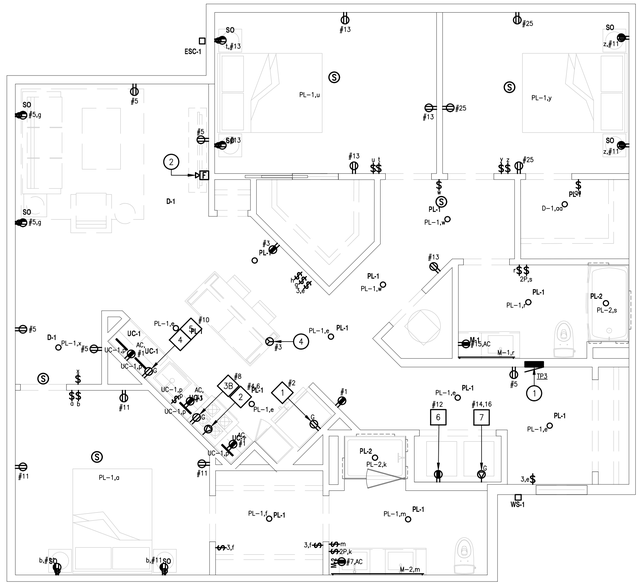


1 / A3.03A — BUILDING 1
UNIT TYPE BD3.2 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"



2 / A3.03A — BUILDING 1
UNIT TYPE C3 ELECTRICAL PLAN
SCALE: 1/4" = 1'-0"

GENERAL DWELLING UNIT ELECTRICAL PLAN NOTES

- ALL LIGHT FIXTURES, CEILING FANS, AND BATHROOM/KITCHEN EXHAUST FANS SHALL BE CONNECTED TO CIRCUIT **#9** IN RESPECTIVE UNIT LOAD CENTER UNLESS SHOWN OTHERWISE.
- REFER TO SHEET **#E3-00** FOR ALL UNIT DETAILS, INCLUDING APPLIANCE SCHEDULES.
- FLOOR RECEPTACLES MAY BE INSTALLED IN LIEU OF WALL RECEPTACLES SHOULD WINDOWS/GLAZING OR SLIDING DOOR RECESSED POCKETS INTERFERE WITH WALL MOUNTING AT ANY LOCATION WITHIN THE DWELLING UNIT. FLOOR RECEPTACLES SHALL BE INSTALLED WITHIN 18" OF WINDOWS/WALLS OR PER NEC #210.52(A)(3).

CODED NOTES

1. LOCATION OF NEW TENANT CIRCUIT BREAKER LOAD CENTER. REFER TO TYPICAL UNIT CIRCUIT BREAKER PANEL SCHEDULE ON SHEET **#E3-00** FOR ADDITIONAL INFORMATION. ALL LIVING AREA BRANCH CIRCUITS SHALL BE PROVIDED WITH ARC-FAULT CIRCUIT INTERRUPTER PROTECTION IN ACCORDANCE WITH N.E.C. ARTICLE #210.12(A).
2. THIS FIRE ALARM DEVICE SHALL BE CONNECTED TO THE BASE BUILDING FIRE ALARM SYSTEM. VERIFY FINAL LOCATION WITH A.H.J. AND FIRE MARSHAL.
3. LOCATION OF STRUCTURED MEDIA ENCLOSURE "SME". REFER TO DETAIL ON DRAWING **#E3-00** FOR ADDITIONAL INFORMATION. PROVIDE POWER FROM NEAREST LIVING AREA BRANCH CIRCUIT.
4. COORDINATE FINAL LOCATION OF JUNCTION BOX FOR FUTURE KITCHEN COUNTER RECEPTACLE WITH G.C. AND OWNERSHIP. ROUTE AN EMPTY 1"C. WITH PULLWIRE TO NEAREST SMALL APPLIANCE BRANCH CIRCUIT (#1 OR #3) FOR FUTURE WIRING. THE FINAL INSTALLATION SHALL BE IN COMPLIANCE WITH NEC SECTION 210.52(C)(3).

Seal

NOT FOR CONSTRUCTION

Issue:	Date:	By:
SCHEMATIC DESIGN	01.07.26	BAC
DESIGN DEVELOPMENT	03.05.26	BAC

whs engineering
2012 West 25th Street, Suite 900
Cleveland, OH 44113
Phone: 216 227 8505

OSU AIRPORT WEST
BUILDING A

3455 W. Dublin Granville Rd, Columbus OH, 43235

Drawing Title
TYPICAL DWELLING UNIT ELECTRICAL PLANS

Date	Project Number
03.05.2026	24-065
Drawn by: DJS	Drawing Number
Checked by: KDS	E3.03A
Approved: GSS	

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